Empirical Research Methods for Operations Management

A Practical Handbook of Theory, Causal Inference, Econometrics, and Survey Research

Jia Yu

Version 1.0 | 2026

An independently compiled research handbook based on methodological and empirical literature in Operations Management and related fields.

Preface

Empirical research in Operations Management (OM) increasingly draws on a broad range of theoretical perspectives, data sources, research designs, and quantitative methods. Researchers are often required not only to identify an appropriate theoretical foundation, but also to determine how empirical data should be collected and analyzed, how potential sources of bias should be addressed, and which econometric or statistical techniques are appropriate for a particular research question.

This handbook provides a structured methodological map for empirical research in Operations Management. It brings together commonly used organizational theories, secondary-data research designs, causal inference approaches, econometric methods, and survey-based analytical techniques within a single reference framework. Rather than treating these topics as isolated methodological tools, the handbook emphasizes the relationships among the research question, data structure, identification problem, underlying assumptions, and choice of empirical method.

The handbook is organized around five recurring questions:

1. What theoretical perspective is appropriate for the research question?
2. What type of data and research design can provide relevant empirical evidence?
3. What threats to identification or inference may arise?
4. Which analytical method is appropriate under the characteristics of the research design and data?
5. What assumptions, diagnostics, and limitations should be considered when interpreting the results?

Accordingly, the discussion of individual methods focuses not only on their definitions and technical procedures, but also on four practical dimensions: when a method may be appropriate, the assumptions required for its use, its major limitations, and the diagnostic or validation procedures that should accompany the analysis.

The handbook covers both secondary-data and survey-based empirical research. Topics include organizational theories commonly employed in OM research; event studies and secondary-data resources; selection bias and quasi-experimental designs; endogeneity and related econometric remedies; instrumental-variable and panel-data approaches; difference-in-differences, matching, and related causal inference methods; as well as reliability, validity, structural equation modeling, common method bias, moderation, and mediation in survey research.

Examples from the Operations Management literature are used throughout the handbook to illustrate how different methodological approaches have been applied in empirical studies. These examples are intended to connect methodological concepts with actual research practice and to highlight the considerations involved in selecting and implementing an empirical strategy.

This handbook is intended as a practical research reference and methodological guide, rather than a substitute for specialized textbooks in econometrics, statistics, causal inference, or structural equation modeling. Many of the methods discussed here involve assumptions and technical issues that require more detailed treatment than can be provided in a single handbook. Readers are therefore encouraged to consult the original methodological literature and specialized texts when implementing a particular method in their own research.

Table of Contents

Part One: Organizational Theory

Transaction Costs Economics

1. Introduction

Transaction cost theory examines whether firms should make something or buy it instead (Coase, 1937; Williamson, 1998). It is often advantageous for firms to enter into trading or other types of agreements with other companies.

The costs of transactions are usually not measured directly but are estimated using proxies of critical dimensions of transactions (Jobin, 2008). Transaction costs include negotiating, monitoring, and enforcing contracts (Hill, 1990) and the costs of planning, adapting, and monitoring task completion (Williamson, 1985).

2. Assumption of the Theory

The theory assumes that two trading companies are risk neutral; deal with each other basically as equals; have extensive business experience; and employ specialized managerial, legal, technical, and financial experts. With these assumptions, rather than focus on differences in the trading partners (such as experienced versus naïve), the theory focuses on differences in contracting issues between the trading partners and on the costs involved in those contracts (Williamson, 1998).

3. Transaction

Transaction is the basic unit of analysis in transaction cost theory. A transaction has occurred when a good or service is transferred across an organizational boundary. All transactions contain conflict, mutuality, and order (Williamson, 2002).

There is a tripartite classification of transactions: (1) bargaining (2) managing (3) rationing (Commons, 1934). Bargaining transactions transfer ownership of wealth between equals by voluntary agreements. Managerial transactions create wealth by commands of legal superiors and authorities. Rationing transactions apportion the benefits and burdens of wealth creation processes through the actions of legal superiors and authorities.

According to the theory, transactions differ in a number of ways, such as the degree to which each party's relationship-specific assets are involved, the amount of uncertainty about the other party's actions and about the future in general, the complexity of the trading agreement, and the frequency with which transactions occur. These differences help a firm decide which governance structures are preferable.

4. Asset Specificity

Asset specificity is very important. It refers to (1) the extent to which assets that support a transaction are tailored to it and (2) the amount of opportunity costs for using these same assets for the next-best alternative, or for alternative users, should the transaction be prematurely terminated (Williamson, 1985).

There are four types of asset specificity: site, physical, human, and dedicated. Site specificity refers to highly immobile assets that remain in place to save transportation and inventory costs

(Williamson, 1985). Physical specificity refers to equipment and machinery that are specific to this relationship. Human specificity refers to human capital or other employee education and training that is specific to this relationship. Dedicated specificity refers to substantial investments that were made only for this transaction and have no value outside this transaction. When one party has specific assets, then it triggers opportunistic behaviors, which puts a firm at risk and requires costly contractual safeguards to deter those negative behaviors (Poppo & Zenger, 1998).

5. Arrangement in Transaction

Transactions must be governed, designed, and carried out by institutional arrangements or contracts between firms. Forms of governance structures have been described as existing along a spectrum from market prices at one end to a fully integrated firm at the other end. Forms of governance structures have been described as existing along a spectrum from market prices at one end to a fully integrated firm at the other end. There are many different types of arrangements between firms, such as complex contracts, partial ownership, hybrid modes and so on (Shelanski & Klein, 1995).

If parties to transactions are bilaterally dependent on each other, meaning that neither party can easily make alternative arrangements, then the parties are vulnerable. In response, the parties create value-preserving governance structures among the parties. These structures infuse order, which helps mitigate conflict and allows the firms to realize mutual gain. Transaction cost theory examines how trading companies protect themselves from the hazards associated with exchange relationships with other companies (Williamson, 1975, 1985). According to the theory, trading partners choose the most cost-effective arrangement or agreement that offers the best protection for their relationship-specific investments. Firms with lower transaction costs are higher performers than are firms with higher transaction costs (Williamson, 1985).

6. Governance Structure

The central idea in transaction cost theory is that organizations change because managers seek to economize their transaction costs (Williamson, 1985). The theory examines the costs of transactions in one governance structure versus another. Organizations will perform the best when they use governance structures that are the least expensive possible.

6.1.Collaborative Arrangement

The collaborative arrangements is also called relational governance, or alliances (Dyer, 1997). Relational or alliance governance exchanges may be more beneficial and practicable than other types of governance, such as when the market fails. However, relational governance exchanges may be hard to legally enforce as they are often open ended and require such mechanisms as trust, mutual dependence, parallel expectations, and fairness to sustain them. Geyskens, Steenkamp, and Kumar (2006) found strong support for transaction cost theory for both make-versus-buy and ally-versus-buy decisions.

Transaction cost research also examines how firms align their transactions so as to pursue multiple goals. Firms often assign different goals for different organizational units. These various goals can lead to complicated decisions about whether to make, buy, or use allies. Multiple decisions that are made by multiple entities can result in organizational transaction misalignment, which can impede organizational performance (Bidwell, 2010).

6.2.Integration

One of the main concepts in transaction cost theory is integration. The theory examines the costs involved in the merger of firm A with firm B. The benefits of firm A merging with firm B derive from the ability of the manager of firm A to give orders to the manager of firm B (a quantity mode) (Coase, 1937). Conversely, if firm A does not merge with (integrate with) firm B, then the manager of firm A must pay or have a contract with the manager of firm B to influence the manager of firm B to perform (a price mode). When two firms integrate, they move from a price mode to a quantity mode. When the quantity mode is more advantageous than the price mode, then the two firms will be more likely to integrate. There are conditions when the quantity mode is less efficient than the price mode. Under these circumstances, such as costs of increased bureaucracy and greater likelihood of managerial error in larger firms (Hart, 1988), the firms will tend not to integrate.

7. TCE in OM Field

In the field of OM, TCE is used to explain the motivation of companies to outsource. Low transaction cost is a key driver for a firm's outsourcing (or "buy") decision (Coase, 1937). With the help of global sourcing, firms are more capable to differentiate products by exploiting unique resources (Teece, 1986), increasing firms' bargaining power with their suppliers (Lampel & Giachetti, 2013), and reducing costs (Jiang et al., 2007; Lampel & Giachetti, 2013).

TCE also emphasizes the difficulties in global sourcing caused by transaction risk and coordination costs perspectives (Clemons et al., 1993; Grover & Malhotra, 2003). Transaction risk disrupts the global supply chain (Williamson, 2008) and has a negative impact on operational continuity and efficiency (Grover & Malhotra, 2003), while coordination costs are related to actions taken to promote information sharing, production rationalization, and process standardization (Clemons et al., 1993; Lampel & Giachetti, 2013). In the field of OM, two streams of literature has been developed, which are supply chain risk management and supply chain diversification. A theoretical framework, drawn on supply chain risk and supply chain diversification, are constructed to explain how trade friction, created by tariff trade barriers, affects the operational performance of domestic firms which source from the affected countries, as well as how various supply chain characteristics and strategies can moderate the impact of such trade friction (Fan et al., 2022).

References:

Bidwell, M. (2010). Problems deciding: How the structure of make-or-buy decisions leads to transaction misalignment. *Organization Science*, 21, 362–379.

Chiles, T. H., & McMackin, J. F. (1996). Integrative variable risk preferences, trust, and transaction cost economies. *Academy of Management Review*, 21, 73–99.

Coase, R. H. (1937, November). The nature of the firm. *Economica*, 4, 386–405. Commons, J. R.

(1934). Institutional economics: Its place in political economy. *Madison: University of Wisconsin Press*.

Dyer, J. H. (1997). Effective interfirm collaboration: How firms minimize transaction costs and maximize transaction value. *Strategic Management Journal*, 18, 535–556.

Dyer, J. H., & Chu, W. (2003). The role of trustworthiness in reducing transaction costs and improving performance: Empirical evidence from the United States, Japan, and Korea. *Organization Science*, 14, 57–68.

Fan, D., Zhou, Y., Yeung, A. C. L., Lo, C. K. Y., & Tang, C. (2022). Impact of the U.S.–China trade war on the operating performance of U.S. firms: The role of outsourcing and supply base complexity. *Journal of Operations Management*, *68*(8), 928–962. https://doi.org/10.1002/joom.1225

Geyskens, I., Steenkamp, J.-B. E. M., & Kumar, N. (2006). Make, buy, or ally: A transaction cost theory meta-analysis. *Academy of Management Journal*, 49, 519–543.

Ghoshal, S., & Moran, P. (1996). Bad for practice: A critique of the transaction cost theory. *Academy of Management Review*, 21, 13–47.

Hart, O. D. (1988, Spring). Incomplete contracts and the theory of the firm. *Journal of Law, Economics, and Organization*, 14, 119–139.

Hill, C. W. L. (1990). Cooperation, opportunism, and the invisible hand: Implications for transaction cost theory. *Academy of Management Journal*, 15, 500–513.

Jiang, S., Yeung, A. C. L., Han, Z., & Huo, B. (2023). The effect of customer and supplier concentrations on firm resilience during the COVID -19 pandemic: Resource dependence and power balancing. *Journal of Operations Management*, *69*(3), 497–518. https://doi.org/10.1002/joom.1236

Jobin, D. (2008). A transaction cost-based approach to partnership performance evaluation. *Evaluation*, 14, 437–465.

Macher, J. T., & Richman, B. D. (2008). Transaction cost economics: An assessment of empirical research in the social sciences. *Business and Politics*, 10(1), 1–63.

Poppo, L., & Zenger, T. (1998). Testing alternative theories of the firm: Transaction cost, knowledge-based, and measurement explanations for make-or-buy decisions in information services. *Strategic Management Journal*, 19, 853–877.

Poppo, L., & Zenger, T. (2002). Do formal contracts and relational governance function as substitutes or complements? *Strategic Management Journal*, 23, 707–725.

Shelanski, H. A., & Klein, P. G. (1995, October). Empirical research in transaction cost economies: A review and assessment. *Journal of Law, Economics*, and Organization, 11, 335–361.

Williamson, O. E. (1975). Markets and hierarchies: Analysis and antitrust implications. *New York: Free Press.*

Williamson, O. E. (1985). The economic institutions of capitalism. *New York: Free Press*.

Williamson, O. E. (1998). Transaction cost economics: How it works; where it is headed. *De Economist*, 146, 23–58.

Williamson, O. E. (2002). The theory of the firm as governance structure: From choice to contract. *Journal of Economic Perspectives*, 16(3), 171–195.

Williamson, O. E. (2010, June). Transaction cost economics: The natural progression. *American Economic Review*, 100, 673–690.

Institutional Theory

1. Introduction

Institutional theory addresses the central question of why all organizations in a field tend to look and act the same (DiMaggio & Powell, 1983). The core concept of institutional theory is that organizational structures and processes tend to acquire meaning and achieve stability, rather than on the basis of their effectiveness and efficiency in achieving desired ends, such as the mission and goals of the organization (Lincoln, 1995).

2. Isomorphism

Institutional theory points out that institutions are an important component in the environment. And institutions are defined as “regulative, normative, and cognitive structures and activities that provide stability and meaning for social behavior” (Scott, 1995). Institutions have a constraining influence on organizations, called isomorphism, that forces organizations in the same population to resemble other organizations that face the same set of environmental conditions (Hawley, 1968).

Institutions exert three types of isomorphic pressure on organizations: coercive, normative, and mimetic (DiMaggio & Powell, 1983). Coercive isomorphism refers to pressure from entities who have resources on which an organization depends. Mimetic isomorphism refers to the imitation or copying of other successful organizations when an organization is uncertain about what to do. Normative isomorphism refers to following professional standards and practices established by education and training methods, professional networks, and movement of employees among firms.

3. Legitimacy

According to the theory, new organizational forms emerge once they are viewed by society as legitimate (Aldrich & Fiol, 1994). Legitimacy refers to the extent to which an organization’s actions are socially accepted and approved by various internal and external stakeholders (Kostova et al., 2008) and are consistent with widely held norms, rules, and beliefs (Sonpar, 2009).

There are three types of legitimacy that are pragmatic legitimacy, moral legitimacy, and cognitive legitimacy. Pragmatic legitimacy rests on the self-interested calculations of an organization's most immediate audiences (Suchman, 1995). It reflects direct exchange and influence relations between a focal organization and specific constituents; it is generally the easiest form of legitimacy to manipulate. Cognitive legitimacy can be broadly defined as how well organizations execute their activities from their stakeholder's point of view (Suchman, 1995).

Moral legitimacy rests not on judgments about whether a given activity benefits the evaluator, but rather on judgments about whether the activity is "the right thing to do." (Suchman, 1995). It reflects a positive normative evaluation of the organization and its activities. Moral legitimacy reflects a prosocial logic that differs fundamentally from narrow self-interest. And it takes one of three forms: evaluations of outputs and consequences, evaluations of techniques and procedures, and evaluations of categories and structures (Scott & Meyer, 1991). A fourth form, evaluations of leaders and representatives, is rarer but nonetheless conceptually important.

The nature of conformism by organizations varies, depending on whether the organization seeks primarily pragmatic, moral, or cognitive legitimacy. When organizations submit to institutional pressures and conform to social norms for certain organizational structures and processes, they are rewarded by earning increased legitimacy, resources, and survival capabilities for their operations (Oliver, 1997; Yang & Konrad, 2010).

4. Institutionalized Activity

Institutional theory posits that institutionalized activities occur due to influences on three levels: individual, organizational, and interorganizational (Oliver, 1997). On the individual level, managers follow norms, habits, customs, and traditions, both consciously and unconsciously (Berger & Luckmann, 1967). On the organizational level, shared political, social, cultural, and belief systems all support following traditions of institutionalized activities. On the interorganizational level, pressures from government, industry alliances, and expectations from society define what is socially acceptable and expected organizational behavior, which pressures organizations to look and act the same (DiMaggio & Powell, 1983).

Institutional theorists argue that many organizational actions are so taken for granted that managers no longer question why a specific action was started or why a specific action should continue (Oliver, 1997). In contrast, organizations act in creative ways to change their institutional environments, and this act is labeled as "institutional entrepreneurship."

5. Strategy and Actor

In response to institutional pressures and expectations to conform, organizations can adopt the following strategies (in ascending order of active organizational resistance): acquiescence, compromise, avoidance, defiance, or manipulation. Institutional entrepreneurs are actors who create new organizations or transform existing ones (DiMaggio, 1988; Garud, Hardy, & Maguire, 2007). The actors can be individuals, groups, organizations, or groups of organizations, but they must both initiate and implement divergent changes (Battilana, Leca, & Boxenbaum, 2009).

6. Development of Institutional Theory

The new (neo) institutional theory developed from the old one. Classical institutional theorists examined such issues as coalitions, competing values, influence, power, and informal structures (Greenwood & Hinings, 1996), cited as a "old" approach. The new (neo) institutional theorists examine organizations at the field level amid both competitive and cooperative exchanges with other organizations and focus on legitimate and "taken for granted" structures and processes.

References:

Aldrich, H., & Fiol, M. (1994). Fools rush in? The institutional context of industry creation. *Academy of Management Review*, 19, 645–670.

Battilana, J., Leca, B., & Boxenbaum, E. (2009). How actors change institutions: Towards a theory of institutional entrepreneurship. *Academy of Management Annals*, 3, 65–107.

Berger, P. L., & Luckmann, T. (1967). The social construction of reality. New York: Doubleday.

Dacin, M. T., Goodstein, J., & Scott, W. R. (2002). Institutional theory and institutional change. *Academy of Management Journal*, 45, 45–57.

Deephouse, D. L. (1999). To be different, or to be the same? It's a question (and theory) of strategic balance. *Strategic Management Journal*, 20, 147–166.

DiMaggio, P. (1988). Interest and agency in institutional theory. In L. Zucker (Ed.), *Institutional patterns and culture* (pp. 3–32). Cambridge, MA: Ballinger.

DiMaggio, P., & Powell, W. W. (1983). The iron cage revisited: Institutional isomorphism and collective rationality in organizational fields. *American Sociological Review*, 48, 147–160.

Garud, R., Hardy, C., & Maguire, S. (2007). Institutional entrepreneurship as embedded agency: An introduction to the special issue. *Organization Studies*, 28, 957–969.

Greenwood, R., & Hinings, C. R. (1996). Understanding radical organizational change: Bringing together the old and the new institutionalism. *Academy of Management Review*, 21, 1022–1054.

Hardy, C., Lawrence, T. B., & Grant, D. (2005). Discourse and collaboration: The role of conversations and collective identity. *Academy of Management Review*, 30, 58–77.

Hawley, A. (1968). Human ecology. In D. L. Sills (Ed.), *International encyclopedia of the social sciences* (pp. 328–337). New York: Macmillan.

Heugens, P.P.M.A.R., & Lander, M. W. (2009). Structure! Agency! (and other quarrels): A meta-analysis of institutional theories of organizations. *Academy of Management Journal*, 52, 61–85.

Koelble, T. A. (1995). The new institutionalism in political science and sociology. *Comparative Politics*, 27, 231–243.

Kostova, T., Roth, K., & Dacin, M. T. (2008). Institutional theory in the study of multinational corporations: A critique and new directions. *Academy of Management Review*, 33, 994–1006.

Kraatz, M., & Zajac, E. (1996). Exploring the limits of new institutionalism: The causes and consequences of illegitimate organizational change. *American Sociological Review*, 61, 812–836.

Landman, J. (1993). *Regret: The persistence of the possible*. New York: Oxford University Press.

Lincoln, J. R. (1995, March). [Review of the book *the new institutionalism in organizational research.] Social Forces*, 73, 1147–1148.

Oliver, C. (1991). Strategic responses to institutional processes. *Academy of Management Review*, 16, 145–179.

Oliver, C. (1997). Sustainable competitive advantage: Combining institutional and resource-based views. *Strategic Management Journal*, 18, 697–713.

Peters, B. G. (2000). Institutional theory: Problems and prospects. *69 Political Science Series*. Vienna, Austria: Institute for Advanced Studies.

Phillips, D. J., & Zuckerman, E. (2001). Middle-status conformity: Theoretical restatement and empirical demonstration in two markets. *American Journal of Sociology*, 107, 379–429.

Phillips, N., Lawrence, T. B., & Hardy, C. (2004). Discourse and institutions. *Academy of Management Review*, 29, 635–652. Scott, W. R. (1995). Institutions and organizations.

McGrawHill. Selznick, P. (1996). Institutionalism “old” and “new.” *Administrative Science Quarterly*, 41, 270–277.

Sonpar, K., Pazzaglia, F., & Kornijenko, J. (2009). The paradox and constraints of legitimacy. *Journal of Business Ethics*, 95, 1–21.

Suddaby, R. (2010). Challenges for institutional theory. *Journal of Management Inquiry*, 19, 14–

20.

Yang, Y., & Konrad, A. M. (2010). Understanding diversity management practices: Implications of institutional theory and resource-based theory. *Group & Organization Management*, 36, 6–38.

Resource-Based Theory

1. Introduction

Resource-based theory examines performance differences of organizations based on their resources (Peteraf & Barney, 2003), focusing on the efficiency-based differences. The theory seeks to explain how organizations maintain unique and sustainable positions in competitive environments (Hoopes, Madsen, & Walker, 2003). And the central idea in resource-based theory is that organizations compete against others based on their resources and capabilities (Barney, 1991; Wernerfelt, 1984). An organization's competitors can be identified by the similarity of their products, resources, capabilities, and substitutes (Peteraf & Bergen, 2003).

This theory makes two main assumptions: (1) organizations within an industry may differ in their resources, and (2) these resources may not be perfectly mobile across organizations, so organizational differences in resources can be very long lasting (Barney, 1991). Meanwhile, this theory assumes that organizational decisions to select and accumulate resources are economically rational and subject to limited information, biases and prejudices, and causal ambiguity (Oliver, 1997). Causal ambiguity means that it is not known exactly how a resource leads to above-average performance for an organization. Another key assumption in resource-based theory is that it focuses on an enterprise level, or business level, of analysis (Peteraf & Barney, 2003).

2. Resources and Capability

Resources, defined as everything that could be thought of as a strength for an organization (Wernerfelt, 1984), include any tangible or intangible assets that are semipermanently owned by the organization (Caves, 1980). An organization's resources are seen as strengths that help the organization to better compete and to accomplish its vision, mission, strategies, and goals (Porter, 1981). A capability is not observable and is therefore intangible, it cannot be valued, and it moves only as part of the unit where it is housed (Makadok, 2001).

3. Sustainable Competitive Advantage

The desirable position for an organization is to create a unique resource situation that makes it more difficult for its rivals to compete (Wernerfelt, 1984). An organization's competitive position relative to other organizations is based on its collection of unique resources and relationships (Rumelt, 1974). An organization has a competitive advantage when it uses a profitable, value-creating strategy that is not being used by competing organizations (Barney, 1991). If competing organizations cannot learn about that strategy and copy it, then an organization has a sustainable competitive advantage (SCA).

Organizational SCA derives from the resources and capabilities that an organization controls that are valuable, rare, inimitable, and nonsubstitutable (VRIN) (Barney, 1991). Resources are valuable when they help an organization create or implement strategies that improve its efficiency and effectiveness. Resources are rare when more organizations want the resource than are able to obtain it. Resources are inimitable and nonsubstitutable when they are immobile and expensive to imitate or replicate. An organization must have the ability to absorb and utilize its resources in order to obtain a sustainable competitive advantage (Barney & Clark, 2007; Conner, 1991).

4. The Focus of Resource-Based Theory

The theory focuses on the resources and capabilities controlled by an organization that underlie performance differences across organizations, which is different from other theories that focus on the dyad level (boss and supervisee), the group level, or the industry level. Also, resource-based theory is not a substitute for other industry-level analytic tools.

In this theory, the performance differences are deemed as earnings differentials attributable to resources having different levels of efficiency (Barney, 1991; Peteraf, 1993). Organizational efficiency means that a firm has lower costs and can create greater value and net benefits compared to inefficient firms. Efficiency is measured in terms of net benefits, or the benefits to an organization that are left after the firm's costs are subtracted.

Sustainable competitive advantages and disadvantages can occur immediately. Resource-based theory did not initially focus on whether resources were static or changing.

References:

Armstrong, C. E., & Shimizu, K. (2007). A review of approaches to empirical research on the resource-based view of the firm. *Journal of Management*, 33, 959–986.

Barney, J. B. (1991). Firm resources and sustained competitive advantage. *Journal of Management*, 17, 99–120.

Barney, J. B., & Clark, D. N. (2007). Resource-based theory: Creating and sustaining competitive advantage. *New York: Oxford University Press*.

Bowman, C., & Ambrosini, V. (2000). Value creation versus value capture: Towards a coherent definition of value in strategy. *British Journal of Management*, 11, 1–15.

Bromiley, P., & Fleming, L. (2002). The resource-based view of strategy: A behavioral critique. *In M. Augier & J. G. March (Eds.), Change, choice and organization: Essays in memory of Richard M. Cyert* (pp. 319–336).

Cheltenham, England: Elgar. Caves, R. E. (1980). Industrial organization, corporate strategy and structure. *Journal of Economic Literature*, 58, 64–92.

Conner, K. R. (1991). A historical comparison of resource-based theory and five schools of thought within industrial organization economics: Do we have a new theory of the firm? *Journal of Management*, 17, 121–154.

Conner, T. (2002). The resource-based view of strategy and its value to practicing managers. *Strategic Change*, 11, 307–316.

Fiol, C. M. (2001). Revisiting an identity-based view of sustainable competitive advantage. *Journal of Management*, 27, 691–699.

Helfat, C. E., & Peteraf, M. A. (2003). The dynamic resource-based view: Capability lifecycles. *Strategic Management Journal*, 24, 997–1010.

Hoopes, D. G., Madsen, T. L., & Walker, G. (2003). Guest editor's introduction to the special issue: Why is there a resource-based view? Toward a theory of competitive heterogeneity. *Strategic Management Journal*, 24, 889–902.

Kraaijenbrink, J., Spender, J.-C., & Groen, A. J. (2010). The resource-based view: A review and assessment of its critiques. *Journal of Management*, 36, 349–372.

Lockett, A., Thompson, S., & Morgenstern, U. (2009). The development of the resource-based view of the firm: A critical appraisal. *International Journal of Management Reviews*, 11, 9–28.

Makadok, R. (2001). Towards a synthesis of the resource-based and dynamic-capability views of rent creation. *Strategic Management Journal*, 22(5), 387–402.

Miller, D. (2003). An asymmetry-based view of advantage: Towards an attainable sustainability. *Strategic Management Journal*, 24, 961–976.

Newbert, S. L. (2007). Empirical research on the resource-based view of the firm: An assessment and suggestions for future research. *Strategic Management Journal*, 28, 121–146.

Oliver, C. (1997). Sustainable competitive advantage: Combining institutional and resource-based views. *Strategic Management Jou*rnal, 18, 697–713.

Peteraf, M. A. (1993). The cornerstones of competitive advantage: A resource-based view. *Strategic Management Journal*, 14, 179–191.

Peteraf, M. A., & Barney, J. B. (2003). Unraveling the resource-based triangle. *Managerial and Decision Economics*, 24, 309–323.

Peteraf, M. A., & Bergen, M. E. (2003). Scanning dynamic competitive landscapes: A market-based and resource-based framework. *Strategic Management Journal*, 24, 1027–1041.

Porter, M. (1981). The contributions of industrial organization to strategic management. *Academy of Management Review*, 6, 609–620.

Priem, R. L., & Butler, J. E. (2001). Is the resource-based "view" a useful perspective for strategic management research? *Academy of Management Review*, 21, 22–40.

Rumelt, R. P. (1974). Strategy, structure, and economic performance. *Cambridge, MA: Harvard University Press*.

Teece, D., Pisano, G., & Shuen, A. (1997). Dynamic capabilities and strategic management. *Strategic Management Journal*, 18, 509–533.

Wernerfelt, B. (1984). A resource-based view of the firm. *Strategic Management Journal*, 5, 171–180.

Knowledge-Based Theory

1. Introduction

The main idea of the knowledge-based theory of the firm is that organizations exist in the way that they do because of their ability to manage knowledge more efficiently than is possible under other types of organizational structures (Conner, 1991; Kogut & Zander, 1992, 1993, 1996; Conner & Prahalad, 1996; Foss, 1996; Grant, 1996a, 1996b; Madhok, 1996; Nahapiet & Ghoshal, 1998; Nickerson & Zenger, 2004).

Organizations are social entities that use and store internal knowledge, competencies, and capabilities that are vital for the firm's survival, growth, and success (Hakanson, 2010). The theory emphasizes the organizational need for superior coordination and integration of learning by employees inside the organization (Kogut & Zander, 1992; Nelson & Winter, 1982).

2. Definition of Knowledge

There are many different definitions of knowledge. And some researchers distinguish information with knowledge, while others not. According to Winkin (1996), data are turned into information, information is turned into knowledge, and then knowledge is confronted with wisdom. Gorman (2002) classified knowledge into four types: declarative (knowing what), procedural (knowing how), judgment (knowing when), and wisdom (knowing why). Balconi et al. presented a synthesis of some typologies: know-what, know-why, know-how, and know-who.

This theory makes a strong distinction between <u>tacit knowledge</u> (what a person knows only inside his or her own mind) and <u>explicit knowledge</u> (what is in the public domain) (Nelson & Winter, 1982; Polanyi, 1966).

<u>Tacit knowledge</u> is a valuable resource for organizations because it cannot be easily acquired, and trying to copy it is often costly, assuming that someone with the desired knowledge can even be located. Tacit knowledge cannot be easily written down and documented (or codified), so it can only be learned through observation of experts and subsequent practicing of skills (Kogut & Zander, 1992; Grant, 1996b). <u>Articulation</u> is the process through which <u>tacit knowledge is made explicitly known to everyone</u>. <u>Codification</u> is the process through which <u>articulated knowledge is fixed, recorded, standardized, and disseminated to people in the organization</u> (Hakanson, 2007).

In contrast, some researchers believe that <u>tacit knowledge cannot be articulated</u> (Grant & Baden-Fuller, 1995; Reed & DeFillipi, 1990). Some researchers argue that once tacit knowledge is articulated, then it stops being knowledge and becomes merely data (Soo et. al., 2002). Others argue that all <u>tacit knowledge can be translated into explicit knowledge</u> (Schulz & Jobe, 2001). Hakanson (2007) created a typology that defined important terms in the theory<u>: explicit knowledge</u> (know-why and know-what), internalized knowledge (explicit knowledge that is not being used), procedural knowledge (knowledge of skills and capabilities), and <u>tacit knowledge</u> (articulate and inarticulate).

3. Relationship between Knowledge and Organizational Performance

How organizations manage their stores of knowledge can determine their success or failure. For example, firms that are more effective than other organizations at finding, absorbing, and exploiting new knowledge from both their internal and external environments will tend to perform better than their competition (Martin-de-Castro, Delgado-Verde, Lopez-Saez, & Navas-Lopez, 2011). Liebeskind (1996) argued that firms that can protect their explicit knowledge will perform better than those that can't protect it. Organizations can protect their knowledge by designing jobs where individuals can't see the "whole picture" of a process, using employment contracts and confidentiality agreements to slow the spread of company secrets, and imposing costs on employees for leaving the company, such as through deferred compensation (pension plans, stock options etc.).

The theory assumes that organizations are all heterogeneous, knowledge-bearing entities that apply knowledge to the production of their goods and services (Foss, 1996). And the theory also assumes that knowledge—created, stored, and used—is the most strategically important of an organization's resources (Grant, 1996b). The theory assumes that knowledge is created, stored, and used by individuals and not by organizations.

Coordinating and integrating this knowledge held by diverse individuals is a difficult task for managers. Grant (1996b) described four mechanisms for integrating specialized knowledge held by individuals: (1) rules and directives (procedures, plans, policies, and practices); (2) sequencing (time-patterned schedules); (3) routines (complex organizational patterns of behavior); and (4) group problem solving and decision making (social communication involving discussing, sharing, and learning and then taking action). These four mechanisms for coordinating and integrating individual-level knowledge all depend on the existence of "common knowledge."

Common knowledge refers to those elements of knowledge of which everyone in the organization should be aware. Common knowledge is important in an organization because it enables everyone to share knowledge that is not common. Different types of common knowledge include language, symbolic communication (literacy, numeracy, software programs), shared specialized knowledge, shared meaning (shared metaphors, analogies, and stories), and recognition and mutual adjustment with other employees (Grant, 1996b). Complex hierarchies in organizations can impede the sharing of common knowledge when knowledge is stored in separate and distant levels within the hierarchy.

4. Intellectual Capital

In this knowledge-based economy, some researchers also examine a firm's knowledge assets, or intellectual capital (Dean & Kretschmer, 2007). A firm's intellectual capital includes human capital, structural capital, and relational capital (Martin-de-Castro et al., 2011). Intellectual capital is related to a firm's ability to create and apply its knowledge base.

References:

Adler, P. S. (2001). Market, hierarchy, and trust: The knowledge economy and the future of capitalism. *Organization Science*, 12, 215–234.

Ancori, B., Bureth, A., & Cohendet, P. (2000). The economics of knowledge: The debate about codification and tacit knowledge. *Industrial and Corporate Change*, 9, 255–287.

Balconi, M., Pozzali, A., & Viale, R. (2007). The "codification debate" revisited: A conceptual framework to analyze the role of tacit knowledge in economies. *Industrial and Corporate Change*, 16, 823–849.

Brown, J. S., & Duguid, P. (2001). Knowledge and organization: A socialpractice perspective. *Organization Science*, 12, 198–213.

Cabrita, M. R., & Bontis, N. (2008). Intellectual capital and business performance in the Portuguese banking industry. *International Journal of Technology Management*, 43, 212–237.

Conner, K. R. (1991). A historical comparison of resource-based theory and five schools of thought within industrial organization economics: Do we have a new theory of the firm? *Journal of Management*, 17, 121–154.

Conner, K. R., & Prahalad, C. K. (1996). A resource-based theory of the firm: Knowledge versus opportunism. *Organization Science*, 7, 477–501.

Dean, A., & Kretschmer, M. (2007). Can ideas be capital? Factors of production in the postindustrial economy: A review and critique. *Academy of Management Review*, 32, 573–594.

Felin, T., & Hesterly, W. S. (2007). The knowledge-based view, nested heterogeneity, and new value creation: Philosophical considerations on the locus of knowledge. *Academy of Management Review,* 32, 195–218.

Foss, N. J. (1996). Knowledge-based approaches to the theory of the firm: Some critical comments. *Organization Science*, 7, 470–476.

Foss, N. J. (2003). Bounded rationality and tacit knowledge in the organizational capabilities approach: An assessment and re-evaluation. *Industrial and Corporate Change*, 12, 185–201.

Gorman, M. E. (2002). Types of knowledge and their roles in technology transfer. *Journal of Technology Transfer*, 27, 219–231.

Grant, R. M. (1996a). Prospering in dynamically competitive environments: Organizational capability as knowledge integration. *Strategic Management Journal*, 7, 375–387.

Grant, R. M. (1996b, Winter). Toward a knowledge-based theory of the firm. *Strategic Management Journal*, 17(Special Issue), 109–122.

Grant, R. M., & Baden-Fuller, C. (1995). A knowledge-based theory of inter-firm collaboration. Academy of Management Best Papers Proceedings (pp. 17–21). *New York: Academy of Management*.

Hakanson, L. (2007). Creating knowledge: The power and logic of articulation. *Industrial and Corporate Change*, 16, 51–88.

Hakanson, L. (2010). The firm as an epistemic community: The knowledge-based view revisited. *Industrial and Corporate Change*, 19, 1801–1828.

Kogut, B. (2000). The network as knowledge: Generative rules and the emergence of structure. *Strategic Management Journal*, 21, 405–425.

Kogut, B., & Zander, U. (1992). Knowledge of the firm, combinative capabilities, and the replication of technology. *Organization Science*, 3, 384–397.

Kogut, B., & Zander, U. (1993). Knowledge of the firm and the evolutionary theory of the multinational corporation. *Journal of International Business Studies*, 24, 625–645.

Kogut, B., & Zander, U. (1996). What firms do? Coordination, identity and learning. *Organization Science*, 7, 502–518.

Liebeskind, J. P. (1996). Knowledge, strategy, and the theory of the firm. *Strategic Management Journal*, 17, 93–107.

Madhok, A. (1996). The organization of economic activity: Transaction costs, firm capabilities and the nature of governance. *Organization Science*, 7, 577–590.

Martin-de-Castro, G., Delgado-Verde, M., Lopez-Saez, P., & Navas-Lopez, J. E. (2011). Towards an intellectual capital view of the firm: Origins and nature. *Journal of Business Ethics*, 98, 649–662.

Nahapiet, J., & Ghoshal, S. (1998). Social capital, intellectual capital and the organizational advantage. *Academy of Management Review*, 23, 242–266.

Nelson, R. R., & Winter, S. G. (1982). An evolutionary theory of economic change. Cambridge, *MA: Harvard University Press*.

Nickerson, J. A., & Zenger, T. R. (2004). A knowledge-based theory of the firm—the problem-solving perspective. *Organization Science*, 15, 617–632.

Nonaka, I. (1994). A dynamic theory of organizational knowledge creation. *Organization Science*, 5, 14–37.

Phelan, S. E., & Lewin, P. L. (2000). Arriving at a strategic theory of the firm. *International Journal of Management Reviews*, 2, 305–323.

Polanyi, M. (1966). The tacit dimension. *New York: Doubleday*.

Reed, R., & DeFillipi, R. J. (1990). Causal ambiguity, barriers to imitation, and sustainable competitive advantage. *Academy of Management Review*, 15, 88–102.

Schulz, M., & Jobe, A. (2001). Codification and tacitness as knowledge management strategies: 、 An empirical exploration. *Journal of High Technology Management Research*, 12, 139–165.

Simon, H. A. (1991). Bounded rationality and organizational learning. *Organization Science*, 2, 125–134.

Soo, C. T., Devinney, D., Midgley, D., & Deering, A. (2002). Knowledge management: Philosophy, processes and pitfalls. *California Management Review*, 44, 129–150.

Tsoukas, R. (1996, Winter). The firm as a distributed knowledge system: A constructionist approach. *Strategic Management Journal*, 17(Special Issue), 11–25.

Williamson, O. E. (1999). Strategy research: Competence and governance perspectives. *Strategic Management Journal*, 20, 1087–1108.

Winkin, Y. (1996). Anthropologie de la communication. De la theorie au terrain (Anthropology of communication. In the theory of the field.) *Brussells, Belgium: De Boeck Université*.

Social Exchange Theory

1. Introduction

The major idea in social exchange theory is that parties enter into and maintain exchange relationships with others with the expectation that doing so will be rewarding (Blau, 1968; Gouldner, 1960; Homans, 1958). The theory is limited to examining actions that are contingent on rewarding reactions from others (Blau, 1964), and examines two-sided, mutually contingent, and mutually rewarding processes called "transactions" and relationships called "exchanges" (Emerson, 1976).

The theory assumes that self-interested parties transact or exchange with self-interested others in order to accomplish outcomes that neither could achieve on his or her own (Lawler & Thye, 1999), and that these exchanges would cease as soon as they are not perceived to be mutually rewarding by both parties (Blau, 1994).

According to the theory, each party has something of value that the other wants. The two parties decide what to exchange and in what quantities. The exchange of benefits or giving something to a recipient that is more valuable to the recipient than it is to the giver, is the underlying basis for human behavior (Homans, 1961). The resources exchanged can be economic or social or both. Economic resources include tangible items. Social resources include intangible items. And the value of outcomes received during a social exchange is decided by the beholder (third-party). The most rewarding outcomes in social exchange relationships (for example, social approval and respect) do not have any material value for which a price could be determined.

And reciprocity, or repaying obligations to another, is one of the best-known exchange rules in social exchange theory.

2. Economic and Social Exchanges

There are similarities and differences between economic and social exchanges. Social and economic exchanges are similar in that they both include expectations that future returns will be received for current contributions made.

However, in economic exchanges, the returns on one's investment can be clearly known and specified, whereas in social exchanges, the returns on one's investment are unspecified and often voluntary. Economic exchanges tend to occur on a quid pro quo (this for that) basis, whereas social exchanges do not. Economic exchanges are based on transactions in the short term, while social exchanges are based on relationships in which both parties trust that the other will fairly meet their obligations in the long term (Holmes, 1981). Social exchanges tend to include short-term inequities or asymmetries between the trading parties, whereas economic exchanges tend to be more equitable and symmetrical.

3. Trust in Social Exchange

Social exchange relationships involve uncertainty over whether parties will reciprocate contributions or not. Therefore, trust between parties is an important part of social exchange theory.

Demonstrating trust to the other party may be difficult during the initial stages of exchange. Typically, social exchanges evolve slowly, with lower-value exchanges occurring initially, then larger-value exchanges occurring when higher levels of trust develop. Trust can be generated in two ways: (1) through regular and consistent reciprocation with the other party for benefits received from them, and (2) through the gradual expansion of exchanges with the other party (Blau, 1964).

4. Premises of Social Exchange Theory

The major premises of social exchange theory were derived with a goal of creating a mutually exclusive and exhaustive set of four generalizations: (1) exchange relationships result in economic or social outcomes (or both), (2) a cost-benefit analysis is performed on the outcomes received and compared with the potential costs and benefits of alternative exchange relationships, (3) the receipt of rewarding outcomes over time increases mutual trust and commitment in exchange relationships, and (4) exchange norms and expectancies develop over time from rewarding exchange relationships.

5. Salinification in Social Exchange

Researchers have defined social and economic exchanges as a type of choice behavior, although no formal bargaining or written contracts are involved. Parties freely perform cost-benefit analyses regarding current or potential social exchanges (Molm, 1990). The satisfaction levels of the parties in the exchanges become the prime determinants of whether future exchanges will occur or not. However, parties do not make these considerations in isolation. Instead, a party's social network can help support or disrupt future exchanges. However, social exchange theory has tended to view party satisfaction as the primary influence over exchange maintenance and to treat social sanctions as a secondary influence.

6. Development of Social Exchange Theory

Social exchange research has evolved from two different traditions: the individualistic and the collectivistic (Makoba, 1993). The individualistic perspective stresses the individual psychological and economic self-interests involved in the exchange (Blau, 1964; Homans, 1961). The collectivistic perspective emphasizes the importance of the social needs of the group or of society (Befu, 1977). According to the collectivistic approach, society is assumed to have its own existence, and individuals are assumed to exist for the benefit of society.

References:

Befu, H. (1977). Social exchange. *Annual Review of Anthropology*, 6, 255–281.

Blau, P. M. (1955). The dynamics of bureaucracy: A study of interpersonal relations in two government agencies. *Chicago: University of Chicago Press*.

Blau, P. M. (1960, May). A theory of social integration. *American Journal of Sociology*, 65, 545–556.

Blau, P. M. (1964). Exchange and power in social life. Hoboken, *NJ: Wiley*.

Blau, P. M. (1968). Interaction: Social exchange. *International Encyclopedia of the Social Sciences*, 7, 452–458.

Blau, P. M. (1994). Structural context of opportunities. *Chicago: University of Chicago Press*.

Burgess, R., & Neilsen, J. (1974). An experimental analysis of some structural determinants of equitable and inequitable exchange relations. *American Sociological Review*, 39, 427–443.

Coyle-Shapiro, J. A.-M., & Conway, N. (2004). *The employment relationship through the lens of social exchange theory*. In J.

Coyle-Shapiro, L. M. Shore, M. S. Taylor, & L. E. Tetrick (Eds.), *The employment relationship: Examining psychological and contextual perspectives* (pp. 5–28). Oxford: Oxford University Press.

Cropanzano, R., & Mitchell, M. S. (2005). Social exchange theory: An interdisciplinary review. *Journal of Management*, 31, 874–900.

Cropanzano, R., Rupp, D. E., Mohler, C. J., & Schminke, M. (2001). *Three roads to organizational justice. In J. Ferris (Ed.)*, Research in personnel and human resources management (Vol. 20, pp. 1–113).

Greenwich, CT: JAI. Ekeh, P. P. (1974). Social exchange theory. Cambridge, *MA: Harvard University Press*.

Emerson, R. M. (1962, February). Power-dependence relations. *American Sociological Review*, 27, 31–41.

Emerson, R. M. (1976). Social exchange theory. *Annual Review of Sociology*, 2, 335–362.

Gouldner, A. W. (1960). The norm of reciprocity: A preliminary statement. *American Sociological Review*, 25, 161–178.

Holmes, J. G. (1981). *The exchange process in close relationships: Microbehavior and* *macromotives. In M. J. Lerner & S. C. Lerner (Eds.),* The justice motive in social behavior (pp. 261–284). New York: Plenum.

Homans, G. C. (1958). Social behavior as exchange. *American Journal of Sociology*, 63, 597–606.

Homans, G. C. (1961). Social behavior: Its elementary forms. *New York: Harcourt Brace. Lawler*, E. J., & Thye, S. R. (1999). Bringing emotions into social exchange theory. *Annual Review of Sociology*, 25, 217–244.

Levi-Strauss, C. (1969). *The elementary structure of kinship*. Boston: Beacon Press.

Makoba, J. W. (1993). Toward a general theory of social exchange. *Social Behavior and Personality*, 21, 227–240.

Mauss, M. (1967). *The gift: Forms and functions of exchange in archaic societies*. New York: Norton.

Molm, L. G. (1990). Structure, action, and outcomes: The dynamics of power in social exchange. *American Sociological Review*, 55, 427–447.

Molm, L. G., & Takahashi, N. (2003). In the eye of the beholder: Procedural justice in social exchange. *American Sociological Review*, 68, 128–152.

Popper, K. R. (1959). *The logic of scientific discovery*. New York: Harper & Row.

Rudner, R. (1966). *Philosophy of social science. Englewood Cliffs*, NJ: Prentice Hall. Sahlins, M. D. (1965). On the sociology of primitive exchange. In M. Banton (Ed.), *The relevance of models for social anthropology* (pp. 139236). London: Tavistock.

Thibaut, J. W., & Kelley, H. H. (1959). *The social psychology of groups*. Hoboken, NJ: Wiley.

Social Capital Theory

1. Introduction

The main idea in social capital theory is that people gain both tangible and intangible resources at the individual, group, and organizational level through social interactions and connections with others (Bourdieu, 1986; Coleman, 1988; Lin, 2001; Putnam, 2000). A key focus in the theory is that social capital resources are embedded within, available through, and derived from social networks of interconnected people, groups, or nations (Bolino, Turnley, & Bloodgood, 2002; Inkpen & Tsang, 2005).

2. Sources of Social Capital

The sources of social capital that can be obtained lie with the <u>network</u> of which the individual, the group, or the nation is a member. Social capital is <u>based on the position or location of the member with the member's network of social relations</u>, so it is different from other types of capital. <u>Social structure has three dimensions: (1) market relations, (2) hierarchical relations, and (3) social relations</u> (Adler & Kwon, 2002). <u>Market relations</u> refer to <u>the bartering or monetary exchange of goods and services</u>. <u>Hierarchical relations</u> refer to <u>exchanging material and security for obedience to authority</u>. <u>Social relations</u> refer <u>to tacit, symmetrical, ongoing mutual exchanges of gifts and favors</u>.

Resources made available through social networks come from two main sources (Portes & Landolt, 2000). The first source of social capital is <u>altruistic</u>: (1) giving resources to others because of moral obligations and (2) giving resources to others to maintain solidarity in the same community or region. Altruistic donations are not expected to be paid back. The second source of social capital is instrumental: (1) person-to-person exchanges and (2) larger social-structure resource transactions (such as a loan from a bank). Instrumental resource exchanges are expected to be paid back. There is trust among the parties in these exchanges because the community has the power to enforce that trust.

3. Benefits Provided by Social Capital

The possession of social capital can provide many benefits, including more career success, better executive compensation, easier access to jobs, a richer pool of recruits for companies, higher levels of product innovation, more resource exchange, lower turnover rates, lower organizational failure rates, firm growth, enhanced entrepreneurship and the start-up of companies, stronger relationships with suppliers, and higher levels of interfirm learning. Therefore, the possession of social capital is quite advantageous and lucrative for individuals, groups, organizations, and nations (Adler & Kwon, 2002; Florin, Lubatkin, & Schulze, 2003).

4. Negative Consequences Provided by Social Capital

There are also many negative consequences that can be brought by social capital: (1) exclusion of outsiders, (2) excess demands on group members, (3) restrictions of individual freedoms, and (4) downward leveling norms (blocking members of minority groups from upward mobility) (Portes & Landolt, 1996; Portes & Sensenbrenner, 1993).

Social capital networks tend to be inward focused and offer benefits only to members at the expense of outsiders. Such arrangements can give rise to homogeneous groups, "good old boy" networks, and discriminatory practices (Ritchie & Gill, 2007). There can also be negative aspects of social capital at the organizational level. The ties to other business units can be helpful but also can be expensive to maintain.

5. Dimensions of Social Capital

There are six dimensions of social capital: (1) community values (Hanifan, 1916); (2) collective action, social structure, and realization of interests (Coleman, 1988, 1990); (3) trust, reciprocity, and cooperation (Putnam, 2000); (4) individual and group relationship resources (Bourdieu, 1986); (5) civic engagement and voluntary associations (Putnam, 2000); and (6) social ties and networks (Granovetter, 1973).

From these six dimensions, <u>two overarching and competing categories of social capital</u> have been created. They are (1) resource social capital and (2) normative social capital. Resource social capital mainly refers to mutually shared resources, networks, and social ties. Normative social capital includes norms; trust; reciprocity; civic engagement; and values among friends, family, and community.

References:

Adler, P. S., & Kwon, S.-W. (2002). Social capital: Prospects for a new concept. *Academy of Management Review*, 27, 17–40.

Arrow, K. (1999). *Observations on social capital. In P. Dasgupta & I. Serageldin (Eds.),* Social capital: A multifaceted perspective (pp. 3–5).

Washington, DC: World Bank. Baker, W. E. (1990). Market networks and corporate behavior. *American Journal of Sociology*, 96, 589–625.

Belliveau, M., O'Reilly, C., & Wade, J. (1996). Social capital at the top: Effects of social similarity and status on CEO compensation. *Academy of Management Journal*, 39, 1568–1593.

Bolino, M. C., Turnley, W. H., & Bloodgood, J. M. (2002). Citizenship behavior and the creation of social capital in organizations. *Academy of Management Review*, 27, 505–522.

Bourdieu, P. (1986). The forms of capital. In J. G. Richardson (Ed.), *Handbook of theory and research for the sociology of education* (pp. 241258). New York: Greenwood Press.

Bourdieu, P., & Wacquant, L. (1992). *Invitation to reflexive sociology*. Chicago: University of

Chicago Press.

Colclough, G., & Sitaraman, B. (2005). Community and social capital: What is the difference? *Sociological Inquiry*, 75, 474–496.

Coleman, J. S. (1988). Social capital in the creation of human capital. *American Journal of Sociology*, 94(Supplement), S95–S120.

Coleman, J. S. (1990). *Foundations of social theory. Cambridge, MA: Belknap Press*. DeFilippis, J. (2002). Symposium on social capital: An introduction. Antipode, 34, 790–795.

Durkheim, E. (1960). *The division of labor in society*. New York: Free Press. (Original work published 1902) Fine, B. (2001). Social capital versus social theory: Political economy and social science at the turn of the millennium. London: Routledge.

Fischer, C. S. (2005). Bowling alone: What's the score? *Social Networks*, 27, 155–167.

Fischer, H. M., & Pollock, T. G. (2004). Effects of social capital and power on surviving transformational change: The case of initial public offerings. *Academy of Management Journal*, 47, 463–481.

Florin, J., Lubatkin, M., & Schulze, W. (2003). A social capital model of high-growth ventures. *Academy of Management Journal*, 46, 374–384.

Foley, M., & Edwards, R. (1999). Is it time to divest in social capital? *Journal of Public Policy*, 19, 141–173.

Fukuyama, F. (1995). *Trust: Social virtues and the creation of prosperity*. London: Hamilton.

Fulkerson, G. M., & Thompson, G. H. (2008). The evolution of a contested concept: A meta-analysis of social capital definitions and trends (1988–2006). *Sociological Inquiry*, 78, 536–557.

Granovetter, M. S. (1973). The strength of weak ties. *American Journal of Sociology*, 78, 1360–

1380.

Hanifan, L. J. (1916, September). The rural school community center. *Annals of the American Academy of Political and Social Science*, 67, 130–138.

Hansen, M. T. (2002). Knowledge networks: Explaining effective knowledge sharing in multiunit companies. *Organization Science*, 13, 232–248.

Haynes, P. (2009). Before going any further with social capital: Eight key criti- cisms to address. (Ingenio working paper series: Instituto de Gestión de la Innovación y del Conocimiento). Valencia: La Universidad Politécnica de Valencia. Inkpen, A. C., & Tsang, E. W. K. (2005). Social capital, networks, and knowledge transfer. *Academy of Management Review*, 30, 146–165.

Jacobs, J. (1965). *The death and life of great American cities*. London: Penguin Books.

Kostova, T., & Roth, K. (2003). Social capital in multinational corporations and a micro-macro model of its formation. *Academy of Management Review*, 28, 297–317.

Lin, N. (2001). *Social capital: A theory of social structure and action*. Cambridge: Cambridge University Press.

Loury, G. C. (1977). A dynamic theory of racial income differences. *In P. A. Wallace & A. M*.

LaMonde (Eds.), *Women*, minorities and employment discrimination (pp. 153–186). Lexington,

MA: Lexington Books. Maraffi, M, (1994). [Review of the book Making democracy work]. *American Journal of Sociology*, 99, 1348–1349.

Mix, T. L. (2011, May). Rally the people: Building local-environmental justice grassroots coalitions and enhancing social capital. *Sociological Inquiry*, 81, 174–194.

Morlino, L. (1995). Italy's civic divide. *Journal of Democracy*, 6(1), 173–177.

Nahapiet, J., & Ghoshal, S. (1998). Social capital, intellectual capital, and the organizational advantage. *Academy of Management Journal*, 23, 242–266.

Portes, A. (1998). Social capital: Its origins and applications in modern sociology. *Annual Review of Sociology*, 24, 1–24.

Portes, A., & Landolt, P. (1996). The downside of social capital. *American Prospect*, 26, 18–22.

Portes, A., & Landolt, P. (2000). Social capital: Promise and pitfalls of its role development. *Journal of Latin American Studies*, 32, 529–547.

Portes, A., & Sensenbrenner, J. (1993). Embeddedness and immigration: Notes on the social determinants of economic action. *American Journal of Sociology*, 98, 1320–1350.

Putnam, R. (1993). The prosperous community: Social capital and public life. *American Prospect*, 13, 35–42.

Putnam, R. (1995). Bowling alone: America's declining social capital. *Journal of Democracy*, 6(1), 65–78.

Putnam, R. (2000). *Bowling alone: The collapse and revival of American community*. New York: Simon & Schuster.

Putnam, R., Leonardi, R., & Nanetti, R. (1993). *Making democracy work: Civic traditions in modern Italy*. Princeton, NJ: Princeton University Press.

Ritchie, L. A., & Gill, D. A. (2007). Social capital theory as an integrating theoretical framework in technological disaster research. *Sociological Spectrum*, 27, 103–129.

Simmel, G. (1971). Georg Simmel on individuality and social forms (D. N. Levine, Ed.).Chicago: University of Chicago Press. Tönnies, F. (1957). Community and society (C. P. Loomis, Ed.). East Lansing: Michigan State University Press. Uzzi, B. (1997). Social structure and competition in interfirm networks: The paradox of embeddedness. *Administrative Science Quarterly,* 42, 35–67.

Stakeholder Theory

1. Introduction

The main idea in stakeholder theory is that organizations should focus on meeting a broader set of interests than just amassing shareholder wealth. According to this theory, firms should not only focus on the firm's financial performance but also focus on their social performance, such as the corporate social responsibility (CSR). They should understand, respect, and meet the needs of all of those who have a stake in the actions and outcomes of the organization. According to stakeholder theory, involving stakeholders in corporate decisions is considered an ethical requirement and a strategic resource, both of which help provide organizational competitive advantages (Cennamo, Berrone, & Gomez-Mejia, 2009; Plaza-Ubeda, de Burgos-Jimenez, & Carmona-Moreno, 2010).

And a central idea of stakeholder theory is that some corporate decision-making power and benefits should be taken away from the shareholders and given to the stakeholders (Stieb, 2008). Meanwhile Greenwood (2007) states that stakeholder theory is morally neutral, in that engaging stakeholders and committing to considering their needs do not automatically obligate a firm to act in stakeholders' best interests.

2. Stakeholder

2.1.Definition of Stakeholder

Stakeholders are individuals or groups who can affect, or are affected by, the actions and results of an organization (Freeman, 1984). However, <u>stakeholder theory</u> usually <u>narrows the definition of a stakeholder to major, legitimate individuals or groups</u>.

If an organization were to focus on meeting the interests of those who have extremely different interests, then the organization might not survive economically (Mitchell, Agle, & Wood, 1997). Therefore, the stakeholder theory only considers stakeholders whose interest is closely related with the firm's operations or corporate objectives.

2.2.Categories of Stakeholder

There are three categories of stakeholders: <u>internal, external, and distal</u> (Sirgy, 2002). Internal stakeholders include employees, executive staff, firm departments, and the board of directors. External stakeholders include shareholders, suppliers, creditors, the local community, and the environment. Distal stakeholders include rival firms, consumer and advocacy groups, government agencies, voters, and labor unions.

2.3.Interest of Stakeholders

The interests of the stakeholders are valuable to the organization for their own sake and not because addressing their interests could benefit any other group, such as the shareholders of the firm. Kaler (2003) developed a typology of stakeholder theories and concluded that there are only two permissible types: (1) theories in which firms have perfect responsibilities toward both shareholders and nonshareholders and (2) theories in which firms have perfect responsibilities toward shareholders, but imperfect ones toward nonshareholders.

3. Perspectives of Stakeholder Theory

Stakeholder theory can be categorized from three points of view: descriptive, instrumental, and normative (Donaldson & Preston, 1995). The descriptive perspective simply states that organizations have stakeholders. The role of organizations is to satisfy a wide range of stakeholders and not just the shareholders. Research has shown that a significant number of firms practice shareholder management, which involves balancing the needs of organizations with the needs of stakeholders (for example, Clarkson, 1991).

The instrumental perspective is that firms that consider their stakeholders' interests will be more successful than those that do not. The assumption is that organizations that practice stakeholder management, assuming all other variables are held constant, will be relatively successful in terms of profitability, stability, growth, and so on.

The normative perspective examines why firms should consider their stakeholders. This perspective has been the predominant view, or main core, in stakeholder theory (Donaldson & Preston, 1995). According to the normative perspective, stakeholders are individuals or groups who have legitimate interests in substantive aspects of the firm (Donaldson & Preston, 1995). Stakeholders are defined by their own interests in the organization, whether the organization has any corresponding interest in the stakeholders.

4. Stakeholder Integration

Stakeholder integration includes the integration of knowledge of stakeholders and their demands (Maignan & Ferrell, 2004). Knowledge of stakeholders refers to uncovering salient stakeholders and prioritizing their demands (Rowley, 1997). Firms must pay primary attention to stakeholders with power, legitimacy, urgent demands, or some combination of these. Interaction with stakeholders should involve mutually satisfying, reciprocal relationships among shareholders and the organization.

References:

Agle, B. R., Donaldson, T., Freeman, R. E., Jensen, M. C., Mitchell, R. K., & Wood, D. J. (2008). Dialogue: Toward superior stakeholder theory. *Business Ethics Quarterly*, 18, 153–190.

Altman, J. A., & Petkus, J. R. (1994). Toward a stakeholder-based policy process: An application of the social marketing perspective to environmental policy development. *Policy Sciences*, 27, 37–51.

Buono, A. F., & Nichols, L. T. (1985). Corporate policy, values and social responsibility. *New York: Praeger*.

Cennamo, C., Berrone, P., & Gomez-Mejia, L. R. (2009). Does stakeholder management have a dark side? *Journal of Business Ethics*, 89, 491–507.

Clarkson, M. B. E. (1991). Defining, evaluating, and managing corporate social performance: A stakeholder management model. In J. E. Post (Ed.*), Research in corporate social performance and policy* (Vol. 12, pp. 331–358). Greenwich, CT: JAI Press.

Clarkson, M. B. E. (1995). A stakeholder framework for analyzing and evaluating corporate social performance. *Academy of Management Review*, 20, 92–117.

Clifton, D., & Amran, A. (2011). The stakeholder approach: A sustainability perspective. *Journal of Business Ethics*, 98, 121–136.

Donaldson, T., & Preston, L. E. (1995). The stakeholder theory of the corporation: Concepts, evidence, and implications. *Academy of Management Review*, 20, 65–91.

Egels-Zanden, N., & Sandberg, J. (2010, January). Distinctions in descriptive and instrumental stakeholder theory: A challenge for empirical research. *Business Ethics*, 19, 35–49.

Freeman, R. E. (1984). Strategic management: A stakeholder approach. Boston: Pitman. Freeman, R. E. (2002). *Stakeholder theory of the modern corporation. In T. Donaldson & P. Werhane (Eds.)*, Ethical issues in business: A philosophical approach (7th ed., pp. 38–48). Englewood Cliffs, NJ: Prentice Hall.

Freeman, R. E. (2004). The stakeholder approach revisited. *Zeitschrift für Wirtschaftsund Unternehmensethik*, 5, 228–241.

Grafe-Buckens, A., & Hinton, A.-F. (1998). Engaging the stakeholder: Corporate views and current trends. *Business Strategy and the Environment*, 7, 124–133.

Green, A. O., & Hunton-Clarke, L. (2003). A typology of stakeholder participation for company environmental decision-making. *Business Strategy and the Environment*, 12, 292–299.

Greenwood, M. (2001, Spring). The importance of stakeholders according to business leaders.

Business and Society Review, 106, 29–49.

Greenwood, M. (2007). Stakeholder engagement: Beyond the myth of corporate responsibility. *Journal of Business Ethics*, 74, 315–327.

Hambrick, D. C., & Finkelstein, S. (1987). Managerial discretion: A bridge between polar views of organizational outcomes. In L. L. Cummings & B. M. Staw (Eds.), *Research in organizational behavior* (Vol. 9, pp. 369–406).

Greenwich, CT: JAI Press. Harrison, J. S., Bosse, D. A., & Phillips, R. A. (2010). Managing for stakeholders, stakeholder utility, and competitive advantage. *Strategic Management Journal*, 31, 58–74.

Hillman, A. J., & Keim, G. D. (2001). Shareholder value, shareholder management, and social issues: What's the bottom line? *Strategic Management Journal*, 22, 125–139.

Kaler, J. (2003). Differentiating stakeholder theories. *Journal of Business Ethics*, 46, 71–83.

Key, S. (1999). Toward a new theory of the firm: A critique of stakeholder "theory." *Management Decision*, 37, 317–328.

Maignan, I., & Ferrell, O. (2004). Corporate social responsibility and marketing: An integrative framework. *Academy of Marketing Science*, 32, 3–19.

Mitchell, R., Agle, B. R., & Wood, D. J. (1997). Toward a theory of stakeholder identification and salience: Defining the principles of who and what really counts. *Academy of Management Review*, 22, 853–886.

Payne, S., & Calton, J. (2004). Exploring research potentials and applications for multi-stakeholder learning dialogues. *Journal of Business Ethics*, 55, 71–78.

Phillips, R. A., Berman, S. L., Elms, H., & Johnson-Cramer, M. E. (2010). Strategy, stakeholders, and managerial discretion. *Strategic Organization*, 8, 176–183.

Plaza-Ubeda, J. A., de Burgos-Jimenez, J., & Carmona-Moreno, E. (2010). Measuring stakeholder integration: Knowledge, interaction and adaptational behavior dimensions. *Journal of Business Ethics*, 93, 419–442.

Rowley, T. J. (1997). Moving beyond dyadic ties: A network theory of stakeholder influences. *Academy of Management Review*, 22, 887–910.

Sirgy, M. J. (2002). Measuring corporate performance by building on the stakeholder model of business ethics. *Journal of Business Ethics*, 35, 143–162.

Stieb, J. A. (2008). Assessing Freeman's stakeholder theory. *Journal of Business Ethics*, 87, 401–414.

Walsh, J. P. (2005). Taking stock of stakeholder management. *Academy of Management Review*, 30, 426–438.

Part Two: OM Research Based on Secondary Data

Event Study

1. Definition

The event study method is developed to measure the effect of an unanticipated event on stock prices. Using this method, a researcher determines whether there is an "abnormal" stock price (i.e., abnormal return) effect associated with an **unanticipated** event (McWILLIAMS & Siegel, 1997). And the inference of the significance of the event can be made vis this determination.

The event study method helps researchers assess the financial impact of changes in corporate policy. In management, this method is used to judge the effects of <u>endogenous corporate events and the effects of exogenous events</u>. It can assess the impact of managerial decision making to the performance of firms.

There are two key factors influence the conclusions obtained under this framework: (1) Obey the assumptions (2) Reasonable research design

2. Measure of Variable

Accounting based measures and stock price can be used as measures for abnormal return of companies.

2.1. Accounting Based Measure

The accounting based measure can be used in event study as the measures for variables and these measures are generally evaluated relative to an industry benchmark (Barber & Lyon, 1996). Barber and Lyon (1996) suggest that, for event studies based on financial data, nonparametric tests, such as nonparametric Wilcoxon signed-rank (Yeung et al., 2011), are more appropriate than the parametric t-statistics.

2.2. Stock Price

The event study can be based on the stock price changes. And stock prices are assumed to reflect the true value of firms, because they are assumed to reflect the discounted value of all future cash flows and incorporate all relevant information (Li et al., 2022; McWILLIAMS & Siegel, 1997).

3. Model

The standard approach is based on estimating a market model for each firm and then calculating abnormal returns. These abnormal returns are assumed to reflect the stock market's reaction to the arrival of new information.

The method is as follows: The rate of return on the share price of firm i on day t is expressed as

$$R_{it} = \alpha_i + \beta_i R_{mt} + \varepsilon_{it}$$

Where

R_{it} = the rate of return on the share price of firm i on day t

R_{mt}= the rate of return on a market portfolio of stocks (such as Standard & Poor's 500 or a market index) on day t

α= the intercept term
β = the systematic risk of stock i
ε_{it}=the error term, with $E(\varepsilon_{it}) = 0$

And the estimates of daily abnormal returns (AR) for the ith firm is

$$AR_{it} = R_{it} - (a_i + b_i R_{mt})$$

Where a_i and b_i are the ordinary least squares (OLS) parameter estimates obtained from the regression of R_{it} on R_{mt} over an estimation period (T) preceding the event. The abnormal return AR_{it} is the difference between the observed return after the event and the predicted normal return, as a direct measure of the (unexpected) change in securityholder wealth associated with the event.

Many researchers also compute a standardized abnormal return (SAR), where the abnormal return is standardized by its standard deviation:

$$SAR_{it} = AR_{it}/SD_{it}$$

With $SD_{it} = \{S_i^2 \times [1 + 1/T(R_{mt} - R_m)^2 / \sum_{t=1}^{T}(R_{mt} - R_m)^2]\}^{\frac{1}{2}}$
Where S_i^2 is the residual variance from the market model as computed for firm i. R_m is the mean return on the market portfolio calculated during tbe estimation period, and T is the number of days in the estimation period. The standardized abnormal returns can then be cumulated over a number of days, k (the event window), to derive a measure of the cumulative abnormal return (CAR) for each firm:

$$CAR_i = (1/k^{\frac{1}{2}}) \sum_{t=1}^{k} SAR_{it}$$

A standard assumption is that the values of CAR_i are independent and identically distributed. Based on this assumption, the average standardized cumulative abnormal returns across n firms (ACAR) over the event window can be computed as:

$$ACAR_t = 1/n \times 1/[(T-2)/(T-4)]^{\frac{1}{2}} \sum_{t=1}^{k} CAR_{it}$$

The test statistic used to assess whether the average cumulative abnormal return is significantly different from zero (its expected value) is:

$$Z = ACAR_t \times n^{\frac{1}{2}}$$

If significant, the cumulative abnormal return is assumed to measure the average effect of the event on the value of the n firms. That is, the significance of the abnormal return allows the researcher to infer that the event had a significant impact on the values of the firms (McWILLIAMS & Siegel, 1997).

4. Assumptions

To use event study to obtain effective conclusions, several assumptions should be satisfied. If these assumptions are violated, the empirical results may be biased and imprecise, and, therefore, basing conclusions on them is problematic (McWILLIAMS & Siegel, 1997). The assumptions are:

(1) **Markets are efficient**: Stock prices incorporate all relevant information that is available to market traders. An event is anything that results in new relevant information. Long event window may cause the violation of the assumption of market efficiency. If so, it may be

reasonable to assume that information is revealed to investors slowly over a period of time, and a researcher must explain why the effect would not be realized within a short period of time.

(2) **Unanticipated event**: an event is announced in the press.

(3) **No confounding effects during the event window**: a researcher has isolated the effect of an event from the effects of other events, no confounding effects from other events. The longer the event window, the more difficult it is for researchers to claim that they have controlled for confounding effects.

5. Research Design issues:

Research design issues affect the results obtained with this framework (McWILLIAMS & Siegel, 1997). The important research design issues are:

(1) **Sample size**: Large samples—normality assumptions; Small samples—"bootstrap" methods.
For a small sample, researchers should use "bootstrap" methods, which do not require the normality assumptions that are relied upon with large samples (Barclay & Litzenberger, 1988).

(2) **Nonparametric tests to identify outliers**: The test statistics employed in event studies is sensitive to outliers. A small sample magnifies the impact of any one firm's returns on the sample statistic. Researchers should adjust the event study technique, or be especially careful to identify outliers, when dealing with small samples. Two nonparametric tests are binomial Z statistic and Wilcoxon signed rank test.

(3) **The length of the event window and justification of the length**: A long event window severely reduces the power of the test statistic, Z. This reduction leads to false inferences about the significance of an event. A short event window can better capture the significant effect of an event. The event window should be as short as possible. It should be long enough to capture the significant effect of the event, but short enough to exclude confounding effects. The nature of the event being studied should determine the length of the event window used (Ryngaert & Netter, 1990).

(4) **Confounding effects**: Long event windows greatly exacerbate the difficulty of controlling for confounding effects. Failing to control for confounding events causes serious doubts about the validity of the empirical results and calls into question any conclusions drawn.

(5) **Explanation of the abnormal returns**: explain the abnormal returns by showing that the cross-sectional variation in the returns across firms is consistent with a given theory.

6. Length of the Event Window

6.1.Short-Horizon Event Study

A three-factor model (Fama & French, 1993) to estimate abnormal returns. Specifically, this model assumes a linear relationship between the return of any stock and three factors (i.e., company size, company price-to-book ratio, and market risk) over time (Lo et al., 2018). And Fama–French three-factor model can use regression (ordinary least squares estimation) and the mean abnormal returns for day t, abnormal returns for firm i on day t are cumulative abnormal returns calculated.

And moderating effects can also be considered in short-horizon event study by developing a two-stage model (Lo et al., 2018). At stage 1, the cumulative abnormal returns are regressed against all the control variables and the residuals are obtained from the first stage. At the stage 2, the residuals obtained from the first stage are regressed against moderating variables.

6.2.Long-Term Event Study

The nature of the event being studied should determine the length of the event window used (Ryngaert & Netter, 1990). The length of the long windows usually is longer than 2 days, which is the standard length of windows. If the long windows are used in the event study, then it should be justified because of uncertainty about when information was revealed.

The reliability of long-term event study (long-horizon method) is usually questioned. That is many researchers warning that the long-horizon methos lacks of reliability (Kothari & Warner, 2007). The short-horizon methods (short-term study), which are relatively straightforward and trouble-free. The short-term event study can provide clean inference results. And researchers have more confidence and put more weight on the results of short-horizon tests than long-horizon tests due to the confounding events (Yeung et al., 2011).

One the use of long windows is justified, techniques should be used to control for confounding events. And such techniques are to be used only when longer windows are necessary and properly justified. The techniques are:

(1) Foster (1980) suggested four methods to control for confounding events: (1) excluding firms with confounding events from the sample; (2) partitioning the sample by grouping firms with the same confounding events; (3) excluding firms from the sample on the day of the confounding event; and (4) subtracting the financial impact of the confounding event when estimating the sample's abnormal returns.
(2) Control variable is used to control confounding event (Salinger, 1992). In this method, control variable is used to control for what would have happened absent the event. The return on market portfolio, the return on an industry portfolio (if the event did not affect other firms in the industry), the change in a commodity price (oil) (e.g., oil companies) can be suitable control factors used in the event study. And the better the controls for what would have happened absent the event, the higher the power of an event study.

7. Statistical Test in Event Study

The statistical tests commonly used in event studies are the parametric paired-sample t-test, and the nonparametric Wilcoxon signed-rank (WSR) and sign tests (Yeung et al., 2011).

WSR test can take the magnitude of the observations into account, yet it is not seriously affected by extremes (Yeung et al., 2011). That is WSR test is less influenced by outliers than the parametric t-test.

Meanwhile, multiply statistical tests are used in the same research to ensure the robustness of the results, for example WSR test and significance test are selected as the major method to obtain the result, at the same time, t-test is also performed to ensure the robustness of findings (Lo et al., 2014).

8. Procedures of Event Study

Step 1: Determine when it is appropriate to use the event study method. When an event is likely to have a financial impact, is unanticipated by the market, and provides new information to the market, it is appropriate to use the method. When an event is likely to have a financial impact, is unanticipated by the market, and provides new information to the market, it is appropriate to use the method.

Step 2: Outline a theory that justifies a financial response to this new information. This step would include the a priori prediction of the sign of the effect, based on the theory outlined.

Step 3: The third step would be to identify the event's dates and a set of firms that experienced the event.

Step 4: Choose an appropriate event window. The appropriate window should be very short, from 1 to 2 days. And the standard event window is 2 days, the day of the event and the day prior to it, assuming these are trading days. For an **unanticipated** event, the first day on which the market can trade on the information is the event day itself. The use of long windows, longer than 2-day length, should be justified in terms of uncertainty about the impact of the event. That is the researchers may not know when the event was revealed.

Step 5: Eliminate firms from a sample if other event financially relevant to them occurred during the chosen window. Relevant events include such things as unexpected dividend or earnings announcements, takeover bids, merger negotiations, changes in key executives, restructuring, joint ventures, major contract awards, significant labor disputes, significant liability suits, and announcements of major new products. When long windows can be justified because of uncertainty about when information was revealed, techniques can he used to control for confounding events. Such techniques are to be used only when longer windows are necessary and properly justified.

Step 6: Compute the daily (or cumulative) abnormal returns accrued during the event window, using the standard methodology, described in the part "Model", and to test the significance of the abnormal return.

Step 7: Report the percentage of negative returns and the binomial Z or the Wilcoxon test statistic, or both.

Step 8: Include additional information if a sample of fewer than 30 firms was employed. This additional information includes the identification and measurement of the influence of outliers and the results of bootstrapping techniques.

Step 9: Outline a theory that explains the cross-sectional variation in abnormal returns and to test this theory econometrically.

Step 10: Report firm names and event dates in a data appendix, to facilitate replication and extension

9. Advantages and Advantages

There are many advantages of event study method. First, the event study method is relatively easy to implement, because the only data necessary are the names of publicly traded firms, event dates, and stock prices. Second, the event study is based on stock price, and this obviates the need to analyze accounting-based measures of profit, which have been criticized because they are often not very good indicators of the true performance of firms. For example, managers can manipulate accounting profits because they can select accounting procedures. Stock prices are not as subject to manipulation by insiders. Stock prices are supposed to reflect the true value of firms, because they are assumed to reflect the discounted value of all future cash flows and incorporate all relevant information. Therefore, the event study method, based on stock price should measure the financial impact of a change in corporate policy, leadership, or ownership more effectively than a methodology based on accounting returns.

There are also some limitations of event study method. First, an analysis of stock price effects may not be the most appropriate method for testing the impact of unexpected event. The use of this method researchers to an analysis of a strictly firm-level measure of performance. This constraint may be problematic when the impact caused by the unexpected event is not at the firm level, instead the influence may be at the other level, such as the factory level. Under this situation, this method will not work, and alternative methods should be used.

Short-term and long-term event study: The long-term event study lacks of reliability due to the potential confounding events that occur in the event window. In contrast, short-term event study is relatively straightforward and trouble-free. So, the short-term event study can provide more reliable conclusion and results than the long-term event study. Besides, the short-horizon tests can provide clean evidence with efficiency, but interpretation of long-horizon results is problematic. The long-horizon tests are highly susceptible to the joint-test problem, and have low power (Kothari & Warner, 2007).

References:

Barber, B. M., & Lyon, J. D. (1996). Detecting abnormal operating performance: The empirical power and specification of test statistics. *Journal of Financial Economics*, *41*(3), 359–399. https://doi.org/10.1016/0304-405X(96)84701-5

Barclay, M. J., & Litzenberger, R. H. (1988). *Announcement effects of new equity issues and the use of intraday price data*.

Fama, E. F., & French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, *33*(1), 3–56. https://doi.org/10.1016/0304-405X(93)90023-5

Foster, G. (1980). Accounting policy decisions and capital market research. *Journal of Accounting and Economics*, *2*(1), 29–62. https://doi.org/10.1016/0165-4101(80)90014-2

Kothari, S. P., & Warner, J. B. (2007). Econometrics of Event Studies**We thank Espen Eckbo, Jon Lewellen, Adam Kolasinski, and Jay Ritter for insightful comments, and Irfan Safdar and Alan Wancier for research assistance. In *Handbook of Empirical Corporate Finance* (pp. 3–36). Elsevier. https://doi.org/10.1016/B978-0-444-53265-7.50015-9

Li, H., Lam, H. K. S., Ho, W., & Yeung, A. C. L. (2022). The impact of chief risk officer appointments on firm risk and operational efficiency. *Journal of Operations Management*, *68*(3), 241–269. https://doi.org/10.1002/joom.1175

Lo, C. K. Y., Tang, C. S., Zhou, Y., Yeung, A. C. L., & Fan, D. (2018). Environmental Incidents and the Market Value of Firms: An Empirical Investigation in the Chinese Context. *Manufacturing & Service Operations Management*, *20*(3), 422–439. https://doi.org/10.1287/msom.2017.0680

Lo, C. K. Y., Pagell, M., Fan, D., Wiengarten, F., & Yeung, A. C. L. (2014). OHSAS 18001 certification and operating performance: The role of complexity and coupling⋆. *Journal of Operations Management*, *32*(5), 268–280. https://doi.org/10.1016/j.jom.2014.04.004

McWILLIAMS, A., & Siegel, D. (1997). EVENT STUDIES IN MANAGEMENT RESEARCH: THEORETICAL AND EMPIRICAL ISSUES. *Academy of Management Journal*. https://doi.org/10.2307/257056

Ryngaert, & Netter. (1990). Shareholder Wealth Effects of the 1986 Ohio Antitakeover Law Revisited: Its Real Effects. *The Journal of Law, Economics, and Organization*. https://doi.org/10.1093/oxfordjournals.jleo.a036989

Salinger, M. (1992). Value Event Studies. *The Review of Economics and Statistics*, *74*(4), 671. https://doi.org/10.2307/2109381

Yeung, A. C. L., Lo, C. K. Y., & Cheng, T. C. E. (2011). Behind the Iron Cage: An Institutional Perspective on ISO 9000 Adoption and CEO Compensation. *Organization Science*, *22*(6), 1600–1612. https://doi.org/10.1287/orsc.1110.0652

Selection Bias

1. Definition

Selection bias occurs when the **sample is not randomly selected**. And selection bias (i.e., selection effect) refers to situations where research bias is introduced due to factors related to the study's participants. Selection bias occurs when **the selection of subjects** into a study (or their likelihood of remaining in the study) causes a result that is **systematically different to the target population**. It can be introduced via the methods used to select the population of interest, the sampling methods, or the recruitment of participants.

Selection bias is a form of systematic error. Systematic differences between participants and non-participants or between treatment and control groups can limit the ability of researchers to compare the groups, leading to erroneous empirical results and incorrect conclusions regard the veracity of theoretical assertions.

2. Types of Selection Bias

(1) **Sampling bias** (i.e., ascertainment bias) occurs when some members of the intended population are less likely to be included than others. As a result, the **sample is not representative of the population**.
(2) **Attrition bias** occurs when participants **who drop out of a study** are systematically **different from** those who **remain**.
(3) **Self-selection bias** arises when **individuals decide entirely for themselves** whether they want to participate in the study (or **whether to accept the treatment**). Due to this, participants may differ from those who don't—for example, in terms of motivation.
(4) **Survivorship bias** is a form of logical error that leads researchers who study a group to **draw conclusions** by **only focusing on examples of successful individuals** (the "survivors") rather than the group as a whole.
(5) **Nonresponse bias** is observed when people **who don't respond to a survey** are **different in significant ways from those who do**. Non-respondents may be unwilling or unable to participate, leading to their under-representation in the study.
(6) **Under-coverage bias** occurs when **some members of your population are not represented in the sample**. It is common in convenience sampling, where you recruit a sample that's easy to obtain.

Difference between Selection Bias and Sample Self-Selection Bias:
Selection bias is a broad term used to describe different situations in which the selection of study participants leads to an error (bias). As a result, the study population is not representative of the target population. Usually, selection bias is a result of an error in the study design.

Self-selection bias is a subtype of selection bias. It specifically refers to research bias caused by study participants themselves and occurs when individuals volunteer to be part of a research study.

3. Endogeneity Caused by Selection Bias

Selection bias is as a sub-form of the omitted-variable bias issue since the selection process represents an excluded variable that manifests in the error term and correlates with the endogenous choice construct and the outcome variable (Antonakis et al., 2010). Under such situation, the independent variable is correlated with the error term, violating the exogenous assumption of regression model and causing the endogeneity issues.

Selection-based endogeneity has two main forms: sample-selection and self-selection biases (Clougherty et al., 2016). **Sample-selection bias** occurs when samples <u>cannot be representative of a true population</u> and thus threaten both internal and external validity. The sample-selection bias occurs when non-random sample selection methods are used to estimate causal relationship. In other words, sample-selection bias occurs in empirical situations where the outcome variable is only observed for a portion of the true sample since a censoring in the data is present. And the two main reasons for the occurrence of the bias are when the observational subjects make decisions so that a subset of a particular population is not observed and when samples of observational data involve some selection by analysts and data processors (Heckman, 1979).

Sample self-selection means that the studied subjects decide whether to accept the treatment in research by themselves. Selection on observables will not cause endogeneity bias, if the observable selection variables are properly controlled. However, <u>selection on unobservable that correlate with the dependent variable and observable independent variables will cause the endogeneity issues</u>.

4. Solutions to Selection Bias

4.1.Matching Method

Matching method is a quasi-experimental method in which the researcher uses statistical techniques to construct an artificial control group by matching each treated unit with a non-treated unit of similar characteristics. This method solves the selection bias by matching the observable features between the treatment group and the control group. Therefore, matching method can mitigate endogeneity caused by omitted variables. And it is useful for estimating the impact of a program or event for which it is not ethically or logistically feasible to randomize. There are two kinds of matching methods: (1) Direct Matching Method (2) Propensity Score Method (PSM).

Direct matching method is a matching method that matches each treated unit with a non-treated unit of based on similar observable features, directly. It is an ideal matching method because it ensures that the observable features of control group and treatment group are same. However, when the dimensions of features increase, it will be very difficult or even impossible to perform matching process directly based on all features. Then, the PSM can be used.

PSM removes selection bias by creating a quasi-control group. Using a set of subjects' characteristics in a <u>probit regression</u>, this technique pairs every subject in the treatment group with a statistical twin subject from a large set of subjects in control group that do not accept treatment to form the quasi-control group. These statistical twins can be used for comparison to examine the treatment effects. To construct the twins, a proper matching algorithm is required. And the common used matching algorithms in PSM are nearest-neighbor matching, radius matching (i.e. 'caliper' matching) and kernel matching. And the selection of matching algorithm should base on the trade-off between variance and bias caused by matching.

4.2.Heckman's Two-Step Procedure

Heckman's two-step approach deals with selection bias:

Step 1: Run a probit regression as the selection equation to predict the conditional distribution of the treatments with a set of covariates that capture the relevant attributes. Often all control variables and moderators of the study are used for this purpose.
Step 2: Add the resulting inverse Mills ratio (IMR) to the final model (Qiu et al., 2022).

4.3. Natural Experiments

Natural experiment is a unique way of forming treatment and control group to address the sample selection bias during the ex-ante research design stage. This approach is based on occurrence of an intervention such as change in regulatory or a crisis that only affects a limited number of subjects, hence the researcher can form the treatment group based on affected subjects, and treat non-affected subjects as a control group (Fan et al., 2022)

4.4. Regression Discontinuity Design

The regression discontinuity design is a statistical approach to find an indicating factor through which a researcher can assign an observation in the sample to either the treatment or the control group in examining the treatment effects.

The major idea of this method is to find a factor that can describe how an observation becomes part of the treatment group and seeks to exploit the cut-off point for this identified factor (Reeb et al., 2012). The discontinuity in this method refers to identifying the threshold or the cutoff point that can distinguish the treatment from the control group (Zaefarian et al., 2017). This method allows researchers to compare subjects that are just above the cutoff point against those subjects that are marginally below the cut-off point, which is like the propensity score model.

4.5. Other Methods

<u>Selection on observables</u> can be removed by treating the observables as the control variable in the regression model. And selection bias can be avoided as the research recruits and retains the sample population.

(1) For **non-probability sampling designs**, such as <u>observational studies</u>, try to make the **control group** as **comparable** as possible **to the treatment group**. This method is called **matching**. Researchers **match each treated unit with a non-treated unit** of **similar characteristics**. This helps estimate the impact of a program or event for which it <u>is not ethically or logistically feasible to randomize</u>.
(2) In **experimental research**, **selection bias** can be minimized by proper use of **random assignment**, ensuring that neither researchers nor participants know which group each participant is assigned to. Otherwise, knowledge of group assignment can taint the data.
(3) **Sampling bias** can be avoided by carefully <u>defining the target population</u> and using **probability sampling** whenever possible. This ensures that **all eligible participants** have an **equal chance of being included in the sample**.

Other methods that can be used to solve general endogeneity issues can also solve selection bias. The methods are:

1. Instrumental Variables: Two-Stage Least Squares (2SLS)
2. Instrumental Variables: Three-Stage Least Squares (3SLS)
3. Instrument-Free Approaches: Higher Moments
4. Instrument-Free Approaches: Identification through Heteroscedasticity

5. Instrument Free Approaches: Latent Instrumental Variables
6. Instrument Free Approaches: Copulas
7. Generalized Method of Moments

The details of these methods are discussed at the chapter "Endogeneity".

References:

Antonakis, J., Bendahan, S., Jacquart, P., & Lalive, R. (2010). On making causal claims: A review and recommendations. *The Leadership Quarterly*, *21*(6), 1086–1120. https://doi.org/10.1016/j.leaqua.2010.10.010

Clougherty, J. A., Duso, T., & Muck, J. (2016). Correcting for Self-selection Based Endogeneity in Management Research: Review, Recommendations and Simulations. *Organizational Research Methods*, *19*(2), 286–347. https://doi.org/10.1177/1094428115619013

Fan, D., Zhou, Y., Yeung, A. C. L., Lo, C. K. Y., & Tang, C. (2022). Impact of the U.S.–China trade war on the operating performance of U.S. firms: The role of outsourcing and supply base complexity. *Journal of Operations Management*, *68*(8), 928–962. https://doi.org/10.1002/joom.1225

Heckman, J. J. (1979). Sample Selection Bias as a Specification Error. *Econometrica*, *47*(1), 153. https://doi.org/10.2307/1912352

Qiu, L., Liu, R., Jin, Y., Ding, C., Fan, Y., & Yeung, A. C. L. (2022). Impact of credit default swaps on firms' operational efficiency. *Production and Operations Management*, *31*(9), 3611–3631. https://doi.org/10.1111/poms.13788

Reeb, D., Sakakibara, M., & Mahmood, I. P. (2012). From the Editors: Endogeneity in international business research. *Journal of International Business Studies*, *43*(3), 211–218. https://doi.org/10.1057/jibs.2011.60

Difference-in-Difference

1. Definition

The Difference-in-Difference (DID) approach, a quasi-experiment design, is used to assess the causal effect of an event by comparing the set of units where the event happened (treatment group) in relation to units where the event did not happen (control group) (Fan et al., 2022).

The logic behind DID is that if the event never happens, the differences between treatment and control groups should stay the same overtime. In details, if the treated and the nontreated groups are subject to the same time trends, and if the treatment has had no effect in the pre-treatment period, then an estimate of the "effect" of the treatment in a period in which it is known to have none, can be used to remove the effect of confounding factors to which a comparison of post-treatment outcomes of treated and nontreated may be subject to (Lechner, 2010).

DID is an effective tool to apply when randomization on the individual level is not possible because the unobserved differences between treatment and control groups are the same overtime without the treatment in DID. It can control the unobserved variables that bias estimates of causal effects, aided by longitudinal data. And DID is suitable when using research designs based on controlling for confounding variables or using instrumental variables is deemed unsuitable, and at the same time, pre-treatment information is available (Lechner, 2010). And DID can be implemented with repeated cross-sections, panel data, or retrospective cross-section data. And it can be used for both time-constant and time-varying variables. <u>DID can mitigate endogeneity caused by omitted variables, errors-in-variables and simultaneity</u>.

2. Assumptions of Difference-In-Difference

To effectively apply DID in research, four assumptions should be fulfilled. The assumptions are:
(1) Exchangeability- Intervention unrelated to outcome at baseline (allocation of intervention between treatment group and control group was not determined by outcome)
(2) Positivity-Treatment/intervention and control groups have Parallel Trends in outcome. And this assumption is also known as the Parallel Trend Assumption.
(3) Stable Unit Treatment Value Assumption (SUTVA)-Composition of treatment and control groups is stable for repeated cross-sectional design.
(4) SUTVA-No spillover effects

To ensure the internal validity of DID models, the parallel trend assumption must be met. An effective method to prove that this assumption is fulfilled is to visually assess whether the pre-treatment trends between the treated and control groups evolve in parallel (Mithas et al., 2022). An effective way to convince readers about this assumption is to visually assess whether the pre-treatment trends between the treated and control groups evolve in parallel. Test can be performed for the parallel in the pre-trends in an event study analysis or relative time model, where the mean difference in the outcomes between the treated and control units is estimated period-by-period, with several periods before and after the treatment (Marcus & Sant'Anna, 2021). It has also been proposed that the smaller the time period tested, the more likely the assumption is to hold. Violation of parallel trend assumption will lead to biased estimation of the causal effect.

3. Model of Difference-in-Difference (with Additional Regressor)

$$Y_{it} = \beta_3 Treat_i \times After_t + \gamma X_{it} + \alpha_i + Year_t + e_{it}$$

The **intervention (event) is exogeneous, therefore $\boldsymbol{E(e_{it}|Treat_i, After_t) = 0}$.** And β_3, the coefficient of interaction term $Treat_i \times After_t$ is the DID estimator. X_{it} , as the observable variable for the individual i at time point t, is the control variable. The control variable, i.e., X_{it}, must be the variable that is not influence by the intervention. The individual fixed effects α_i control for the time-constant characteristics between the treatment group and control group. And the time fixed effects $Year_t$ control for time-varying features between the treatment group and control group. Under this situation, a hidden assumption is that the difference between treatment and control groups should stay the same overtime (parallel trend assumption is satisfied), so this difference can be controlled by the same time fixed effects $Year_t$. If this assumption is satisfied, the inclusion of the control variable X_{it} will not change the estimation of the coefficient. X_{it} only decomposes a "noise" part from the disturbance term (error term), reducing the variance of the estimated value and making the estimator more efficient. (The control variable is the additional regressor that may be or may not be included in DID)

If the estimated value of the coefficient changes, after the inclusion of the control variable X_{it}, this means the parallel trend assumption is not satisfied. This indicates an potential omitted-variables endogeneity issue. There may exist unobservable and time-constant variable(s) that is correlated with independent variable(s) and dependent variables.

The control variable X_{it} used in the DID model must not be the variable that is not influenced by the event. The selection of the control variable for the DID model can be difficult, researchers should make judgement based on theory and extant literature.

3.1. Least Square Dummy Variable Estimator

Least squares dummy variables (LSDV) estimator can handle the dynamic panel bias, it works only for balanced panels (Roodman, 2009) and is unsuitable for our unbalanced sample with some samples having more observations than the others (Lam et al., 2016).

4. Advantage and Disadvantage of Difference-In-Difference Method

There are many advantages of DID. First, the result can be intuitive interpreted. The result obtained from DID can be easily interpreted by research based on intuition. Second, a relatively clean casual effect can be obtained using observational data if assumptions are met. Third, DID methods can use individual and group level data, which is very flexible. Forth, the control groups can start at different levels of the outcome, because DID focuses on change rather than absolute levels. Finally, DID accounts for change due to factors other than intervention. In other words, DID can control confounding variables in the research.

This method also has many limitations. First, DID require baseline data and a control group. The data before the intervention and the control group may not be available. Second, if some key assumptions are not fulfilled, this method cannot be applied. If the allocation of the intervention is decided by baseline outcome, DID cannot be used. And if the control groups have different outcome trend with the treatment groups, the method cannot be applied. Meanwhile, if the composition of groups before and after the intervention are not stable, the method cannot be used.

References:

Fan, D., Zhou, Y., Yeung, A. C. L., Lo, C. K. Y., & Tang, C. (2022). Impact of the U.S.–China trade war on the operating performance of U.S. firms: The role of outsourcing and supply base complexity. *Journal of Operations Management*, *68*(8), 928–962. https://doi.org/10.1002/joom.1225

Lam, H. K. S., Yeung, A. C. L., & Cheng, T. C. E. (2016). The impact of firms' social media initiatives on operational efficiency and innovativeness. *Journal of Operations Management*, *47–48*(1), 28–43. https://doi.org/10.1016/j.jom.2016.06.001

Lechner, M. (2010). The Estimation of Causal Effects by Difference-in-Difference Methods Estimation of Spatial Panels. *Foundations and Trends® in Econometrics*, *4*(3), 165–224. https://doi.org/10.1561/0800000014

Common Proxy

Construct	Name of Measure	Detail of Measure
Operational Efficiency	Return on Assets (ROA)	Measured as operating income before depreciation, interest, and taxes divided by total asset
Operational Efficiency	Inventory days	The days sales of inventory (DSI) is a financial ratio that indicates the average time in days that a company takes to turn its inventory, including goods that are a work in progress, into sales.
Operational performance	total factor productivity = multi-factor productivity	Measured as the ratio of aggregate output (e.g., GDP) to aggregate inputs
Business unit profitability	business success(profitability)	Profitability for a given business unit (such as a line of staplers)
	Industry efficiency level	Industry median ROA (two-digit SIC code).
Market concentration of an industry - the size of firms in relation to the industry they are in- and is used to determine market competitiveness.	Industry competitiveness, or the Herfindahl index (Herfindahl Hirschman Index (HHI))	The sum of squared market shares of the firms in an industry.
	Firm size	The natural logarithm of a firm's total assets.
	Firm size	The natural logarithm of a firm's sales
	Firm size	The natural logarithm of the number of employees employed by the firm
	Industry size	The natural logarithm of all the firms' total assets in the same industry
	Industry size	Industry output
	Industry size	Number of firms
	Industry size	Number of employees
	Firm technology intensity	The ratio of a firm's R&D expenses to its sales.
	Firm labor productivity	The ratio of a firm's operating profit (before depreciation and tax) to its number of employees.

	Firm labor intensity	The ratio of a firm's employee number to its total assets
	R&D intensity	It is defined as expenditures by a firm on its research and development (R&D) divided by the firm's sales
A key indicator used to monitor resources devoted to science and technology	R&D expenditure	R&D expenditure
Knowledge impact and knowledge diversity	Patent data	Number of patent
	Firm Age	calculating the number of years since the establishment of the firm.
	Leverage	the ratio of total debts to total assets.
	Tangibleness	proportion of tangible assets to total assets
	R&D intensity	the ratio of R&D expenditures to total assets.
Operational efficiency	labor productivity	
Operational efficiency	inventory turnover	
Operational efficiency	Inventory efficiency	days of supply
Operational efficiency	Inventory efficiency	the ratio of sales to average inventory
Operational efficiency	Profitabiligy	returen-on-assets (ROA)
Operational efficiency	Fixed asset turnover	the ratio of sales to PP&E
	Capital expenditure	a firm's capital expenditure normalized by sales
	R&D expenditure	annual value of R&D expenditure
	Horizontal complexity	the firm's number of first-tier suppliers
	Level of outsourcing	reversed the scale of vertical integration
	Spatial complexity	the number of countries or regions where a firm's suppliers were located.
	Firm age	the natural logarithm of the number of years since a firm's initial public offering [IPO]
	Firm advertising expense	a firm's advertising expenses divided by sales

	Market-to-book ratio	a firm’s market value divided by its book value
	Cash holdings	a firm’s cash divided by total assets
	Retained earnings	a firm’s retained earnings divided by total assets
	Tangible assets	a firm’s property, plant, and equipment divided by total assets
	Working capital	equals a firm’s working capital divided by total assets
credit risk	credit rating	a dummy variable that equals 1 if a firm has an institutional stockholder; otherwise, 0)
credit risk	Investment grades	dummies assigned to each investment grade level) to capture firms’ credit risk.
Operational efficiency	Stochastic Frontier Estimation (SFE) Method	
Financial performance	Tobin’s Q	A ratio of the firm’s market value to the replacement cost of its assets

Availability of Datasets

1. Primary Data and Secondary Data

Primary Data: Data that has been generated by the researcher through surveys, interviews, experiments, specially designed for understanding and solving the research problem at hand.

Secondary Data: Using existing data generated by others, such as large government institutions, healthcare facilities etc. It is the data that have been already collected by and readily available from other sources. This kind of data is usually extracted from data sets.

1.1. Advantages and Disadvantages of Secondary Data

There are many advantages of using secondary data:

(1) It is economical. It saves efforts and expenses.
(2) It is time saving.
(3) It helps to make primary data collection more specific since with the help of secondary data, we are able to make out what are the gaps and deficiencies and what additional information needs to be collected.
(4) It helps to improve the understanding of the problem.
(5) It provides a basis for comparison for the data that is collected by the researcher.

There are also many disadvantages of using secondary data:

(1) Secondary data might not have all the information needed to answer the research questions being addressed.
(2) Accuracy of secondary data is not known.
(3) Data may be outdated.

The use of secondary data can range from exploration to theory testing. secondary data usually have a number of limitations. They can be used alone or in combination with primary data.

2. Data Source of Secondary Data

2.1. Public Information

Public information includes announcements and articles; stock price data; public accounting data; analysts' opinion.

As for the announcements and articles, data can be collected from Business publications (use Factiva Database), Wall Street Journal, Dow Jones News Service, Businesswire, PRNewswire, Database of Newspapers and Newswires and sources of information about Chinese companies.

As for stock price data, Center for Research in Security Prices (CRSP) is available. The related data, such stock prices, dividends return, is available at daily, monthly and annual frequency since 1925.

As for public accounting data, the financial statements disclosed by listed companies are available. And the COMPUSTA also provides such data.

2.2.Other Data Source

Private accounting data is also available by using the accounting records in the accounting system of firms. Analysts' opinion, such as the industrial research reports from Wind, is also available. Data may also be gathered from the company information, trade associations and publications, as well as the government. Data created by the researchers from company records or survey of managers can also be used. The research may also run a Web crawler to gather information from websites.

As for the data from government, many open data sets are available. Open Government Data (OGD) provides data on business information, registers, patent and trademark information, public tender databases, legal information etc.

3. Data Sets

Compustat: Corporate investment and financial data for US and non-US companies. It includes annual and quarterly accounting data, earnings announcement dates, business segment data (segment levels) and some stock data.

Bloomberg: It is the world's primary distributor of financial data and a top news provider. It provides business and markets news, data, analysis, and video to the world, featuring stories from Businessweek and Bloomberg News. Stock information is also available.

Thomson One: It provides financial data and market information, such as information of companies' merge and acquisition.

The Thomson Financial Securities Data Corporation (SDC) Database: Providng detailed information on different types of joint ventures. This database compiles information from different publicly available sources.

Datastream: global data on patents, brand value, number of consumer controversies directly linked to companies' products or services, product recall, eco-design products, energy footprint reduction, organic-product initiatives, among others.

Open Government Data (OGD): Global open government data. It provides data on business information, registers, patent and trademark information, public tender databases, legal information etc.

Factiva: Factiva contains news articles from top media outlets such as The Wall Street Journal, The New York Times, as well as hundreds of other sources. And it covers the announcements of social media initiatives deployed on both public and private social media platforms, thus using Factiva avoids the possible bias arising from focusing on public social media platforms (e.g., Facebook and Twitter) only.

Bloomberg's SPLC Database: longitudinal supply chain data, such as information of suppliers and customers.

FactSet Revere: It collects panel supply chain data from various sources such as SEC 10-K annual filings, investor presentations, press releases, and corporate actions to cover a wide range of supply chain information.

Fresard-Hoberg-Philips database: It provides vertical integration data.

IPE: It gathers information on environmental incidents from various government departments or news service agencies at different levels and maintains an easily accessible database. Although the IPE database provides the date for each government announcement of an environmental incident, this date is not necessarily the earliest date that the public became aware of the news.

Political polling data is the repeated cross-sectional data, in which political preferences are measured by a series of surveys of randomly selected potential voters, where the surveys are taken at different dates and each survey has different respondents. (A repeated cross-sectional data set is a collection of cross-sectional data sets, where each cross-sectional data set corresponds to a different period. The subjects observed in each data sets at different time point are different.)

4. Availability of Secondary Data

Existing company data have the advantage of being easy to assemble, but those data might not have all the information needed to answer the research questions being addressed, so researchers will often augment existing data with additional data collected for the purpose of their research project. To answering the research question, researchers may need to construct a database within a company.

If the secondary data is not available, primary data should be considered. Data can be collect from direct observation of products, direct observation of processes augmented by discussions with managers, experiments in a laboratory, experiments in companies, survey and interview.

As for OM empirical research, some confidential data of companies, such as the factory level information, may not be available.

Meanwhile, there is a decrease in the use of cross-sectional data and event data, while an increase trend in using panel data. The former has some limitations in addressing endogeneity while the latter is difficult to obtain. The unavailability of data for an important explanatory variable could be one of the reasons for omitted variable bias, and the emergence of firm-specific business and financial databases could somehow facilitate business researchers to overcome this issue and capture relevant and reasonable number of firm-specific characteristics in their econometric analysis.

5. Evaluation of Secondary Data

Evaluation means the following four requirements must be satisfied:

(1) Availability- It has to be seen that the kind of data you want is available or not. If it is not available then you have to go for primary data.
(2) Relevance- It should be meeting the requirements of the problem. For this we have two criterion: (a) Units of measurement should be the same. (b) Concepts used must be same and currency of data should not be outdated.

(3) Accuracy- In order to find how accurate the data is, the following points must be considered: (a) Specification and methodology used. (b) Margin of error should be examined (c)The dependability of the source must be seen.

6. Procedure to Evaluate Secondary Data

(1) Applicability of research objective.

(2) Cost of acquisition.

(3) Accuracy of data.

Moderating Factors

1. Definition:

A moderator variable is a qualitative or quantitative variable that affects the relationship between a predictor variable (X) (i.e., independent variable) and an outcome variable (Y) (i.e., dependent variable). Moderator variables commonly affect the strength of the relationship between X and Y (Gellman & Turner, 2013). In some cases, a moderator can affect the direction of association between X and Y(Gellman & Turner, 2013).

2. Test of Moderation Effect

To make illustration, the following linear regression model with moderating factor is used.

$$Y = aX + bM + cXM + e \tag{1}$$

The Eq. (1) can be rewritten as:

$$Y = bM + (a + cM)X + e \tag{2}$$

For fixed M, this is a linear regression of Y on X. The relationship between Y and X is characterized by the regression coefficient $a + cM$, which is a linear function of M. <u>M is the moderator variable and the magnitude of the moderating effect is measured by c.</u>

<u>To infer that a variable is a moderating variable, there must be a significant statistical interaction between the predictor and the moderator (i.e. p < .05).</u> For the model Eq. (1), the analysis of the moderating effect focuses on testing and estimating c. If c is significant, then the null hypothesis ($H_o: c = 0$) will be rejected and the moderating effect of M is significant.

Moderating effect analysis for two types of variables are considered. The first type is for observable variables, that is the dependent variable, independent variable(s) and the moderator variable(s) are all observables. The second type is for variables, including, independent variable(s) and the moderator variable(s), that contain at least one latent variable (i.e., unobservable variable).

2.1. Test for Observable Variables

Variables X and M can be classified into two categories. On is the category variables, including definite and sequential variables. The other category is continuous variables, including fixed-distance and fixed-ratio variables. When fixed-order variables have more values and are evenly spaced, they can also be treated approximately as continuous variables. The methos are summarized as follow:

Moderator Variable (M)	Independent Variable (X)	
	Category Variable	Continuous Variable
Category Variable	Analysis of variance (ANOVA) with interaction effect of two factors. Interaction effect is moderated effect.	Grouped regression: The sample data is grouped based on the value of M. Regress Y on X. If the difference in regression coefficients is significant, the moderating effect is significant. If the difference in regression coefficients is significant, then

		the moderating effect is significant.
Continuous Variable	Pseudo-variables were used for the independent variables, centering the independent and moderator variables. Centering means subtracting the mean value from every value of a variable. Performing hierarchical regression analysis for $Y = aX + bM + cXM + e$ (1) Regress Y on X and M to obtain R_1^2 (2) Regress Y on X, M and XM to obtain R_2^2. If R_2^2 is significantly higher than R_1^2, then the moderating effect is significant. Or, perform the regression coefficient test of XM. if it is significant, then the moderating effect is significant.	Centering the independent variable (X) and the moderator variable (M). Performing hierarchical regression analysis for $Y = aX + bM + cXM + e$ (1) Regress Y on X and M to obtain R_1^2 Regress Y on X, M and XM to obtain R_2^2 . If R_2^2 is significantly higher than R_1^2, then the moderating effect is significant. Or, perform the regression coefficient test of XM. if it is significant, then the moderating effect is significant. Besides considering the interaction effect of XM, the higher-order interaction effect terms can be considered (e.g., XM^2 for the nonlinear moderating effect; MX^2 for moderating effect of the curvilinear regression).

When using **regression analyses** to detect a moderator effect, multicollinearity problem can be solved by "centering" the predictor (independent variable) and moderator variables. Centering simply means that these variables are transformed by subtracting each variable by its sample mean (e.g., X-mean of X).

Three-Stage Least Squares (3SLS): 3SLS is an instrumental variables estimator for structural equations in which at least one equation contains endogenous independent variables. It is like the 2SLS estimation, with the difference being <u>that moderator variables are used as instrumental variables</u> to obtain residuals for the endogenous independent variable. The detailed steps are:

Step 1: Regress each predictor on all moderators, confirm significant relationship between moderatos and the predictor, and obtain residuals for the predictor.
Step 2: Replace each predictor with the corresponding residual and regress dependent variable on obtained residuals.
Step 3: Add the interaction terms to the model.

2.1.1. Examples in OM

Lo et. al. (2014) uses Hierarchical Linear Modeling (HLM) estimations with ROA as a dependent variable and investigate the moderating effect of firms' technological knowledge characteristics on the impact of innovative technology products on firm's financial performance.

And Yiu et. al. (2020) tests the moderating effects in a dynamic panel data (DPD) model by including the moderating factors as the interaction in the regression model and the coefficients reflect the moderating effects.

2.2. Test for Latent Variables

Moderating effects analysis for latent variables needs to use structural equation models (SEM). The measurement of latent variables introduces measurement error, so latent variables are continuous. Moderated effects models with latent variables usually only consider two scenarios: either the moderator is a categorical variable, and the independent variable is a latent variable, or both the moderator and the independent variable are latent variables.

When **the moderator variable is a categorical variable**, group analysis in SEM is used. The steps are: First, the regression coefficients obtained from the two-group analysis using SEM is restricted to be equal, yielding a chi-square value and corresponding degrees of freedom. Then, this restriction is removed and re-estimating the regression coefficients based on the same model to obtain another chi-square value and corresponding degrees of freedom. The previous chi-square is subtracted from the subsequent chi-square to obtain a new chi-square whose degrees of freedom are the difference between the degrees of freedom of the two models. If the chi-square test is statistically significant, the moderating effect is significant.

When **both the moderator and independent variables are latent variables**, there are many different methods of analysis. For example, the constrained approach of Algina and Moulder (2001) allows to ignore the intercepts and latent means by centering the product of indicators (Steinmetz et al., 2011). This method is palpable for the normal distribution. And the generalized product indicator (GAPI) method of Wall and Amemiya (Wall & Amemiya, 2001), also for the non-normal distribution. Both these two methods are constrained approach and require non-linear constraints, which are cumbersome and error prone.

And there is unconstrained approach proposed by Marsh et al., (2007), which does not require any constraint, greatly simplifying the procedure and easily to be applied in research. This is the making it one of the most convenient methods.

References:

Gellman, M. D., & Turner, J. R. (Eds.). (2013). *Encyclopedia of Behavioral Medicine*. Springer New York. https://doi.org/10.1007/978-1-4419-1005-9

Lo, C. K. Y., Pagell, M., Fan, D., Wiengarten, F., & Yeung, A. C. L. (2014). OHSAS 18001 certification and operating performance: The role of complexity and coupling⋆. *Journal of Operations Management*, *32*(5), 268–280. https://doi.org/10.1016/j.jom.2014.04.004

Marsh, H. W., Wen, Z., Hau, K.-T., Little, T. D., Bovaird, J. A., & Widaman, K. F. (2007). Unconstrained Structural Equation Models of Latent Interactions: Contrasting Residual- and Mean-Centered Approaches. *Structural Equation Modeling: A Multidisciplinary Journal*, *14*(4), 570–580. https://doi.org/10.1080/10705510701303921

Steinmetz, H., Davidov, E., & Schmidt, P. (2011). Three Approaches to Estimate Latent Interaction Effects: Intention and Perceived Behavioral Control in the Theory of Planned Behavior. *Methodological Innovations Online*, *6*(1), 95–110. https://doi.org/10.4256/mio.2010.0030

Wall, M. M., & Amemiya, Y. (2001). Generalized Appended Product Indicator Procedure for Nonlinear Structural Equation Analysis. *Journal of Educational and Behavioral Statistics*, *26*(1), 1–29. https://doi.org/10.3102/10769986026001001

Yiu, L. M. D., Lam, H. K. S., Yeung, A. C. L., & Cheng, T. C. E. (2020). Enhancing the Financial Returns of R&D Investments through Operations Management. *Production and Operations Management*, *29*(7), 1658–1678. https://doi.org/10.1111/poms.13186

Endogeneity

1. Definition

Endogeneity happens when an independent variable is correlated with the disturbance term (i.e., error term) in regression modeling. And the consideration of endogeneity should be based on substantive theoretical considerations (Ketokivi & McIntosh, 2017). If the endogeneity issue is ignored, the biased estimation results can be obtained, leading to distort causal relationship, and thus the incorrect conclusions and managerial insights. And it is important to understand that endogeneity is based on substantive, theoretical considerations instead of on statistical theory(Ketokivi & McIntosh, 2017).

2. Sources of Endogeneity

Generally speaking, there are three sources of endogeneity: (1) omission of variables (2) errors-in-variables (3) simultaneous causality (Wooldridge, 2002; Zaefarian et al., 2017). For panel data, the different sources of endogeneity may include unobserved heterogeneity, simultaneity and dynamic endogeneity (Wintoki et al., 2012; Yiu et al., 2020).

2.1.Omission of Variables

The omission of variables in regression models can cause endogeneity. The omission of variables is generally caused by the unavailability of data. When the omitted variable is correlated with the dependent variable and the independent variable, then the independent variable should be associated with the disturbance term since all omitted and unobservable variables are included in the disturbance term. Under this situation, the assumption of exogeneity is violated and the endogeneity occurs (Kennedy, 2008; Wooldridge, 2002).

Omitting selection is a conventional type of the endogeneity caused by omission of variable (Qiu et al., 2022). Sample self-selection based on observables can be controlled by using control variables. Unfortunately, the sample self-selection based on unobserved variables that are correlated with the dependent and independent variables can cause the endogenous problems. This situation can result in the observed values of dependent variables of the sample are different from non-sample parts on a nonrandom basis (Fan et al., 2022; Qiu et al., 2022; Yeung et al., 2011).

2.2.Errors-In-Variables

Errors-in-variables is also a source that can cause endogeneity. The endogeneity occurs when variables are inappropriately measured and their true values are still unobserved (Wooldridge, 2002). Measurement errors occurs when the constructs in the theoretical model are not measured perfectly, or non-comprehensiveness of the data collection method since missing data is also deemed as a type of measurement error (Kennedy, 2008). Besides, if the variables that the researchers collect data for differ from the variables that truly affect the phenomena to be studied, the errors-in-variable will also negatively influence the results of the research (Wooldridge, 2002).

Measurement error in dependent variables will not cause endogeneity. However, measurement errors in independent variables may cause sever endogenous issue depend on the assumptions about the measurement error (Wooldridge, 2002). The first assumption is that the measurement error and the observed independent variable are uncorrelated, and that the error term of the model is uncorrelated with the actual (unobserved) and the observed independent variable. In this case, estimation yields consistent coefficients. The second assumption is also known as the 'classical errors-in-variable (CEV) assumption'. Under this assumption, the measurement error is not correlated with the actual (unobserved) independent variable, and the disturbance term in the regression model is also uncorrelated with the actual(observed) independent variable. Under this situation, the observed independent variable is correlated with the measurement error, and thus it is correlated with the disturbance term, violating the fundamental assumption for OLS model and resulting in endogeneity. In other words, the inconsistent estimation of the coefficient, which is biased towards zero, will be obtained from the regression model. And the size of this bias depends on the variance of the actual independent variable relative to the variance in the measurement error (Zaefarian et al., 2017).

2.3.Simultaneous Causality

Simultaneous causality, also known as reverse causality, can also cause endogeneity, when one (or more) independent variable is jointly determined with the dependent variable (Wooldridge, 2002). In other words, the independent variables and dependent variables cause each other at the same time and the causal effects for them are reciprocal and independent and dependent variables at the same time cause each other (Wooldridge, 2002). For panel data, the problems related with the simultaneous causality are known as the dynamic endogeneity (Yiu et al., 2020), where the past values of dependent variables influence the current value of the independent variable.

3. Solutions to Endogeneity

Because the disturbance term in endogeneity issue is unobservable, there is no direct method to statistically test whether the endogenous variable is correlated with the disturbance term. Therefore, only indirect tests and precautionary measures can assist the research to make the decision of how to mitigate endogeneity in regression modeling.

The endogeneity issue can be mitigated at two stages, including the research design stage and the statistical analysis stage. At the research design stage, researchers can extensively reviewing literature and providing comprehensive research designs that could help to apply appropriate statistical tools (Ketokivi & McIntosh, 2017). As for the statistical analysis, research should ensure the data used in the research is rigorously invested by statistical techniques (Ketokivi & McIntosh, 2017) Besides the research design, there are additional factors that need to be considered when deciding the appropriate method to mitigate endogeneity in research, such as data collection instrument, sample size, complexity of the model, and underlying theory and research context. In this report, the solutions to endogeneity will be summarized and stated for the two stages, i.e., the research design stage and the statistical analysis stage.

3.1. Solutions at Research Design Stage

Research design is a strategy for answering the research question using empirical data. And the research design includes the following decisions:

(a) Overall research objectives and research approach
(b) Type of research: primary research or secondary research
(c) Sampling method
(d) Data collection methods
(e) Data analysis methods

To address the endogeneity issue in OM emprical research, theoretical, empirical and pragmatic considerations should be made. And the solutions that can be used to mitigate endogeneity at the research design stage are summarized and analyzed below.

3.1.1. Theoretical Prescription

There are some theoretical steps for researchers to mitigate the endogeous issues in their research. First, the research should <u>list all the potential endogeneity concerns implied by the theoretical model</u>, and <u>decide how significant a threat each of the listed concerns generate to empirical analysis</u> (Ketokivi & McIntosh, 2017). This procedure require a thorough understanding of the theoretical model. Researchers should understand the structure of the model and to interpret the assumptions embedded in the structural limitations of the model, and explore how and why the relevant factors of interest are interrelated (Ketokivi & McIntosh, 2017). Second, the research should <u>identify potential omitted variables at the model specification stage</u> (Ketokivi & McIntosh, 2017). All possible potential factors that should be included in the model should be identified based on theory and extant literaturey. Researchers should make judement about whether there is any omitted variable exist form the theoretical aspect when building the theoretical model for the research. Unfortunately, there may be factors, such unobservable factors, that canno be identified. Then other remidies are required, and they will be discussed in the following parts.

3.1.2. Empirical Remedy

After the theoretical and substantive concerns be handled, empirical remedies can be considered by researchers to solve the endogeneity. In this report, some common approaches are discussed.

3.1.2.1.Experimental Research

In experimental research, researchers systematically intervene in a process and measure the outcome. And to carry out an experiment, researchers should vary the independent variable, accurately measrue the dependent variable, and and control for confounding variables. This method can generate more clear casual inference than other non-experiment research method. The validity of research depends on the experimental design and different experimental design has different level of internal and external validity. There are four diferent types of experiments that are laboratory experiment, field experiment, quasi-experiment and natural experiment. A laboratory experiment takes place in a laboratory environment under controlled conditions. In a laboratory experiment, researchers can chose the conditions, such as place, time and other participants. And the laboratory experiment is a true experiment. For a true experiment, three criteria should be met. The criteria includes (1) Manipulation of treatment conditions (with treatment and control groups) (2) Random assignment of participants to treatments (3) Control of all confounding variables.

A field experiment is one, where the researcher conducts an experiment by manipulating an independent variable and measuring the effects on the dependent variable in a natural environment. In a field experiment, researchers still try control confounding variables, but it’s just happening in a natural environment-the field. The aim of a field experiment is to obtain as much as raw data as possible in the natural world. A natural experiment is one where the independent variable is naturally occurring. i.e. it hasn’t been manipulated by the researcher. A quasi-experiment aims to evaluate interventions but that do not use randomization. Similar to randomized trials, quasi-experiments aim to demonstrate causality between an intervention and an outcome. The main difference between the natural experiment and the quasi-experiment is that the criterion for assignment is selected by the researcher in the quasi-experiment while the assignment occurs 'naturally,' without the researcher's intervention in a natural experiment. Field researchs are not quasi-experiment, but most the quasi-experiments are conducted in the field and hence are field experiments.

In OM empirical research, natural experiment or quasi-experiment are more commonly used to solve the endogeneity than other experiment design because these two types of experiments can obtain conclusion from environment that is more realistic. In other words, the natural experiment and quasi-experiment have better external validity than other experiment design. There are approaches to mitigate the endogeneity caused by self-selection bias by studying the variable of interest before and after a particular intervention has occurred. Then main challenge to apply natural experiment or quasi-experiment in OM empirical research is that once a intervention has occurred, plent of factors may change and thus the comparison of before and after interventioin situations is not useful to draw on meaningful conclusion. To solve this problem, researchers should control for this by using a set of subjects that are not influenced by the intervention as a control group. And then, they should compare the difference in the group that is influenced by the intervention, also known as the treatment group, to the difference in the control group (Fan et al., 2022). This approach is often called as the difference-in-difference (DID) test.

In the research in OM emprical field, a natural experiment to understand the influence of treatment (or intervention) by comparing the observations from the treatment group and control group. Propensity score matching (PSM) is used to develop the matched pairs. The matching procedures ensured the treatment and control firms were highly similar in terms characteristics (confounding factors) other than the variabales of interest. And then, DID test is performed to make casual inference (Fan et al., 2022). The details of DID test and PSM will be disscussed in the corresponding parts of them.

Unfortunately, there are many problems to perfrom an experimental research in the operations management research. First, if the experiments are carried out in the field rather than in the laboratory, the researchers may not be able to manipulate the insterested variables and exmaine the related effects. Besides, the field experimental research also encounter endogeneity when the variables of insterest may be difficult to make exogenous by manipulation. A longitudinal field experiment can be employed to mitigate the endogeneity issue, however it may be impossible to implement due to the reality restrictions and ethic concerns. Second, it is difficult to comfirm that something practically relevant is being estimated since it is merely an experiment.

In general, the main disadvantage of the experimental research is the concerns of external validity instead of the endogeneity. The conclusion obtained from experimental research may not be generalized to more realistic and common situations.

3.2.1.2. Instrumental Variables Techniques

The instrumental variables (IVs) are variables that are uncorrelated with the disturbance but are correlated with the endogenous independent variable in the original regression model (Murray, 2006). And IVs do not influence the dependent variable directly, instead they influence it indirectly through the endogeous independent variables (Rossi, 2014). The fundamental idea of IVs is to decompose the variations in the endogenous independent variable with the help of IVs by concentrating on the variations in the endogenous independent variable that are uncorrelated with the disturbance term in the model and disregarding the variations that bias the estimation (Zaefarian et al., 2017). The insturmental variables techniques can mitigate there types of endogeneity, including omitted-variable endogeneity, endogeneity caused by errors-in-variables and endogeneity caused by simultaneous causality.

To effectively use IVs to mitigate endogeneity, valid IVs should be identified. There are two conditions based on which the validity of IVs can be judged. These two conditions are relevance and exogeneity (Bartels, 1991; Kennedy, 2008; Murray, 2006). Relevance refers to the degree to which an instrumental variable is sufficiently correlated with the endogenous independent variable. And an instrumental variable that has a high correlation with the endogenous independent variable is defined as the strong instrumental variable. In contrast, an instrumental variable that has a low correlation with the endogenous independent variable is defined as the weak instrumental variable. Exogeneity is defined as the degree to which an instrumental variable and the distrubance term in the model are uncorrelated (Murray, 2006).

There are many disadvantages of IVs to be used in OM empirical research to solve endogeneity. First, it may be impossible to find a valid instrumental variable that are correlated with the endogeneous variable but not correlated with the disturbance. The incorporation of disqualified instrumental variable may not solve the endogenous issues but make the estimations obatined from the regression model become even worse. Second, the precision of instrumental variable regression estimates is lower than that of OLS estimates (Baum, 2006). Although the use of instrumental variable can mitigate estimation inconsistency, however, the estimation will become less accurate. If only weak instruments are available, the loss of precision will be severe and instrumental variable regression estimates may have no improvement over OLS estimates. This efficiency loss can be controlled by using a large sample. Third, most of the justification for the use of instrumental variable regression is asymptotic and the regression technique may not work well in small samples. Using instrumental variables can cause the increase of the finite-sample bias. Finally, the instrumental variable techniques are based on the assumption that at least one of the instrumental variable is exogeneous. This means that at least some of them are untestable. And there is no way to ensure that this assumption definitly holds in research.

Although the endogeneity concerns can be addressed by employing conventional IV techniques that use external variables as instruments, prior research has demonstrated the difficulty of obtaining such strictly exogenous instruments externally (Roodman 2009, Wintoki et al. 2012;Yiu et al., 2020). If there is a large number of endogenous variables included in research and the limited availability of appropriate external datasets that cover the sample sujects, IV techniques are not applicable (Yiu et al., 2020). The system GMM with internal instruments can solve such kinds of situation, which will be discussed in the section of GMM.

3.2.1.3. Examine Variance and Change Over Time

Panel data contains multiply subjects. It is a subset of longitudinal data where observations are for the same subjects each time. And two relevant variants of longitudinal data: observational and quasi-experimental data.

3.2.1.3.1. Longitudinal, Observational Data

There are two different types of model in panel data, i.e. the static panel data model and the dynamic panel model. The static panel data model includes fixed effects (FE) model and random effects (RE) model, which can mitigate endogeneity caused by omitted variable. And the dynamic panel model includes generalized method of moments (GMM) model, which can mitigate endogeneity caused by omitted variable, errors-in-variables and simultaneous causality.

Panel data contains multiply subjects. It is a dataset in which the behavior of each subject is observed at multiple points in time. Thus, it is a subset of longitudinal data. There are two different types of model in panel data, i.e. the static panel data model and the dynamic panel model. The static panel data model includes fixed effects (FE) model and random effects (RE) model, which can mitigate endogeneity caused by omitted variable. And the dynamic panel model includes generalized method of moments (GMM) model, which can mitigate endogeneity caused by omitted variable, errors-in-variables and simultaneous causality.

3.2.1.3.1.1. Static Panel Data Model

The static panel data model does not allow to use lagged values of the dependent variable. And the two static panel data models, i.e. the static panel data and the dynamic panel data model, are discussed in details below.

3.2.1.3.1.1.1.Random Effects Model

To study the time-invariant effects, a model includes time-invariant regressors should be used. And this is called random effects (RE) model. The model of RE is showed as follow:

$$Y_{it} = \alpha + \beta x_{it} + \gamma z_i + e_{it}$$

Where the disturbance term $e_{it} = \boldsymbol{\alpha_i} + u_{it}$, and α_i is an unobservable and time-invariant variable that is correlated with the dependent variable Y_{it}. x_{it} and z_i are independent variables (i.e., regressors), where x_{it} is an observable variable that varies over the time and z_i is an observable time-invariant variable. The key assumption of RE is that the omitted variable α_i is uncorrelated with the included independent variables x_{it} and z_i. This method can mitigate the endogeneity caused by omitted variables.

3.2.1.3.1.1.1.1. When to Use Random Effects Model

If the individual effects are strictly uncorrelated with the independent variables, it may be suitable to use the random distribution across cross-sectional units to model the individual specific constant terms, i.e., α_i in the regression model. This is appropriate if the that sampled cross-sectional units are drawn from a large population.

If the differences across subjects have influence on the dependent variable of theoretical interest, the RE should be included into the model. In a RE model, researchers should clarify whether those individual characteristics, such as α_i in the regression model, may or may not influence the independent variables.

3.2.1.3.1.1.1.2. Advantages and Disadvantages

And advantage of using RE method is that time invariant variables (e.g., geographical contiguity, distance between states) can be included in the regression model. In the fixed effects model, these variables are absorbed by the intercept.

The challenge with the application of RE is that RE must be assumed to be exogenous to all of the predictor variables, which may not be possible in research due to realistic restricions, such as data is not available. And another concern of RE method is that data on some variables (i.e., individual characteristics such as innate ability) may not be available, hence leading to omitted variable bias in the model.

3.2.1.3.1.1.Fixed Effects Model

Fixed effects method employs panel data to control for variables that differ across subjects, but are constant over time. Because adding the FE can alleviate unobserved heterogeneity due to omitted variables, this method can mitigate endogeneity caused by omission of variable. There are two fundamental assumptions of FE. The first assumption is that the omitted variable α_i is correlated with the independent variables. The second assumption is that the unobserved variable α_i is correlated with independent variables but does not differ over time.

The model of FE is showed as follow:

$$Y_{it} = \alpha + \beta x_{it} + \gamma z_i + \boldsymbol{\alpha_i} + u_{it}$$

Where u_{it} is the disturbance term. In the FE model, the unobservable variables are correlated with the independent variables, therefore, the individual effect α_i should not be included in the disturbance term, instead, it should be treat as an explanatory variable. In this way, the FE model control for unmodeled differences that are both stable over time and that may correlate with the predictors of interest. In other words, this model remove all the unmodeled time-invariant effects from the dependent variable, therefore, researchers can assess the net effect of independent variables on the dependent variable. In fixed effects models, the slope coefficient of the population regression line is the same for all subjects, but the intercept of the population regression line varies across subjects (Hidalgo-Mazzei et al., 2019).

3.2.1.3.1.2.1. Estimation

The report discusses two different methods to estimate fixed effects models: (i) within estimator (ii) dummy variable estimator.

3.2.1.3.1.1.1.1.With Estimator

This is the more commonly used estimator for FE models. This estimator is called the "within estimator", as it uses time variation within each cross-section.

3.2.1.3.1.1.1.2. Dummy Variable Regression

When there are a small number of fixed effects to be estimated, it is convenient to just run dummy variable regression for a FE model.

3.2.1.3.1.1.1.3.Two-Way Fixed Effects Model

Two-way fixed effects model is a linear regression with unit and time fixed effects (Imai & Kim, 2021). Many OM empirical research control time fixed effects and individual fixed effects (Jiang et al., 2023)

3.2.1.3.1.2.2. Advantage and Disadvantage
As for the advantage, if researchers only aim to control for the effect of a time-invariant variable, such as industry or country of incorporation, FE will be an effective and appropriate model.

However, there are some disadvantages of FE method. Frist, FE do not work when lagged outcomes are included in the regression. Therefore, the lagged dependent variable should not be used as a regressor in FE model. Second, the FE model therefore cannot include any predictors that are stable over time (Ketokivi & McIntosh, 2017). Because any independent variable of theoretical interest that is stable over time will be highly collinear with the FE, which results in estimation problems.

3.2.1.3.1.2. Fixed Effects or Random Effects?
The **Hausman test** to decide whether to use a fixed effects or random effects model. Procedures are: (1) Run a fixed effects model and save the estimates (2) Run a random effects model and save the estimates (3) Perform the Hausman test.

Both FE and RE models both assume all regressors-including the unobserved effect-to be uncorrelated with the disturbances of the dependent variable. The choice of RE versus FE thus constitutes a tradeoff: FE relaxes the assumption of no correlation between the unobserved effect and the predictor variables, but simultaneously, preempts the modeling of time-invariant predictors (Ketokivi & McIntosh, 2017).

3.2.1.3.1.3. System Model-Generalized Method of Moments (GMM) model
Arellano and Bond (1991) proposed a GMM estimator that uses the lagged values of the endogenous regressors as instruments for the variables in the difference equations, which is commonly known as the Difference GMM estimator (Bapna et al.,2013). Valid instruments, by definition, should be highly correlated with the variables to be instrumented but orthogonal to the error term in order to address the endogeneity concern (Chizema et al., 2015). These requirements result in a set of "moment conditions" that enables the Difference GMM estimator to select the suitable lagged values as valid instruments for the difference equations. Blundell and Bond (1998) showed that the instruments used in the Difference GMM estimator could be weak if the autoregressive process becomes too persistent over time.

Blundell and Bond (1998) developed a new GMM estimator with additional moment conditions in which the lagged differences of the endogenous regressors are used as instruments for the original-level equations, which is known as the System GMM estimator (Bapna et al., 2013). The System Generalized Method of Moments (System GMM) estimation technique that relies on transformations of existing variables rather than the use of external variables to address the endogeneity.

The System GMM estimates a system of two equations simultaneously, i.e., the original-level equation and the transformed difference equation. As the fixed effects are still present in the error term of the original-level equations, an additional assumption is required in order to implement the System GMM estimator. Although the endogenous variables are correlated with the fixed effects in the error term by construction, it is assumed that the correlation is constant over time (i.e., time-invariant) (Arellano & Bover, 1995; Blundell & Bond, 1998). This assumption allows the System GMM estimator to develop differences as instruments to make them exogenous to the fixed effects in the error term and address the endogeneity concern (Roodman, 2009).

The System GMM estimator uses lagged differences as instruments for the original-level equations, in addition to the use of lagged levels as instruments for the transformed difference equations. The introduction of more instruments enables the System GMM estimator to address the concern of weak instruments in the Difference GMM estimator and improve the estimation efficiency dramatically (Roodman, 2009).

The lagged dependent variables are used as instruments to mitigate the dynamic endogeneity for the panel data and these instruments are defined as **internal instrument** (Lam et al., 2016). The GMM model can mitigate endogeneity caused by different source, namely "unobserved heterogeneity, simultaneity and dynamic endogeneity" (Wintoki & Netter, 2012). In other words, GMM model can mitigate endogeneity caused by omitted variables, errors-in-variables and simultaneity.

And system GMM estimator to have much better performance under finite sample properties in terms of bias and root mean squared error (rmse) than that of the difference GMM estimator (Bun & Windmeijer, 2010).

3.2.1.3.1.3.1. The Method of System GMM Model

The System GMM model mitigate endogeneity by "internally transforming the data"– transformation refers to a statistical process where a variable's past value is subtracted from its present value (Roodman, 2009). The internal transformation process reduces the number of observations and thus improves the efficiency of the GMM model (Ullah et al., 2018). There are two kinds of internal transformation methods, which are first-difference transformation (one-step GMM) and second-order transformation (two-step GMM). And these two method can also be utilized as GMM estimators. There are limitations of the first-difference transformation (one-step GMM), such as the loss of too many observations if a variable's recent value is missing in this model. Therefore, to avoid potential data loss owing to the internal transformation problem with the first-step GMM, the use of a second order transformation (two-step GMM) is recommended (Yiu et al., 2020).

For the second-order transformation (two-step GMM), the 'forward orthogonal deviations' is utilized. It means that the two-step GMM model subtracts the average of all future available observations of a particular variable (Roodman, 2009). Because the two-step GMM model can avoid the data loss mentioned above, the two-step GMM model can provide more efficient and consistent estimates for the involved coefficients for a balanced panel dataset (Yiu et al., 2020). And the GMM model relies on internal instruments (lagged values, internal transformation) to address the different sources endogeneity, which is the main difference compared with the instrumental variable techniques that use exogeneous instruments.

Forth, the System GMM estimator addresses the dynamic panel bias directly by instrumenting the lagged dependent variables with variables uncorrelated with the fixed effects in the error term (Roodman, 2009). Fifth, the System GMM estimator is suitable for our data with unbalanced panels. the System GMM estimator appears to be one of "the most robust methodologies for unbalanced panels with endogenous variables". Finally, although the System GMM estimator also employs the IV techniques to deal with the endogeneity issue, it does not rely on external variables that are outside the immediate dataset to construct instruments. Instead, the System GMM estimator constructs instruments internally with the transformation of existing variables, overcoming the difficulty in obtaining exogenous instruments externally (Roodman, 2009; Wintoki et al., 2012).

3.2.1.3.1.3.2. Post-Estimation Test of System GMM

When applying the System GMM model in research, two post-estimation tests need to be performed to determine that an appropriate econometric model is applied. The two tests are: (1) the Sargan test; and (2) the Arellano-Bond test for first order and second-order correlation.

Sargan test determines whether the econometric model is valid or not, and whether the instruments are correctly specified or not. In other words, if the null hypothesis is rejected, the researcher needs to reconsider the model or the instruments used in the estimation process.

To examine the validity of a strong exogeneity assumption, the Arellano-Bond test for no autocorrelation (or no serial correlation) is used under the null hypothesis that the error terms of two different time periods are uncorrelated.

3.2.1.3.1.3.3. Advantage of System GMM

The System GMM mode has many advantages. It offers more efficient and consistent estimates for the coefficients as compared with other estimation techniques. It provides a consistent, asymptotically normal, and efficient estimator among all the estimators that only use the information provided by the moment conditions.

First, GMM is convenient for estimating interesting extensions of the basic unobserved effects model, for example, models where unobserved heterogeneity interacts with observed covariates (Wooldridge, 2001). Second, GMM have better performance that other methods, when it is applied to unobserved effects models when the explanatory variables are not strictly exogenous even after controlling for an unobserved effect. Thirdly, GMM is well suited for obtaining efficient estimators that account for the serial correlation existing in the time-series data of the dependent variable. It can be applied to a model contains a lagged dependent variable along with an unobserved effect.

Comparison with Instrumental Variable Techniques: For instrumental variable techniques, it is difficult to obtain such strictly exogenous instruments externally as pointed out by prior studies (Bardhan et al., 2013; Wintoki et al., 2012). In contrast, System GMM use internal instruments. Therefore, System GMM has several advantages. First, the System GMM controlled for different kinds of endogeneity by including lagged values of the dependent variable (previous value of the dependent variable) as an explanatory variable in the model. The System GMM is well suited for obtaining efficient estimators that account for the serial correlation. In contrast to 2SLS and 3SLS, it does not rely on external exogenous instruments, which in practice may be difficult to identify (Wintoki et al., 2012), but consists of a System of two sets of equations, each with its own internal instruments (Abdallah et al., 2015). Thus, System GMM model is more useful when exogeneous instruments of good quality (i.e., valid and strong) are not available. Second, the System GMM estimator also can be used to generate standard errors that are robust to autocorrelation (in time series or long panel data) in the data (Baum et al., 2007). This cannot be achieved by instrumental variable techniques, because the instrumental variable techniques do not allow the lagged value. Third, the System GMM model can be more efficient (i.e., smaller standard errors) if errors are considered to have arbitrary heteroscedasticity or intra-cluster correlation for over-identified equations. The optimal GMM estimator is asymptotically no less efficient than two-stage least squares under homoskedasticity, and GMM is generally better under heteroskedasticity. Meanwhile, the GMM estimator has similar performance with 2SLS for just-identified models, but better performance than 2SLS for over-identified models. Forth, the System GMM can be applied more often to unobserved effects models when the explanatory variables are not strictly exogenous even after controlling for an unobserved effect. For example, when the fixed effects are eliminate by differencing the observations of adjacent years, if the errors in the first-differenced equation are homoscedastic and serially uncorrelated, the 2SLS estimator is efficient; if not, a GMM estimator can improve upon two-stage least squares.

Comparison with Instrumental Variable Techniques: First, the GMM controlled for different kinds of endogeneity by including lagged values of the dependent variable (previous value of the dependent variable) as an explanatory variable in the model. In contrast to 2SLS and 3SLS, it does not rely on external exogenous instruments, which in practice may be difficult to identify (Wintoki et al., 2012), but consists of a system of two sets of equations, each with its own internal instruments (Abdallah et al., 2015). Thus, GMM model is more useful when exogeneous instruments are not available. Second, the GMM estimator also can be used to generate standard errors that are robust to autocorrelation (in time series or long panel data) in the data (Baum et al., 2007). This cannot be achieved by instrumental variable techniques, because the instrumental variable techniques do not allow the lagged value. Third, GMM model can be more efficient (i.e., smaller standard errors) if errors are considered to have arbitrary heteroscedasticity or intra-cluster correlation for over-identified equations. 2SLS estimator (IV) is a special case of the more flexible Generalized Method of Moments (GMM) estimation that subsumes the 2SLS. In general, the two estimators are likely to generate similar estimates.

Comparison with AR and FE model: The AR and FE models are static panel data model. While, the System GMM is a dynamic panel data model. In other words, an important difference between FE model and GMM is that FE estimation uses 'strict exogeneity' assumptions resulting into a static FE model. For panel data, the past realization of dependent variables may also affect realization of dependent variables of the current year and the simultaneity could potentially violate the strict exogeneity assumptions. Under such situations, traditional FE and AR panel data static models will provide inconsistent and biased results (Wooldridge, 2012). Conversely, the System GMM estimation is more robust in dealing with these sources of endogeneity.

Comparison with DID: When the lagged value of dependent variable are correlated with the fixed effects in the error term or the dependent variable's realization of current value is influenced by the past value, the ordinary least squares (OLS) estimation will obtain inconsistent estimated result (Roodman 2009). In other words, under such situation, the endogeneity issue will be caused due to omitted variables. Although a more advanced least squares dummy variables (LSDV) estimator can address this bias (Kiviet 1995), it does not deal with the other endogeneity concerns such as unobservable heterogeneity and simultaneity arising from the other regressors included in our research (Roodman 2009). Therefore, System GMM should be used.

3.2.1.3.1.3.Advantage and Disadvantage of Panel Data

As for the advantage, the longitudinal data, especially panel data, has some advantages in addressing endogeneity over cross-sectional data.

For the disadvantage, the panel data may not be available in research. The panel datasets combine the characteristics of cross-sectional and time series data. Although, the emergence of business/financial databases (Bloomberg, Datastream, Thomson One, Compustat) provides more flexibility for researchers to gather the panel data, however, these data sets are not primarily designed for OM empirical research. The data provided by them may not answering the research question in OM research. So, the panel data may not be feasible in OM research.

3.2.1.3.1.4.Lagging independent variable

The lagging independent variable method can mitigate the endogeneity caused by simultaneity. It is a lagged endogenous regressor technique, which is very similar to the GMM model. A temporal separation through introducing a time lag between the independent and dependent variables can reduce this bias (Zaefarian et al., 2017). In this method, the dependent variable is measured in a time-lagged fashion and the past value of dependent variable is also included as the regressors in the regression model. In this way, the endogeneity concern stemming from simultaneity effects can be significantly alleviated.

However, this approach also has limitation. For example, it is possible that the value of dependent variable at year t-2 can influence the value of the dependent variable in the year t-1, and these two value are still correlated with the value of the dependent variable in the year t. Under this situation, the lagged collaborative regressors are still endogenous.

3.2.1.3.2. Longitudinal, quasi-experimental data

A quasi-experiment aims to evaluate interventions but that do not use randomization. Similar to randomized trials, quasi-experiments aim to demonstrate causality between an intervention and an outcome. The quasi-experiments allow a a before-and-after type analysis, which enables consistent estimation of the effect of the intervention, given that subjects essentially act as their own controls in terms of stable observed and unobserved factors (Ketokivi & McIntosh, 2017). And the outcome is measured at multiple points before and after the intervention, and the objective to determine whether there is a discernible discontinuity in the time series at the intervention point (Ketokivi & McIntosh, 2017). And the difference-in-difference can be used in the quasi-experiment design to make casual inference.

3.2.1.3.2.1. Difference-in-Difference Method

Difference-in-difference is a quasi-experimental design that using longitudinal data from treatment and control groups to obtain an appropriate counterfactual to estimate a causal effect. It is a method to estimate the effects of new policies. It consists of identifying a specific intervention or treatment. Then, the difference in outcomes after and before the intervention or treatment for groups affected by the intervention to the same difference for unaffected groups should be compared (Bertrand et al., 2004).

In DID, the unobserved differences between treatment and control groups are the same overtime without the treatment. Thus, it is an effective tool to apply when randomization on the individual level is not possible. This approach eliminates biases in post-intervention period comparisons between the treatment and control group that could be the result from permanent differences between those groups, as well as biases from comparisons over time in the treatment group that could be the result of trends due to other causes of the outcome. Therefore, it controls for the unobserved variables that bias estimates of causal effects, aided by longitudinal data, DID can mitigate endogeneity caused by omitted variables, errors-in-variables and simultaneity.

3.2.1.3.2.1.1. Assumptions of Difference-In-Difference

To effectively apply DID in research, four assumptions should be fulfilled. The assumptions are:
(1) Exchangeability- Intervention unrelated to outcome at baseline (allocation of intervention between treatment group and control group was not determined by outcome)
(2) Positivity-Treatment/intervention and control groups have Parallel Trends in outcome. And this assumption is also known as the Parallel Trend Assumption.
(3) Stable Unit Treatment Value Assumption (SUTVA)-Composition of treatment and control groups is stable for repeated cross-sectional design.
(4) SUTVA-No spillover effects

To ensure the internal validity of DID models, the parallel trend assumption must be met. An effective method to prove that this assumption is fulfilled is to visually assess whether the pre-treatment trends between the treated and control groups evolve in parallel (Mithas et al., 2022). An effective way to convince readers about this assumption is to visually assess whether the pre-treatment trends between the treated and control groups evolve in parallel. Test can be performed for the parallel in the pre-trends in an event study analysis or relative time model, where the mean difference in the outcomes between the treated and control units is estimated period-by-period, with several periods before and after the treatment (Marcus & Sant'Anna, 2021). It has also been proposed that the smaller the time period tested, the more likely the assumption is to hold. Violation of parallel trend assumption will lead to biased estimation of the causal effect.

3.2.1.3.2.1.2. Advantage and Disadvantage of Difference-In-Difference Method

There are many advantages of DID. First, the result can be intuitive interpreted. The result obtained from DID can be easily interpreted by research based on intuition. Second, a relatively clean casual effect can be obtained using observational data if assumptions are met. Third, DID methods can use individual and group level data, which is very flexible. Forth, the control groups can start at different levels of the outcome, because DID focuses on change rather than absolute levels. Finally, DID accounts for change due to factors other than intervention. In other words, DID can control confounding variables in the research.

This method also has many limitations. First, DID require baseline data and a control group. The data before the intervention and the control group may not be available. Second, if some key assumptions are not fulfilled, this method cannot be applied. If the allocation of the intervention is decided by baseline outcome, DID cannot be used. And if the control groups have different outcome trend with the treatment groups, the method cannot be applied. Meanwhile, if the composition of groups before and after the intervention are not stable, the method cannot be used. Finally, although least squares dummy variables (LSDV) estimator is able to handle the dynamic panel bias, it works only for balanced panels (Roodman, 2009) and is unsuitable for our unbalanced sample with some samples having more observations than the others (Lam et al., 2016).

3.2.1.3.2.2. Matching Method

Matching method is a quasi-experimental method in which the researcher uses statistical techniques to construct an artificial control group by matching each treated unit with a non-treated unit of similar characteristics. It is a method deal with the selection bias caused by the sample self-selection on observables. This method <u>solve the selection bias</u> by matching the observable features between the treatment group and the control group. Therefore, <u>matching method can mitigate endogeneity caused by omitted variables</u>. And it is useful for estimating the impact of a program or event for which it is not ethically or logistically feasible to randomize. There are two kinds of matching methods: (1) Direct Matching Method (2) Propensity Score Method (PSM).

3.2.1.3.2.2.1. Assumption of Matching Method

There are two assumptions for the matching method. The first assumption is the conditional independence, which states that after conditioning on a set of observed covariates, treatment assignment is independent of potential outcomes. The second assumption is the common support condition, which indicates the regions of stratification share enough members of the treatment and control groups. To effectively apply the matching method, the assumptions should be fulfilled.

3.2.1.3.2.2.2. Direct Matching Method

Direct matching method is a matching method that matches each treated unit with a non-treated unit of based on similar observable features, directly. It is an ideal matching method because it ensures that the observable features of control group and treatment group are same. However, when the dimensions of features increase, it will be very difficult or even impossible to perform matching process directly based on all features. Then, the PSM can be used.

3.2.1.3.2.2.3. Propensity Score Method

Propensity Score Matching (PSM) is a matching method that computes that probability that a unit will enroll in the program. This probability is called the propensity score and is used to match units in the treatment group with unenrolled units of similar propensity scores. There are four stages in PSM, that are: (1) Estimation of propensity score (2) Selection of a matching algorithm (3) Checking for balance in the characteristics of the treatment and control groups (4) Estimating the program effect and interpreting the results.

Estimation of propensity score: The propensity score is also the probability of participation in the program for each unit in the sample. And the logit or probit regression method is used to obtain this score.

Select a matching algorithm: After the propensity scores are estimated, units in the treatment group are then matched with non-beneficiaries with similar propensity. The most used matching algorithms in PSM are:

(1) Nearest-neighbor matching: each unit in the treatment group is matched with the n units in the control group with the closet propensity score, where n is the number of units that can be matched with the treated units. Units in control groups for which there are not unit in the treatment groups with a sufficiently similar score are removed from the sample. And

the same is true for units in treatment groups for which there is not similar unit in control groups. A variation of nearest-neighbor matching matches multiple units in control groups to one single unit in treatment groups. If the unit in the control group can be used repeatedly, the bias decreases while the variance increases. And if the number of units in control groups that can be used to match with each treated unit increases, the bias increases and the variance decreases.

(2) Radius matching: (i.e. 'caliper' matching): A maximum propensity score radius – a 'caliper' – is established, and all units in control groups within the given radius of a unit in treatment groups are matched to that treated unit. When the value of radius increases, the bias increases and the variance decreases.

(3) Kernel matching: For each treated subject, a weighted average of the outcome of all untreated unit is derived. The weights are based on the distance of the propensity score of untreated units to that of the treated subject's, with the highest weight given to those with scores closest to the treated unit. When the bandwidth increases, the bias increases and the variance decreases.

The selection of matching algorithm should be based on the trade-off between the bias and the variance.

3.2.1.3.2.2.3.1. PSM in OM Research

In OM empirical, PSM is commonly used to ensure that the treatment and control firms are highly similar (Petryk et al., 2022). Qiu et al. (2022) use PSM approach aims to calculate the probability (i.e., propensity score) of being treated/accept treatment for their treatment and control firms. They use the nearest neighborhood approach to match each treatment firm to a control firm with the closest propensity in the same industry (two-digit SIC code) in the research. By the nearest neighborhood approach, one control firm is matched to multiple treatment firms, which may cause a double-counting issue. To solve this issue, they remove the treatment firms that could not be matched with any control firms. They calculate the propensity score for each treatment and control firm based on the estimation coefficients- Pseudo-R-squared. And the range of variance inflation factor is between 1.00 and 1.28, indicating that multicollinearity is not a serious concern. And they perform paired t-test for samples after matching to ensure no selection bias in the post matching samples. They conduct paired t-tests on all the independent variables (except industry dummies) used in the probit model. The test results indicated no significant difference between treatment and control firms on these variables ($p > .1$) after PSM. Thus, they concluded that the statistics of treatment and control firms are highly similar.

3.2.1.3.2.2.4. Coarsened Exact Matching

Coarsened exact matching (CEM) is another matching method, wherein continuous and ordinal characteristics are categorized, and values of categorical variables are collapsed (i.e., coarsened). CEM is faster, easier to use, requires fewer assumptions, robust to measurement error, and strictly bounds both the degree of model dependence and the average treatment–effect estimation error through ex ante user choice (Qiu et al., 2022). To implement the CEM method, we need to first coarsen our data, determine an exact match on these coarsened data, and then conduct our analysis on the uncoarsened, matched data.

3.2.1.3.2.2.4.1 Steps to Perform Coarsened Exact Matching

Steps to conduct CEM is described as follows (Blackwell et al., 2009):

Step 1: Begin with the covariates samples or observations X and make a copy of them, which we denote X∗.
Step 2: Coarsen X∗ according to user-defined cut-points, or CEM's automatic binning algorithm.
Step 3: Create one stratum per unique observation of X∗ and place each observation in a stratum.
Step 4: Assign these strata to the original data, X and drop any observation whose stratum does not contain at least one treated and one control unit.

3.2.1.3.2.2.4.2. Advantage of Coarsened Exact Matching
PSM matching is based on an overall score between the samples and control, so even the total propensity scores are similar, the IVs between the two may not be the same and sometime significantly different. CEM offered a better way to control the differences between the sample and controls IVs in the model.

3.2.1.3.2.2.4. Limitations of Matching Method
There are two major limitations. First, extensive datasets are required to properly match units and they rely on broad assumptions that are difficult to prove. In research, there may not be enough data to perform the matching method. Meanwhile, the datasets should have detailed information on baseline characteristics which is not always available. Second, the validity of matching methods relies on the assumption that there are no systematic differences in unobserved features between the treatment units and the matched units in control groups. It is difficult to prove this assumption correct, making matching methods a less robust approach than other methods.

3.2.1.3. Problem with Longitudinal Research Designs
The general problems that are common for different research methods used in the longitudinal research design are discussed in this part. First, it is difficult to obtain panel data. OM researchers often have to build their datasets from scratch. The extant commercial databases are mainly designed for strategy or economics research and may not satisfy the demands of OM research. There are no commercial datasets that contain, such as longitudinal factory-level information about technology and productivity (Ketokivi & McIntosh, 2017). Second, since the intervention or exogenous shocks happened in the past, the OM researcher should be lucky enough to find a meaningful exogenous shock that has an impact of theoretical interest (Ketokivi & McIntosh, 2017).

3.1.2.4. Regression Discontinuity Design
Regression discontinuity (RD) is a method to deal with non-random treatment effects. The major idea of this method is to find a factor that can describe how an observation becomes part of the treatment group and seeks to exploit the cut-off point for this identified factor (Reeb et al., 2012). The discontinuity in this method refers to identifying the threshold or the cutoff point that can distinguish the treatment from the control group (Zaefarian et al., 2017). This method allows researchers to compare subjects that are just above the cutoff point against those subjects that are marginally below the cut-off point. The comparisons between firms that are just below or just above the cut-off point is similar to the propensity score model. Thus, RD can be used to mitigate the endogeneity caused by selection bias.

3.1.2.5. Instrument Free Approaches: Latent Instrumental Variables

The challenges related to identifying valid instruments have resulted in the emergence of instrument-free techniques. The Latent Instrumental Variables (LIV) method is one of the instrument free approaches. In LIV method, the variations in the endogenous independent variable are separated into exogenous (approximated by a latent discrete variable) and endogenous parts (Ebbes et al., 2005).The exogenous part is uncorrelated with the disturbance term, while the endogenous part is probably correlated with the disturbance term. And the exogenous part will be included in the regression model to mitigate the endogeneity caused by omitted variables, errors-in-variables and simultaneity.

There are many advantages of this method. First, this method employs a latent variable model to account for regressor-error dependencies, and addresses the issues of instrument availability, weakness, and validity, because LIV estimator belongs to the family of thrifty instruments estimators that do not require observed instruments (Ebbes et al., 2005; Ebbes et al., 2009). Second, LIV estimator, as a maximum likelihood method, can achieve unbiased estimation. And In the LIV solution, a latent variable model is used to separate the endogenous covariate into systematic parts, whereby one part is uncorrelated with the error term and the other part is possibly correlated with the error term. This allows to achieve an unbiased estimate of the effect of an endogenous covariate on the dependent variable.

3.1.2.6. Instrument Free Approaches: Identification through Heteroscedasticity

The Identification through Heteroscedasticity (IH) estimator is also an instrument free approach. In this method, instruments are obtained in a similar manner to Higher Moments (HM) approach, which means that instruments are built based on available data in general regressor-error dependencies models. But the information of heteroscedasticity is required by IH method (Hogan & Rigobon, 2003). This IH method can be used to mitigate the endogeneity caused by omitted variables, errors-in-variables and simultaneity.

Since the IH approach is a method-of-moments estimators, one limitation of this approach is that it often obtain biased estimation from the model. Because the method of moments is fairly simple and yields consistent estimators under very weak assumptions, though these estimators are often biased.

3.2. Solutions at Statistical Analysis Stage

The statistical analysis stage also called as post-design stage. There are several methods that can be used to mitigate the endogeneity at this stage.

3.2.1.Instrument Free Approaches: Higher Moments

The Higher Moments (HM) approach is also one of the instrument free approaches. In this approach, the instruments are built based on available data in general regressor-error dependencies models, and can be used together with or in the absence of traditional instruments (Erickson & Whited, 2002; Lewbel, 1997). The HM approach can mitigate the endogeneity caused by omitted variables, errors-in-variables and simultaneity.

Since the HM approach is a method-of-moments estimators, one limitation of this approach is that it often obtain biased estimation from the model. Because the method of moments is fairly simple and yields consistent estimators under very weak assumptions, though these estimators are often biased.

3.2.2. Instrument Free Approaches: Copulas

The joint estimation with copulas is another instrument free method to deal with endogeneity issues. A copula is a function that ‘couples’ multivariate distributions to their one-dimensional marginal distribution function and captures the correlation between the independent variable and the disturbance term (Ketokivi & McIntosh, 2017). If the correlation is properly handled, this method can mitigate the endogeneity caused by omitted variables, errors-in-variables and simultaneity.

The likelihood of the structural equation of interest can be maximized by using a density estimation method in the model of the joint distribution of the endogenous variable and the disturbance term. This is achieved by copulas, where the functions that “couple” multivariate distributions to their one-dimensional marginal distribution functions and capture the correlation between the variable and the error (Ketokivi & McIntosh, 2017).

3.2.2. Heckman Two-Step Procedure

Heckman (1979) two-step procedure solves endogeneity caused by selection bias. The selection bias will occur when samples to be non-randomly selected by researchers. The non-random sample is subject to the selection bias that will cause the endogeneity issues.

There are two steps in this method. The first step is to run a probit regression (selection equation) to model the conditional distribution of the treatment with a set of covariates that captures the relevant features (Ketokivi & McIntosh, 2017). And the relevance of the selected set of covariates needs to be theoretically justified. Then, the propensity scores are estimated, which needs to be repeated for each independent variable of the model. The second step is to correct the sample self-selection by integrating the Inverse Mills Ratio (IMR), defined as the ratio of the standard normal density divided by the standard normal cumulative distribution function, into the final regression models before testing hypotheses (Ketokivi & McIntosh, 2017).

This method also has limitations. First, at least one variable with a non-zero coefficient in the selection equation in the first step should not be included in the final equation in the second step (Ketokivi & McIntosh, 2017). This variable is used as an instruments, which may be difficult to find. Second, a dummy vairable only with two possible value of 0 and 1 is used to represent the probability of selection. In the first step of the model, a probit regression model is used to predict the probability of selection, so the predictor need to be a dummy variable that can take two discrete value of 0 and 1. To solve address this problem, one solution is to recode the predictor variable into a dummy one (Ketokivi & McIntosh, 2017). Another possible solution is to use the two-step approach for selectivity-bias correction with a continuous choice variable (Garen, 1984). In method, first, regressing the independent variable against a set of factors that influence the independent variable. The output of this regression model is the choice variable, i.e. the probability of selection. In this procedure, the researchers generally include all control variables, independent constructs, and moderators except for the main dependent variable(s) into the correction regression model to predict the choice variable (Ketokivi & McIntosh, 2017). In the second step, the predicted errors from this correction regression equation should be added in to the second-stage performance equation (Ketokivi & McIntosh, 2017).

Second, not all factors that contribute to the occurrence of the selection bias can be measured in the selection equation. That is the omitted variables can still exist, causing the endogeneity issues. Therefore, other remedies may need to be used complementary to this method, such as PSM (Qiu et al., 2022).

4. Procedure to Solve Endogeneity

There are possible theoretical procedures for researcher to follow to handle the endogeneity. The detailed steps are showed as follows:

1. Where in the model the problem is-Identification of the endogeneity.

First, the endogeneity should be identified from the theory and extant literature, because endogeneity is based not on statistical theory but on substantive, theoretical considerations, and that is why evaluation of endogeneity is ultimately a judgment call (Lu et al., 2018). Researchers should list all the potential endogeneity concerns implied by the model, and then try to determine how serious a threat each of these pose to empirical analysis, from the theoretical aspect (Ketokivi & McIntosh, 2017). For example, endogeneity caused by simultaneity can be identified based on theoretical arguments or extant literature.

Second, endogeneity tests, i.e., statistical tests (e.g., Durbin–Wu–Hausman test), can be performed to identify the endogeneity. If the literature and/or theory suggest that endogeneity may be a concern, endogeneity tests should be performed to examine the extent of endogeneity.

2. What could be causing it-Identification the reason for the endogeneity

The sources of the identified should also be identified from theoretical aspect, first.

3. What can be done about it.

The first step in tackling endogeneity is to specify a theoretically sound model (Ketokivi & McIntosh, 2017; Lu et al., 2018). Researchers should build a theoretically grounded model based on the literature when possible and consult experts with extensive knowledge of the research context to gauge the extent of endogeneity. The endogeneity caused by omitted variables can be controlled by including control variables that are both theoretically and empiricall relevant to the theretical model in research. Meanwhile, endogeneity caused by simultaneity can be identified based on theoretical arguments or extant literature. And the endogeneity caused by errors-in-variables can be controlled by using appropriate and accurate measures according to the extant research.

After this, researchers can rely on empirical techniques to address endogeneity.

References:

Abdallah, W., Goergen, M., & O'Sullivan, N. (2015). Endogeneity: How Failure to Correct for it can Cause Wrong Inferences and Some Remedies: Endogeneity. *British Journal of Management*, *26*(4), 791–804. https://doi.org/10.1111/1467-8551.12113

Bardhan, I., Krishnan, V., & Lin, S. (2013). Business Value of Information Technology: Testing the Interaction Effect of IT and R&D on Tobin's Q. *Information Systems Research*, *24*(4), 1147–1161. https://doi.org/10.1287/isre.2013.0481

Bartels, L. M. (1991). Instrumental and "Quasi-Instrumental" Variables. *American Journal of Political Science*, *35*(3), 777. https://doi.org/10.2307/2111566

Baum, C. F., Schaffer, M. E., & Stillman, S. (n.d.). *Enhanced Routines for Instrumental Variables/Generalized Method of Moments Estimation and Testing*.

Bertrand, M., Duflo, E., & Mullainathan, S. (n.d.). HOW MUCH SHOULD WE TRUST DIFFERENCES-IN-DIFFERENCES ESTIMATES? *QUARTERLY JOURNAL OF ECONOMICS*.

Blackwell, M., Iacus, S., King, G., & Porro, G. (2009). Cem: Coarsened Exact Matching in Stata. *The Stata Journal: Promoting Communications on Statistics and Stata*, *9*(4), 524–546. https://doi.org/10.1177/1536867X0900900402

Bun, M. J. G., & Windmeijer, F. (2010). *The Weak Instrument Problem of the System GMM Estimator in Dynamic Panel Data Models*.

Ebbes, P., Wedel, M., & Böckenholt, U. (2009). Frugal IV alternatives to identify the parameter for an endogenous regressor. *Journal of Applied Econometrics*, *24*(3), 446–468. https://doi.org/10.1002/jae.1058

Ebbes, P., Wedel, M., Böckenholt, U., & Steerneman, T. (2005). Solving and Testing for Regressor-Error (in)Dependence When no Instrumental Variables are Available: With New Evidence for the Effect of Education on Income. *Quantitative Marketing and Economics*, *3*(4), 365–392. https://doi.org/10.1007/s11129-005-1177-6

Erickson, T., & Whited, T. M. (2002). TWO-STEP GMM ESTIMATION OF THE ERRORS-IN-VARIABLES MODEL USING HIGH-ORDER MOMENTS. *Econometric Theory*, *18*(3), 776–799. https://doi.org/10.1017/S0266466602183101

Fan, D., Zhou, Y., Yeung, A. C. L., Lo, C. K. Y., & Tang, C. (2022). Impact of the U.S.–China trade war on the operating performance of U.S. firms: The role of outsourcing and supply base complexity. *Journal of Operations Management*, *68*(8), 928–962. https://doi.org/10.1002/joom.1225

Garen, J. (1984). The Returns to Schooling: A Selectivity Bias Approach with a Continuous Choice Variable. *Econometrica*, *52*(5), 1199. https://doi.org/10.2307/1910996

Heckman, J. J. (1979). Sample Selection Bias as a Specification Error. *Econometrica*, *47*(1), 153. https://doi.org/10.2307/1912352

Hidalgo-Mazzei, D., Berk, M., Cipriani, A., Cleare, A. J., Florio, A. D., Dietch, D., Geddes, J. R., Goodwin, G. M., Grunze, H., Hayes, J. F., Jones, I., Kasper, S., Macritchie, K., McAllister-Williams, R. H., Morriss, R., Nayrouz, S., Pappa, S., Soares, J. C., Smith, D. J., … Stokes, P. R. A. (2019). Treatment-resistant and multi-therapy-resistant criteria for bipolar depression: Consensus definition. *British Journal of Psychiatry*, *214*(1), 27–35. https://doi.org/10.1192/bjp.2018.257

Imai, K., & Kim, I. S. (2021). On the Use of Two-Way Fixed Effects Regression Models for Causal Inference with Panel Data. *Political Analysis*, *29*(3), 405–415. https://doi.org/10.1017/pan.2020.33

Jiang, S., Yeung, A. C. L., Han, Z., & Huo, B. (2023). The effect of customer and supplier concentrations on firm resilience during the COVID -19 pandemic: Resource dependence and power balancing. *Journal of Operations Management*, *69*(3), 497–518. https://doi.org/10.1002/joom.1236

Ketokivi, M., & McIntosh, C. N. (2017). Addressing the endogeneity dilemma in operations management research: Theoretical, empirical, and pragmatic considerations. *Journal of Operations Management*, *52*(1), 1–14. https://doi.org/10.1016/j.jom.2017.05.001

Lam, H. K. S., Yeung, A. C. L., & Cheng, T. C. E. (2016). The impact of firms' social media initiatives on operational efficiency and innovativeness. *Journal of Operations Management*, *47–48*(1), 28–43. https://doi.org/10.1016/j.jom.2016.06.001

Lu, G., Ding, X. (David), Peng, D. X., & Hao-Chun Chuang, H. (2018). Addressing endogeneity in operations management research: Recent developments, common problems, and directions for future research. *Journal of Operations Management*, *64*(1), 53–64. https://doi.org/10.1016/j.jom.2018.10.001

Mithas, S., Chen, Y., Lin, Y., & De Oliveira Silveira, A. (2022). On the causality and plausibility of treatment effects in operations management research. *Production and Operations Management*, *31*(12), 4558–4571. https://doi.org/10.1111/poms.13863

Murray, M. P. (2006). Avoiding Invalid Instruments and Coping with Weak Instruments. *Journal of Economic Perspectives*, *20*(4), 111–132. https://doi.org/10.1257/jep.20.4.111

Petryk, M., Rivera, M., Bhattacharya, S., Qiu, L., & Kumar, S. (2022). How Network Embeddedness Affects Real-Time Performance Feedback: An Empirical Investigation. *Information Systems Research*, *33*(4), 1467–1489. https://doi.org/10.1287/isre.2022.1110

Qiu, L., Liu, R., Jin, Y., Ding, C., Fan, Y., & Yeung, A. C. L. (2022). Impact of credit default swaps on firms' operational efficiency. *Production and Operations Management*, *31*(9), 3611–3631. https://doi.org/10.1111/poms.13788

Reeb, D., Sakakibara, M., & Mahmood, I. P. (2012). From the Editors: Endogeneity in international business research. *Journal of International Business Studies*, *43*(3), 211–218. https://doi.org/10.1057/jibs.2011.60

Rossi, P. E. (2014). Even the Rich Can Make Themselves Poor: A Critical Examination of IV Methods in Marketing Applications. *Marketing Science*, *33*(5), 655–672. https://doi.org/10.1287/mksc.2014.0860

Wintoki, M. B., Linck, J. S., & Netter, J. M. (2012). Endogeneity and the dynamics of internal corporate governance. *Journal of Financial Economics*, *105*(3), 581–606. https://doi.org/10.1016/j.jfineco.2012.03.005

Yeung, A. C. L., Lo, C. K. Y., & Cheng, T. C. E. (2011). Behind the Iron Cage: An Institutional Perspective on ISO 9000 Adoption and CEO Compensation. *Organization Science*, *22*(6), 1600–1612. https://doi.org/10.1287/orsc.1110.0652

Yiu, L. M. D., Lam, H. K. S., Yeung, A. C. L., & Cheng, T. C. E. (2020). Enhancing the Financial Returns of R&D Investments through Operations Management. *Production and Operations Management*, *29*(7), 1658–1678. https://doi.org/10.1111/poms.13186

Zaefarian, G., Kadile, V., Henneberg, S. C., & Leischnig, A. (2017). Endogeneity bias in marketing research: Problem, causes and remedies. *Industrial Marketing Management*, *65*, 39–46. https://doi.org/10.1016/j.indmarman.2017.05.006

Wooldridge, J. M. (2002). *Econometric analysis of cross section and panel data*. Cambridge, MA: MIT Press.

Instrumental Variable

1. Definition

If there are strong theory- or evidence-based arguments that the endogeneity problems exist in the regression model, then most common method to estimate the parameters of interest is through instrumental variable (IV)-based estimation.

The instrumental variables (IVs) are variables that are uncorrelated with the disturbance term but are correlated with the endogenous independent variable in the original regression model (Murray, 2006). The fundamental idea of IVs is to decompose the variations in the endogenous independent variable with the help of IVs by concentrating on the variations in the endogenous independent variable that are uncorrelated with the disturbance term in the model and disregarding the variations that bias the estimation (Zaefarian et al., 2017). In short, IV-based estimation decomposes the observed variations in independent variables into two parts, an exogenous part and an endogenous part, to mitigate endogeneity issues.

The instrumental variables techniques can mitigate three types of endogeneity, including omitted-variable endogeneity, endogeneity caused by errors-in-variables and endogeneity caused by simultaneous causality. Since selection bias can be deemed as a sub-form of the omitted variable bias issue (Antonakis et al., 2010), the instrumental variables techniques can also mitigate the endogeneity caused by selection bias.

2. Conditions For Valid Instruments

Two types of instruments are external instruments and internal instruments. Internal instruments are instrumental variables that are used from the existing econometric model (Roodman, 2009.). For example, the lagged values of the dependent variables are used in dynamic panel data estimation techniques as the instruments to control endogeneity issues in the regression model. Internal instruments are often used in GMM method, so they will be discussed in the chapter discussing GMM method. This chapter mainly discusses external instruments.

For the external instruments, instruments are not allowed to be independent variables in the original regression model. Valid IVs should be identified based on two conditions that are relevance and exogeneity (Bartels, 1991; Fullerton et al., 2014; Murray, 2006). Relevance refers to the degree to which an instrumental variable is sufficiently correlated with the endogenous independent variable. And an instrumental variable that has a high correlation with the endogenous independent variable is defined as the strong instrumental variable. In contrast, an instrumental variable that has a low correlation with the endogenous independent variable is defined as the weak instrumental variable. Exogeneity is defined as the degree to which an instrumental variable and the disturbance term in the model are uncorrelated (Murray, 2006). For valid IVs, they do not influence the dependent variable directly, instead they influence it indirectly through the endogenous independent variables (Rossi, 2014).

2.1.How to Find Good IVs

To ensure the good performance of the IVs techniques, both the choice and strength of a valid instrumental variable (IV) are very important. Because the used of weak for invalid instruments can obtain results from the IV-based estimation model that are no better or even worse than that obtained from models do not handle endogeneity issues, such as OLS.

To find a good IV, researchers should depend on theory (Ullah et al., 2021). IVs can be found either outside the unite of analysis (e.g. within physical or institutional environment), outside the unit of analysis yet being affected by the unit of analysis (e.g. within organizational environment), or within unit of analysis (such as lagged variables, i.e., internal instruments). For IVs find outside the unit, it is easier for them to satisfy the exogeneous condition, but difficult to meet the relevance condition. In contrast, IVs find within the unit (i.e., internal instruments) can easily meet the relevance condition while unlikely to satisfy the exogeneity condition.

Statistical tests can test the exogeneity and relevance conditions of IVs. For the exogeneity condition, the test of over-identifying restrictions can be used to determine the exogeneity of the instrument. For the relevance condition, the first-stage F-statistics can be used to ascertain the relevance of an instrument. Higher F-statistics indicate that the chosen instruments can be considered in the 2SLS. While lower F-statistics indicate a weaker instrument (Semadeni et al., 2014).

3. Application of IVs Techniques

IV-based estimation methods are use when endogeneity problems are asserted to exist in the regression model. Otherwise, OLS can give consistent estimation in a more efficient way, i.e., with lower variance.

For under-identification, where the number of IV is less than the number of endogenous variables, the coefficients of endogenous variables cannot be identified.

IV-based estimation method can be applied only when valid and strong instruments can be found to solve the endogeneity issues identified in the original regression model. If the invalid or weak instruments are used, the estimations obtained will probably not be better or even worse than that obtained by OLS not dealing with the endogeneity. The use of invalid and weak instruments can introduce more harm than benefits. Besides, the IV-based estimation should not be used if the only instruments available are lagged variables because no economic arguments can justify this practice (Rossi, 2014)

4. Implementation of IVs Techniques

4.1. Number of IVs

(1) When the number of instruments equals the number of suspected endogenous variables, the model is just identified.
(2) When the number of instruments is greater than the number of suspected endogenous variables, the model is over-identified.
(3) When the number of instruments is less than the number of suspected endogenous variables, the model is under-identified.

4.2. Different Estimation Method

4.2.1. Indirect Least Squares (ILS)

This method is also called as instrumental estimator method or the ratio of coefficients method. It is a method used to estimate simultaneous equation models that are just-identified. It is a single equation method because it is applied to each equation of the system one by one, which means all the equations are not estimated simultaneously. This method does not apply to overidentified models as it will not give unique estimates of the structural coefficients of the model. The Two-stage least squares (2SLS) are used for estimating over-identified models.

4.2.2. Two Stage Least Square (2SLS)

2SLS is used for just-identified over-identified models. And it can be used if the disturbance term is homoscedastic or heteroscedastic and/or auto correlated. First, it uses IV to compute the estimated values of the suspected endogenous variables, which means an OLS-regression on the instrument (the first stage). Second, it uses those computed values to estimate a linear regression model of the dependent variable, which means the OLS-regression of the fitted values from the first stage regression (the second stage). Since the computed values are based on variables that are uncorrelated with the errors, the results of the two-stage model are optimal. The detailed two steps are shown as follow:

Step 1: Regress the endogenous variable on all chosen instruments, which have previously undergone relevance and exogeneity checks, and obtain the residual for the endogenous variable.
Step 2: Replace the endogenous variable with the corresponding residual and regress the dependent variable on it.

4.2.3. Three-Stage Least Squares (3SLS)

3SLS is an instrumental variables estimator for structural equations in which at least one equation contains endogenous independent variables. It is similar to the 2SLS estimation, with the difference being that moderator variables are used as instrumental variables to obtain residuals for the endogenous independent variable. The detailed steps are:

Step 1: Regress each predictor on all moderators, confirm significant relationship between moderatos and the predictor, and obtain residuals for the predictor.
Step 2: Replace each predictor with the corresponding residual and regress dependent variable on obtained residuals.
Step 3: Add the interaction terms to the model.

4.2.4. IV-GMM Estimator

In econometrics, generalized method of moments (GMM) is one estimation methodology that can be used to calculate IV estimates. This method is used for just-identified and over-identified models with the disturbance term that is homoscedastic or heteroscedastic and/or auto-correlated. GMM method is based on moment functions that depend on observable random variables and unknown parameters, and that have zero expectation in the population when evaluated at the true parameters. GMM estimators contain a test of over-identification, often dubbed Hansen's J-test, as an inherent feature. Therefore, to use IV-GMM estimator, the test of over-identifying restrictions must be performed as post-estimation tests.

If the model is just identified, the performance of 2SLS and GMM method is similar. However, if the model is overidentified, the IV-GMM estimates will differ, and will be more efficient than the robust 2SLS estimates.

For weak instruments, if the assumption of limited information maximum likelihood method is not reasonable, the continuously updated GMM estimator, or CUE estimator can be used.

4.2.5. Limited Information Maximum Likelihood and Fuller-k estimators

For weak instruments, the limited information maximum likelihood (LIML) or Fuller-k estimators can be used to obtain consistent estimation, because they are less biased than 2SLS in finite samples (Staiger and Stock, 1997). However, the less-biased LIML is also much less precise than 2SLS.

LIML estimator is more resistant to weak instruments problems than the IV estimator. Like any maximum likelihood estimator, LIML is invariant to normalization. To use LIML method, the i.i.d. errors assumption should be met.

4.3.Postestimation Tests IV-Based Estimation

4.3.1. Test of Endogeneity: Durban and Wu-Hausman Test

If the perceived endogenous variables turn out to be exogenous, then endogeneity is not a cause for concern, and standard OLS will be more efficient and IV-based estimation method should not be used.

The Durban and Wu-Hausman test can be used to identify whether endogenous problems exist in the regression model. In Durbin and Wu-Hausman test, the null hypothesis states that variables are exogenous. Therefore, if these tests are statistically significant, it would indicate an endogeneity issue in the model.

4.3.2. Test of Relevance (Weak Instrument Test)

The weak instrument tests (Stock-Yogo test/F-statistics) are implemented to judge the explanatory power of the instruments. These tests assess the level of correlation between the additionally included IVs and the endogenous variables.

This process includes the first-stage F-statistics, which are often considered a “more robust and conservative test that should always be reported”. In this F-statistics, the null hypothesis states that the instruments are weak. A higher F-statistics indicates the instruments are stronge, while lower F-statistics indicates weak instruments.

4.3.3. Test of Exogeneity (Test of Over-identifying Restriction)

A test of the overidentifying restriction is conducted to assess the “exogeneity condition” of the instruments. Tests of overidentifying restrictions/ orthogonality condition also confirm whether the instrument or model is correctly specified.

To use 2SLS estimator with homoscedastic errors, this test reports Sargan’s and Basmann’s $\chi 2$ tests for 2SLS without robust variance-covariance (VCE). If Sargan’s and Basmann’s tests are significant, this implies that one or more of the instruments are not valid or that the equation (model) is incorrectly specified.

For 2SLS estimator with heteroskedastic and autocorrelated error terms, Wooldridge’s robust score test of overidentifying restrictions is used.

For GMM estimators, the Hansen J-test is used for heteroskedasticity of errors.

For LIML estimators, the Anderson-Rubin likelihood ratio test and a Basmann F-test are used.

4.4. Weak Instruments

A weak instrument may still be useful when the sample size is sufficiently large, such as over one million observations in sample. Weak instrument issue is more severe in small samples and just-identified models, which make instrumental variable regression estimates unstable and imprecise. The use of multiple weak instrumental variables cannot solve this problem. Therefore, the use of weak instruments is of limited utility especially when sample size is small.

The weak instruments problem can arise even when the first-stage t- and F -tests are significant at conventional levels in a large sample. In the worst case, the bias of the IV estimator is the same as that of OLS, IV becomes inconsistent, and instrumenting only aggravates the problem.

The weak instrument problems should not only be tested but also be solved. The LIML method and Fuller-k estimators can be used, because they are less biased than 2SLS in finite samples. However, the less-biased LIML is also much less precise than 2SLS. The LIML require the satisfaction of the assumption that the errors in the model are independent and identically distributed (i.i.d.). If the assumption of LIML method is not reasonable, the continuously updated GMM estimator, or CUE estimator can be used.

4.5. Steps of Application of IVs

Step 1: Clearly define the research problem and describe the problem mechanism. Set up the base model for the research question and perform OLS regression on the base model to obtain preliminary results. Analysis and explain the possible endogeneity of the model and it causes.

Step 2: Select effective instrumental variables based on theory and explain why the selected IVs are relevance and exogeneous from the theoretical aspect.

Step 3: Perform the IV-based estimation and the necessary statistical tests. And explain the results with caution. The statistical tests are:

(1) Test of Endogeneity
(2) Test of Relevance of IVs
(3) Test of Exogeneity of IVs-Test of Over-Identifying Restriction

If and only if the model is overidentified, with more excluded instruments than included endogenous variables, the exogeneity condition of the IVs can be test. The test should always be performed when it is possible to do so, as it allows researchers to evaluate the validity of the instruments.

And researchers should report R^2 values for the first-stage regression (without IVs and with IVs), as well as values for the "incremental" F-statistics. This describes how R^2 of the first-stage regression changes after additional instruments are included in the model.

Step 4: Compare results obtained from IV-based estimation and that from OLS. Explain why the results are different from different estimation methods.

5. Advantage and disadvantage of IVs

Although IV-based estimation method is one of the most commonly used method to solve endogeneity in empirical research. There are also many disadvantages or limitations of IV-based estimation methods. They are:

(1) The strict limitations on both strength and validity of instrumental variables have made them hard to find (Lu et al., 2018; Rossi, 2014). The use of a weak instrument can caused higher standard errors in estimated results (Semadeni et al., 2014). Moreover, the use of weak or "invalid instruments can cause the estimates to differ even when there is no endogeneity bias" (Rossi, 2014. p 671). The choice and use of poor instrument variable can do more harm than good.

(2) The exogeneity condition cannot be tested directly. This condition can only be tested by tests of over-identifying restrictions. However, it may be difficult to find even just one valid instrument, and multiple instrumental variables to meet the condition to perform the endogeneity test. Because this test can only be performed for over-identified models. And since the exogeneity of IVs cannot be tested directly, there is no way to guarantee that the instruments have actually provided any protection against endogeneity.

(3) An untestable assumption is required that at least one of the instruments satisfies the exclusion restriction. Instrumental variables arises from the fact that all statistical analyses are erected on assumptions, at least some of which are untestable, at least one of the IV is exogeneous. And this assumption may not be reasonable.

(4) Most of the justification for the use of instrumental variable regression is asymptotic and the regression technique may not work well in small samples.

(5) The precision of instrumental variable regression estimates is lower than that of OLS estimates (Baum, 2006). If only weak instruments are available, the loss of precision will be severe and instrumental variable regression estimates may have no improvement over OLS estimates. Efficiency loss can obviously be offset by having a large sample, but with typical sample sizes (in the hundreds or perhaps a thousand), the efficiency loss tends to render the IV approach infeasible.

When good instrumental variable is not available:

1. develop hypotheses that are theoretically grounded and practically relevant.
2. thoroughly describe the nature of the potential endogeneity problem.
3. consider alternative ways to address endogeneity
   efficiency-bias tradeoff: endogeneity removing techniques relying on instruments come at a price of non-negligible efficiency losses in estimation.

In addition to instrumental variable regression, researchers can also include additional control variables or employ fixed effects models (based on panel data) to mitigate endogeneity.

References:

Antonakis, J., Bendahan, S., Jacquart, P., & Lalive, R. (2010). On making causal claims: A review and recommendations. *The Leadership Quarterly*, *21*(6), 1086–1120. https://doi.org/10.1016/j.leaqua.2010.10.010

Bartels, L. M. (1991). Instrumental and “Quasi-Instrumental” Variables. *American Journal of Political Science*, *35*(3), 777. https://doi.org/10.2307/2111566

Fullerton, R. R., Kennedy, F. A., & Widener, S. K. (2014). Lean manufacturing and firm performance: The incremental contribution of lean management accounting practices. *Journal of Operations Management*, *32*(7–8), 414–428. https://doi.org/10.1016/j.jom.2014.09.002

Murray, M. P. (2006). Avoiding Invalid Instruments and Coping with Weak Instruments. *Journal of Economic Perspectives*, *20*(4), 111–132. https://doi.org/10.1257/jep.20.4.111

Roodman, D. (n.d.). *How to do Xtabond2: An Introduction to Difference and System GMM in Stata*.

Rossi, P. E. (2014). Even the Rich Can Make Themselves Poor: A Critical Examination of IV Methods in Marketing Applications. *Marketing Science*, *33*(5), 655–672. https://doi.org/10.1287/mksc.2014.0860

Ullah, S., Zaefarian, G., & Ullah, F. (2021). How to use instrumental variables in addressing endogeneity? A step-by-step procedure for non-specialists. *Industrial Marketing Management*, *96*, A1–A6. https://doi.org/10.1016/j.indmarman.2020.03.006

Zaefarian, G., Kadile, V., Henneberg, S. C., & Leischnig, A. (2017). Endogeneity bias in marketing research: Problem, causes and remedies. *Industrial Marketing Management*, *65*, 39–46. https://doi.org/10.1016/j.indmarman.2017.05.006

GMM and System GMM Method

1. Definition of GMM

The generalized method of moments (GMM) is a generic method to estimate parameters (Hansen, 1982), which is a statistical method that combines observed economic data with the information in population moment conditions to produce estimates of the unknown parameters of the economic model. After obtaining those parameters, the inference about the research question can be performed. And this method is applied in the context where the distribution of the data is unknown, and the model is over-identified.

2. Mechanism of GMM

The common assumption of linear regression model is that the error term (i.e., disturbance term) is uncorrelated with the independent variables. However, if the heteroskedasticity is of unknown form, the assumption of OLS may not be reasonable. GMM does not require the same kind of assumption as OLS. Instead, the fundamental assumption of GMM is that the error term has a zero mean conditional on the covariates and the error term is heteroskedastic (Wooldridge, 2001).

The steps to implement GMM are generally described. First, one must decide which extra moment conditions to add to those generated by the usual zero correlation assumption. Next, having first done ordinary least squares, one must obtain the weighting matrix that is a crucial component to an efficient GMM analysis. The weighting matrix is obtained by inverting a consistent estimator of the variance-covariance matrix of the moment conditions(Wooldridge, 2001).

The GMM estimator minimizes a quadratic form in the sample moment conditions, where the weighting matrix appears in the quadratic form. And the moment conditions with larger variances receive relatively less weight in the estimation, since they contain less information about the population parameters (Wooldridge, 2001). Moment conditions with smaller variances receive relatively more weight. If the moment conditions are correlated, the weighting matrix efficiently combines the moment conditions by accounting for different variances and nonzero correlations.

3. Advantage and Limitation of GMM

There are many advantages of using GMM estimators. First, GMM estimators allow for heteroskedasticity, serial correlation and nonlinearities. It has be proven that GMM estimators can improve over ordinary least squares in the presence of heteroskedasticity of unknown form (Wooldridge, 2001). GMM estimators show that moment conditions could be exploited very generally to estimate parameters consistently under weak assumptions. Second, GMM estimators can be used for over-identified model. Over-identification restriction is very important for the application of instrumental variables techniques using external instruments. Because the exogeneity condition of instruments cannot be test directly and it can only be tested via the tests for over-identifying restriction. For over-identified model, the exogeneity of instruments can be tested. <u>2SLS estimator is a generalized method of moments estimator that uses a weighting matrix constructed under homoskedasticity</u>. The optimal GMM estimator is asymptotically no less efficient than two-stage least squares under homoskedasticity, and GMM is generally better under heteroskedasticity (Wooldridge, 2001). Third, GMM has been applied successfully to estimate certain nonlinear models with endogenous explanatory variables that do not appear additively in an equation (Wooldridge, 2001).

GMM estimators also have limitation. GMM can suffer from finite-sample problems, especially if one unreasonably adds many moment conditions that do not add much information on the studied population (Wooldridge, 2001).

4. Application of GMM

4.1.Cross-Section Applications

For the cross-section regression, GMM estimators have similar performance as the 2SLS estimators. Because the overidentified parameters can be used to improve upon 2SLS in the context of heteroskedasticity of unknown form. And additional gains from using GMM may be small (Wooldridge, 2001). And the most important step in applying instrumental variable methods is finding good instruments. With poor instruments, the efficient GMM estimator is not likely to help much. GMM and 2SLS yield similar results. Under such situations, GMM provides no particular advantage over 2SLS.

However, GMM has been applied successfully to estimate certain nonlinear models with endogenous explanatory variables that do not appear additively in an equation (Wooldridge, 2001).

4.2.Time Series Applications

If the overidentifying restriction is met, the moment conditions can be added with time series by assuming that past values of explanatory variables, or even past values of the dependent variable, are uncorrelated with the error term, even though they do not appear in the model. Using lagged values of dependent and independent variables makes more sense in the context of models estimated under rational expectations. Then, the error term in the equation is uncorrelated with all variables dated at earlier time periods (Wooldridge, 2001). And with time-series data, GMM estimators can also be applied to nonlinear model.

4.3.Panel Data Applications

The standard estimator used to eliminate the potential bias caused by omitted heterogeneity is the fixed effects, or within estimator. The fixed effects (FE) estimator, which is a method of moments estimator based on the data after subtracting off time averages, is popular because it is simple, easily understood, and robust standard errors are readily available (Wooldridge, 2001). For FE estimators, the standard assumptions are that the time-varying errors have zero means, constant variances and zero correlations, all conditional on the observed history of the covariates and on the unobserved effect (Wooldridge, 2001). The first assumption, that the conditional mean of the time-varying errors is zero, implies that the observed covariates in every time period are uncorrelated with the time-varying errors in each time period, which is called strict exogeneity assumption for the covariates is crucial for consistency of the FE estimator. But the assumptions about constant variance and no serial correlation are used primarily to simplify calculation of standard errors. If either heteroskedasticity or serial correlation exists, GMM estimators can be more efficient than the FE estimator, although the likely gains in standard applications are largely unknown. Extra moment conditions are available from the assumption that the covariates in all time periods are assumed to be uncorrelated with each time-varying error.

GMM methods can be easily used in the basic unobserved effects model, for example, models where unobserved heterogeneity interacts with observed covariates. The GMM method is applied more often to unobserved effects models when the explanatory variables are not strictly exogenous even after controlling for an unobserved effect. Under such condition, GMM estimator is usually more efficient than other estimators. If the errors in the first-differenced equation are homoscedastic and serially uncorrelated, the 2SLS estimator is efficient. If not, a GMM estimator can improve upon two-stage least squares.

Another application of GMM in panel data contexts is when a model contains a lagged dependent variable along with an unobserved effect. GMM is well suited for obtaining efficient estimators that account for the serial correlation. And the System GMM is a commonly used method in such situation.

4.3.1. Difference GMM

Arellano and Bond (1991) proposed a GMM estimator that uses the lagged values of the endogenous regressors as instruments for the variables in the difference equations, which is commonly known as the Difference GMM estimator (Bapna et al., 2013). Valid instruments, by definition, should be highly correlated with the variables to be instrumented but orthogonal to the error term in order to address the endogeneity concern (Chizema et al., 2015). These requirements result in a set of “moment conditions” that enables the Difference GMM estimator to select the suitable lagged values as valid instruments for the difference equations. Blundell and Bond (1998) showed that the instruments used in the Difference GMM estimator could be weak if the autoregressive process becomes too persistent over time.

4.3.2. System Generalized Method of Moments (System GMM)

4.3.2.1. Definition of System GMM

To solve the weak instrument problem in Difference GMM, Arellano and Bover (1995), and Blundell and Bond (1998) developed a new GMM estimator with additional moment conditions in which the lagged differences of the endogenous regressors are used as instruments for the original-level equations, which is known as the System GMM estimator (Bapna et al., 2013). The System Generalized Method of Moments (System GMM) estimation technique that relies on transformations of existing variables rather than the use of external variables to address the endogeneity.

The System GMM estimates a system of two equations simultaneously, i.e., the original-level equation and the transformed difference equation. As the fixed effects are still present in the error term of the original-level equations, an additional assumption is required in order to implement the System GMM estimator. Although the endogenous variables are correlated with the fixed effects in the error term by construction, it is assumed that the correlation is constant over time (i.e., time-invariant) (Arellano & Bover, 1995; Blundell & Bond, 1998). This assumption allows the System GMM estimator to develop differences as instruments to make them exogenous to the fixed effects in the error term and address the endogeneity concern (Roodman, 2009).

The System GMM estimator uses lagged differences as instruments for the original-level equations, in addition to the use of lagged levels as instruments for the transformed difference equations. The introduction of more instruments enables the System GMM estimator to address the concern of weak instruments in the Difference GMM estimator and improve the estimation efficiency dramatically (Roodman, 2009).

The lagged dependent variables are used as instruments to mitigate the dynamic endogeneity for the panel data and these instruments are defined as **internal instrument** (Lam et al., 2016). The GMM model can mitigate endogeneity caused by different source, namely “unobserved heterogeneity, simultaneity and dynamic endogeneity” (Wintoki & Netter, 2012). In other words, GMM model can mitigate endogeneity caused by omitted variables, errors-in-variables and simultaneity.

And System GMM estimator to have much better performance under finite sample properties in terms of bias and root mean squared error (rmse) than that of the difference GMM estimator (Bun & Windmeijer, 2010).

4.3.3.2.Key Assumptions of System GMM Model and Statistical Test

To use System GMM model, some fundamental assumptions need to be met. The assumptions are (1) some regressors may be endogenously determined (2) the nature of the relationship is dynamic, implying that current performance is affected by previous ones (3) the idiosyncratic disturbances are uncorrelated across individual (4) some regressors may not necessarily be strictly exogenous; and finally (5) the time periods in panel data, T, may be small. (i.e., “small T, large N.”) (Ullah et al., 2018). A critical assumption for the validity of System GMM requires that instruments are exogenous. In other words, the findings from GMM will not be valid if the instruments are endogenously determined (Ullah et al., 2018) (i.e., when instrumets are correlated with the error term). Therefore, it is important to ensure the validity of the instruments used in System GMM model.

4.3.3.3.The Method of System GMM Model

The System GMM model mitigate endogeneity by “internally transforming the data”– transformation refers to a statistical process where a variable's past value is subtracted from its present value (Roodman, 2009). The internal transformation process reduces the number of observations and thus improves the efficiency of the GMM model (Ullah et al., 2018). There are two kinds of internal transformation methods, which are first-difference transformation (one-step GMM) and second-order transformation (two-step GMM). And these two method can also be utilized as GMM estimators. There are limitations of the first-difference transformation (one-step GMM), such as the loss of too many observations if a variable's recent value is missing in this model. Therefore, to avoid potential data loss owing to the internal transformation problem with the first-step GMM, the use of a second order transformation (two-step GMM) is recommended (Yiu et al., 2020).

For the second-order transformation (two-step GMM), the ‘forward orthogonal deviations’ is utilized. It means that the two-step GMM model subtracts the average of all future available observations of a particular variable (Roodman, 2009). Because the two-step GMM model can avoid the data loss mentioned above, the two-step GMM model can provide more efficient and consistent estimates for the involved coefficients for a balanced panel dataset (Yiu et al., 2020). And the GMM model relies on internal instruments (lagged values, internal transformation) to address the different sources endogeneity, which is the main difference compared with the instrumental variable techniques that use exogeneous instruments.

Forth, the System GMM estimator addresses the dynamic panel bias directly by instrumenting the lagged dependent variables with variables uncorrelated with the fixed effects in the error term (Roodman, 2009). Fifth, the System GMM estimator is suitable for our data with unbalanced panels. the System GMM estimator appears to be one of "the most robust methodologies for unbalanced panels with endogenous variables". Finally, although the System GMM estimator also employs the IV techniques to deal with the endogeneity issue, it does not rely on external variables that are outside the immediate dataset to construct instruments. Instead, the System GMM estimator constructs instruments internally with the transformation of existing variables, overcoming the difficulty in obtaining exogenous instruments externally (Roodman, 2009; Wintoki et al., 2012).

4.3.3.4.Post-Estimation Test of System GMM

When applying the System GMM model in research, two post-estimation tests need to be performed to determine that an appropriate econometric model is applied. The two tests are: (1) the Sargan test; and (2) the Arellano-Bond test for first order and second-order correlation.

Sargan test determines whether the econometric model is valid or not, and whether the instruments are correctly specified or not. In other words, if the null hypothesis is rejected, the researcher needs to reconsider the model or the instruments used in the estimation process.

To examine the validity of a strong exogeneity assumption, the Arellano-Bond test for no autocorrelation (or no serial correlation) is used under the null hypothesis that the error terms of two different time periods are uncorrelated.

4.3.3.5.Advantages and Disadvantages of System GMM

The System GMM mode has many advantages. It offers more efficient and consistent estimates for the coefficients as compared with other estimation techniques. It provides a consistent, asymptotically normal, and efficient estimator among all the estimators that only use the information provided by the moment conditions.

First, GMM is convenient for estimating interesting extensions of the basic unobserved effects model, for example, models where unobserved heterogeneity interacts with observed covariates (Wooldridge, 2001). Second, GMM have better performance that other methods, when it is applied to unobserved effects models when the explanatory variables are not strictly exogenous even after controlling for an unobserved effect. Thirdly, GMM is well suited for obtaining efficient estimators that account for the serial correlation existing in the time-series data of the dependent variable. It can be applied to a model contains a lagged dependent variable along with an unobserved effect.

Comparison with Instrumental Variable Techniques: For instrumental variable techniques, it is difficult to obtain such strictly exogenous instruments externally as pointed out by prior studies (Bardhan et al., 2013; Wintoki et al., 2012). In contrast, System GMM use internal instruments. Therefore, System GMM has several advantages. First, the System GMM controlled for different kinds of endogeneity by including lagged values of the dependent variable (previous value of the dependent variable) as an explanatory variable in the model. The System GMM is well suited for obtaining efficient estimators that account for the serial correlation. In contrast to 2SLS and 3SLS, it does not rely on external exogenous instruments, which in practice may be difficult to identify (Wintoki et al., 2012), but consists of a System of two sets of equations, each with its own internal instruments (Abdallah et al., 2015). Thus, System GMM model is more useful when exogeneous instruments of good quality (i.e., valid and strong) are not available. Second, the System GMM estimator also can be used to generate standard errors that are robust to autocorrelation (in time series or long panel data) in the data (Baum et al., 2007). This cannot be achieved by instrumental variable techniques, because the instrumental variable techniques do not allow the lagged value. Third, the System GMM model can be more efficient (i.e., smaller standard errors) if errors are considered to have arbitrary heteroscedasticity or intra-cluster correlation for over-identified equations. The optimal GMM estimator is asymptotically no less efficient than two-stage least squares under homoskedasticity, and GMM is generally better under heteroskedasticity. Meanwhile, the GMM estimator has similar performance with 2SLS for just-identified models, but better performance than 2SLS for over-identified models. Forth, the System GMM can be applied more often to unobserved effects models when the explanatory variables are not strictly exogenous even after controlling for an unobserved effect. For example, when the fixed effects are eliminated by differencing the observations of adjacent years, if the errors in the first-differenced equation are homoscedastic and serially uncorrelated, the 2SLS estimator is efficient; if not, a GMM estimator can improve upon two-stage least squares.

Comparison with AR and FE model: The AR and FE models are static panel data model. While the System GMM is a dynamic panel data model. In other words, an important difference between FE model and GMM is that FE estimation uses ‘strict exogeneity’ assumptions resulting into a static FE model. For panel data, the past realization of dependent variables may also affect realization of dependent variables of the current year, and the simultaneity could potentially violate the strict exogeneity assumptions. Under such situations, traditional FE and AR panel data static models will provide inconsistent and biased results (Wooldridge, 2012). Conversely, the System GMM estimation is more robust in dealing with these sources of endogeneity.

Comparison with DID: When the lagged value of dependent variable is correlated with the fixed effects in the error term or the dependent variable’s realization of current value is influenced by the past value, the ordinary least squares (OLS) estimation will obtain inconsistent estimated result (Roodman 2009). In other words, under such situation, the endogeneity issue will be caused due to omitted variables. Although a more advanced least squares dummy variables (LSDV) estimator can address this bias (Kiviet 1995), it does not deal with the other endogeneity concerns such as unobservable heterogeneity and simultaneity arising from the other regressors included in our research (Roodman 2009). Therefore, System GMM should be used.

References:

Arellano, M., & Bond, S. (1991). Some Tests of Specification for Panel Data: Monte Carlo Evidence and an Application to Employment Equations. *The Review of Economic Studies*, *58*(2), 277. https://doi.org/10.2307/2297968

Arellano, M., & Bover, O. (1995). Another look at the instrumental variable estimation of error-components models. *Journal of Econometrics*, *68*(1), 29–51. https://doi.org/10.1016/0304-4076(94)01642-D

Bapna, R., Langer, N., Mehra, A., Gopal, R., & Gupta, A. (2013). Human Capital Investments and Employee Performance: An Analysis of IT Services Industry. *Management Science*, *59*(3), 641–658. https://doi.org/10.1287/mnsc.1120.1586

Bardhan, I., Krishnan, V., & Lin, S. (2013). **Research Note** —Business Value of Information Technology: Testing the Interaction Effect of IT and R&D on Tobin's Q. *Information Systems Research*, *24*(4), 1147–1161. https://doi.org/10.1287/isre.2013.0481

Blundell, R., & Bond, S. (1998). Initial conditions and moment restrictions in dynamic panel data models. *Journal of Econometrics*. https://doi.org/10.1016/S0304-4076(98)00009-8

Bun, M. J. G., & Windmeijer, F. (2010). *The Weak Instrument Problem of the System GMM Estimator in Dynamic Panel Data Models*.

Chizema, A., Liu, X., Lu, J., & Gao, L. (2015). Politically connected boards and top executive pay in Chinese listed firms. *Strategic Management Journal*, *36*(6), 890–906. https://doi.org/10.1002/smj.2253

Lam, H. K. S., Yeung, A. C. L., & Cheng, T. C. E. (2016). The impact of firms' social media initiatives on operational efficiency and innovativeness. *Journal of Operations Management*, *47–48*(1), 28–43. https://doi.org/10.1016/j.jom.2016.06.001

Wintoki, M. B., Linck, J. S., & Netter, J. M. (2012). Endogeneity and the dynamics of internal corporate governance. *Journal of Financial Economics*, *105*(3), 581–606. https://doi.org/10.1016/j.jfineco.2012.03.005

Wooldridge, J. M. (2001). Applications of Generalized Method of Moments Estimation. *Journal of Economic Perspectives*, *15*(4), 87–100. https://doi.org/10.1257/jep.15.4.87

Yiu, L. M. D., Lam, H. K. S., Yeung, A. C. L., & Cheng, T. C. E. (2020). Enhancing the Financial Returns of R&D Investments through Operations Management. *Production and Operations Management*, *29*(7), 1658–1678. https://doi.org/10.1111/poms.13186

Quasi-experiment

1. Definition of Quasi-Experiment

Quasi-experiments are study aim to evaluate interventions that do not use randomization, instead, the assignemnt to the treatment condition is based on some criterions, which can be controlled by researchers. That is the randomness is introduced by variations in individual circumstances that make it appear as if the treatment is randomly assigned. The quasi-experiments can demonstrate causality between an intervention and an outcome.

Quasi-experiment have higher external validity than other experiment design, such as the laboratory experiment. The quasi-experiment can mitigate the endogeneity caused by selection bias by studying the variable of interest before and after a particular intervention has occurred.

2. Types of Quasi-Experiment

Then main challenge to apply quasi-experiment in OM empirical research is that once a intervention has occurred, plenty of factors may change and thus the comparison of before and after intervention situations is not useful to draw on meaningful conclusion. To solve this problem, researchers should control for this by using a set of subjects that are not influenced by the intervention as a control group. And then, they should compare the difference in the group that is influenced by the intervention, also known as the treatment group, to the difference in the control group (Fan et al., 2022).

Generally, there are two types of experiments. In the first type is whether an entity receives treatment is viewed as if it is randomly determined. In this case, the causal effect can be estimated by OLS using the treatment as a regressor and DID estimator can be used. In the second type of quasi-experiment, the as-if random variation only partially determines the treatment. In this case, the causal effect is estimated by instrumental variables regression, where the as-if random source of variation provides the instrumental variable. And DID estimator augmented with additional regressor (instrumental variable) can be used.

2.1 Non-equivalent Group Design

In non-equivalent group design (NEGD), the researcher chooses existing groups that appear similar, but where only one of the groups experiences the treatment. In contrast to a true experiment with randomness, the control and treatment groups in the quasi-experiment are not randomly assigned. That is the control and treatment groups are non-equivalent groups, which means that they may differ in other ways causing the endogeneity issues.

To effectively use this quasi-experiment design, researchers need to account for confounding variables by controlling for them in their analysis or by choosing groups that are as similar as possible, similar to matching method.

2.1.1. Regression Discontinuity

Many potential treatments that researchers aim to study are designed around an essentially arbitrary cutoff, where those above the threshold receive the treatment and those below it do not.

Near this threshold, the differences between the two groups are often so minimal as to be nearly nonexistent. Therefore, researchers can use individuals just below the threshold as a control group and those just above as a treatment group.

2.2. Natural Experiments

Although in OM empirical research, natural experiment and quasi-experiment sometimes are used interchangeably, some researchers still distinguish them based on whether researchers can decide the assigement criterions of the control and treatment groups.

In a natural experiment, an external event or situation ("nature") results in the random or random-like assignment of subjects to the treatment group. Although some researchers use random assignments, natural experiments are not considered to be true experiments because they are observational in nature.

Although the researchers have no control over the independent variable, they can exploit this event after the fact to study the effect of the treatment.

3. When to Use Quasi-experiment

Although true experiments have higher internal validity, quasi-experimental design is a useful tool in situations where true experiments cannot be used for ethical or practical reasons.

As for the ethical reasons, sometimes it would be unethical to provide or withhold a treatment on a random basis, so a true experiment is not feasible. As for the practical issues, true experimental design may be infeasible to implement or simply too expensive or too much work is involved in recruiting and properly designing an experimental intervention for an adequate number of subjects to justify a true experiment.

In either case, quasi-experimental designs allow researchers to study the question by taking advantage of secondary data (i.e., data that has already been collected by others for other purposes).

4. Advantages and Disadvantages of Quasi-Experiment

4.1. Limitations

Quasi-experiments face threats to internal and external validity.

4.1.1. Threats to Interanl Validity

(1) Failure of randomization: Without randomization, it can be difficult to verify that all confounding variables have been accounted for, which may cause the omitted-variable endogeneity.

(2) Failure to follow the treatment protocol: In a quasi-experiment, if the treatment protocol is not followed, the assignment ceritions of control and treatment groups as well as the rule of treatment decided by researchers only influence the group assigement and treatment level. If this happened, it can result in the omitted variables included that are not be included in the model, causing the omitted-variables endogeneity.

(3) Attrition: The selection bias will be caused by attrition. If the attrition caused by by personal choices or characteristics, then it can induce correlation between the treatment level and the error term, resulting to the engogeneity caused by selection bias.

(4) Instrument validity in quasi-experiments: If instrumental variables are required to be used in a quasi-experiment, it can be difficult to find a vaild and strong instruments.

In summary, the disadvantages of the quasi-experiment method are: (1) Lower internal validity than true experiments—without randomization, it can be difficult to verify that all confounding variables have been accounted for. (2) The use of retrospective data that has already been collected for other purposes can be inaccurate, incomplete or difficult to access. The retrospective data can be inaccurate, incomplete or difficult to access.

4.2. Advantages

First, the quasi-experiment method provide higher external validity than true experiments, because they often involve real-world interventions instead of artificial laboratory settings. Second, it also provides higher internal validity than other non-experimental types of research, because they allow researchers to better control for confounding variables than other types of studies do. Third, few experimental effects exist in a quasi-experiment. Because the quasi-experiments are not true experiments, there typically is no reason for individuals to think that they are experimental subjects. Thus experimental effects such as the Hawthorne effect generally are not germane in quasi-experiments.

Difference between quasi-experiment methods and other methods: The econometric methods used in this chapter for analyzing quasi-experiments are familiar ones developed in other methods, including OLS, panel data estimation methods, and instrumental variables regression. What differentiates quasi-experiments from the applications in other methods are the way in which these methods are interpreted and the data sets to which they are applied. Quasi-experiments provide econometricians with a way to think about how to acquire new data sets, how to think of instrumental variables, and how to evaluate the plausibility of the exogeneity assumptions that underlie OLS and instrumental variables estimation.

References:

Fan, D., Zhou, Y., Yeung, A. C. L., Lo, C. K. Y., & Tang, C. (2022). Impact of the U.S.–China trade war on the operating performance of U.S. firms: The role of outsourcing and supply base complexity. *Journal of Operations Management*, *68*(8), 928–962. https://doi.org/10.1002/joom.1225

Part Three: Survey Research in OM

Survey Research

1. Introduction

One of the main characteristics of the survey is that it relies on structured instruments to collect information.

The survey research has many advantages and disadvantages. The advantages of the survey research are:

(1) An excellent vehicle for measuring a wide variety of unobservable data
Such as people's preferences (e.g., political orientation), traits (e.g., self-esteem), attitudes (e.g., toward immigrants), beliefs (e.g., about a new law), behaviors (e.g., smoking or drinking behavior), or factual information (e.g., income).

(2) Ideally suited for remotely collecting data about a population that is too large to observe directly.
A large area, such as an entire country, can be covered using mail-in, electronic mail, or telephone surveys using meticulous sampling to ensure that the population is adequately represented in a small sample

(3) Due to their unobtrusive nature and the ability to respond at one's convenience, questionnaire surveys are preferred by some respondents.

(4) Interviews may be the only way of reaching certain population groups such as the homeless or illegal immigrants for which there is no sampling frame available.

(5) Large sample surveys may allow detection of small effects even while analyzing multiple variables, and depending on the survey design, may also allow comparative analysis of population subgroups (i.e., within-group and between-group analysis).

(6) Survey research is economical in terms of researcher time, effort and cost than most other methods such as experimental research and case research.

This research method also has some disadvantages:

(1) Not be appropriate or practical for certain demographic groups such as children or the illiterate (Especially, the questionnaire surveys).

(2) Subject to a large number of biases such as non-response bias, sampling bias, social desirability bias, and recall bias.

2. Data Collection Methods

Selection of data collection methods should be based on needs of the specific survey as well as the time, cost and resource constraints. Mixed approaches may raise difficult issues, the most important of which is represented by the fact that a same respondent may give different answers to the same question administered through different methods. And handling missing data should be a key concern during data collection.

The two data collection methods are questionnaire and interview. And data can be collected via different ways, including mail, telephone, web and face-to-face.

In the questionnaire survey, a questionnaire is a research instrument consisting of a set of questions (items) (unstructured or structured questions) intended to capture responses from respondents in a standardized manner. The design of the survey questionnaire depends on whether the questionnaire is to be administered by telephone interview, on-site interview, on-site using pen-and-paper or by post using pen-and-paper.

And the main data collection methods that are used in the OM empirical research is summarized as follows:

(1) Postal survey: questionnaires are printed and sent by post. The respondents are asked to complete the questionnaire on their own and to send it back.
Advantages: inexpensive; can be completed at the respondent's convenience; can be prepared to give an authoritative impression; ensure anonymity; reduce interviewer bias; and the respondent has more time to consider his/her responses.
Disadvantages: a lower response rate; longer time periods and are more affected by self-selection; lack of interviewer involvement and lack of open-ended questions.

(2) Face-to-face survey: interviewer solicits information directly from a respondent during personal interviews.
Advantages: flexibility in sequencing the questions, details and explanation; the possibility of administering highly complex questionnaires; improved ability to contact hard-to-reach populations; higher response rates; increased confidence that data collection instructions are followed.
Disadvantages: higher costs; interviewer bias; the respondent's reluctance to co-operate; greater stress for both respondents and interviewer; less anonymity.

(3) Telephone surveys involve collecting information through the use of telephone interviews.
Advantages: rapid data collection; low costs; anonymity; large-scale accessibility; higher confidence that instructions are followed.
Disadvantages are: less control over the interview situation; less credibility; lack of visual materials.

(4) Electronic surveys: questionnaire through e-mail or ask respondents to visit a website where the questionnaire can be filled in without the need to return the questionnaire physically.
This type of survey contains the visual design and consideration of computer logic as constraints and opportunities.
Advantage: minimal cost compared with other means of distribution; technologies improve their speed and (user) friendliness.
Disadvantage: potential problems in sampling and controlling the research environment.

The advantages and disadvantages are summarized in the Table 1 (Karlsson, 2016):

Table 1. Comparison of Different Data Collection Methods

Factors influencing coverage and secured information	*Mailed*	*Personal interview*	*Telephone survey*	*E-survey*
Lowest relative cost	2	4	3	1
Highest response rate	4	1	2	3
Highest accuracy of information	2	1	4	3
Largest sample coverage	1	4	3	2
Completeness, including sensitive materials	3	1	2	4
Overall reliability and validity	2	1	3	4
Time required to secure information	4	2	1	3
Ease of securing information	1	4	3	2

Interview A semi-structured interview protocol was used to guide the interviews (Jiang et al., 2023).

References:

Jiang, S., Yeung, A. C. L., Han, Z., & Huo, B. (2023). The effect of customer and supplier concentrations on firm resilience during the COVID -19 pandemic: Resource dependence and power balancing. *Journal of Operations Management*, *69*(3), 497–518. https://doi.org/10.1002/joom.1236

Karlsson, C. (Ed.). (2016). *Research Methods for Operations Management*. Routledge.

Reliability and Validity of Survey Instruments

1. Definition

Reliability and Validity are the criteria to evaluate the quality of measures used in survey. And the assessment of measure quality occurs at different stages of survey research: before data collection; in pilot testing; in ad-hoc analyses to validate the measure; and, finally, after data collection, for hypothesis testing. At all of these stages, measures may be refined and items may be eliminated or modified. The elimination of an item requires the researcher to return to content validity assessment and redo all of the subsequent tests.

2. Reliability

Reliability is concerned with stability and consistency in measurement scores. It indicates dependability, stability, predictability, consistency and accuracy, and refers to the extent to which a measuring procedure yields the same results in repeated trials and it measures the ability to replicate the study (Flynn et al., 1990). Lack of reliability introduces a random error.

It is a prerequisite to establishing validity, but not sufficient (Flynn et al., 1990). Reliability is assessed after data collection, although some actions to reduce reliability problems may be taken while assessing face validity. There are four common methods used in OM research to assess the reliability, which are: (1) test-retest method (2) alternative form method (3) split halves method (4) internal consistency method. The details of these methods are shown by Table 1.

Table 1. Methods for Reliability Assessment

Method	Procedure	Meaning
Test-retest	It calculates the correlation between responses obtained through the same measure applied to the same respondents at different points of time (e.g. separated by two weeks).	It estimates the ability of the measure to maintain stability over time. This aspect is indicative of measure stability and low vulnerability to change in uncontrollable testing conditions and in the state of the respondents.
Alternative form	It calculates the correlation between responses obtained through different measures applied to the same respondents at different points of time (e.g. separated by two weeks).	It assesses the equivalence of different forms for measuring the same construct.
Split halves	It subdivides the items of a measure into two subsets and statistically correlates the answers obtained at the same time to them.	It assesses the equivalence of different sets of items for measuring the same construct.
Internal consistency	It uses various algorithms to estimate the reliability of a measure from measure	It assesses the equivalence, homogeneity and inter-correlation of the items used

	administration at one point in time.	in a measure. This means that the items of a measure should hang together as a set and should be capable of independently measuring the same construct.

The most commonly used reliability indicator in OM survey research is Cronbach's coefficient alpha (Flynn et al., 1990), an internal consistency method, which is the average inter-item correlation among the n measurement items in the instrument under consideration.

2.1. Reliability for Questionnaire

There are a number of methods for measuring various aspects of reliability. When a measure is administered to a group of individuals at two different points in time and their scores at the two times are correlated, that correlation coefficient is a measure of **test-retest reliability**. **Parallel form's reliability** can be established by constructing two equivalent (in terms of means and variances) forms of the same measure and administering them to a common set of subjects at different points in time. **Internal consistency** is important when there is only one form of a measure available, such as most P/OM questionnaires. There should be a high degree of intercorrelation among the items that comprise the measure or summated scale. Using the split-half technique, the items within a summated scale are divided into two subsets and the total scores for the two subsets are correlated. To the extent that the measure is internally consistent, the correlation between total scores for the two subsets will be high. The most widely accepted measure of a measure's internal consistency is Cronbach's Alpha (Cronbach & Meehl, 1955). Alpha is the average of the correlation coefficient of each item with each other item (Nunnally (1978). It is popular because it incorporates every possible split of the scale in its calculation.

Because of the difficulty of establishing reliability and validity, it is very useful for researchers to use summated scales whose reliability and validity have already been demonstrated, when using questionnaire.

3. Validity

Validity is concerned with whether we are measuring what we intend to measure. validity measures two things. First, does the item or scale truly measure what it is supposed to measure? Second, does it measure nothing else? Lack of validity introduces a systematic error (bias). There are four types of validity: (1) face validity (2) content validity (3) criterion validity (4) construct validity.

3.1. Content validity

When the set of items in a multi-item measure has been developed), the researcher should test it for content validity. The content validity of a construct measure can be defined as the degree to which a measure's items represent a proper sample of the theoretical content domain of a construct (Flynn et al., 1990). Or it is the degree to which the measure spans the domain of the construct's theoretical definition (Flynn et al., 1990).

Content validity is a judgement, by experts, of the extent to which a summated scale truly measures the concept that it intended to measure, based on the content of the items. It can only be determined by experts and by reference to the literature.

Two conditions are to be satisfied by a measure's items in order to confirm content validity:
(1) Each item should fall within the concept's theoretical domain.
(2) The set of items should capture the different facets of a construct in a balanced way.

3.2.Face validity
Evaluating the face validity of a measure (i.e. assessing whether the measure 'at face value' seems like a good translation of the theoretical concept) can indirectly assess its content validity (Flynn et al., 1990).

Face validity is a matter of judgement and must be assessed before data collection. Two stage method:
(1) They performed item generation and purification refinement prior to developing their questionnaire and field study.
(2) They focused on the survey instrument, data collection, confirmatory analyses and item refinement. They iterated the first stage four times in order to reach enough agreement between experts.

3.2.1. Methods to assess face validity:
(1) A panel of subject-matter experts (SMEs) who are exposed to individual items and are asked to judge the degree to which these items are representative of the construct's conceptual definition.

    For each i^{th} candidate item in the measure, Lawshe's (1975) <u>content validity ratio (CVR_i)</u> is calculated. $CVR_i = \frac{n_e - \frac{N}{2}}{\frac{N}{2}}$. where n_e is the number of SMEs indicating the measurement item i as 'essential' and N is the total number of SMEs in the panel.
(2) Give judgement on both the list of items and the definition of each **construct or construct dimension**. Subsequently, judges are asked to map the items into the **construct** definitions or, for multifaceted constructs, into the construct dimension definitions (Hardesty and Bearden, 2004).
(3) Cohen's (1960) k index, as well as other inter-rater agreement indexes, is also used to assess face validity and to improve it, if necessary, through identifying which items to discard.

3.3.Criterion-related validity
Criterion-related (predictive) validity investigates the empirical relationship between the scores on a test instrument (predictor) and an objective outcome (the criterion). It is established when the measure <u>differentiates subjects</u> on a criterion it is expected to predict. This can be achieved by establishing the <u>concurrent validity or predictive validity</u>. <u>Concurrent validity</u> is established when the <u>scale discriminates subjects</u> that are known to be <u>different</u>. <u>Predictive validity</u> is the ability of the <u>measure to differentiate among subjects</u> with respect to a future criterion.

3.3.1. Methods to Test Criterion-related Validity

The most commonly used measure of criterion-related validity is a validity coefficient, which is the correlation between predictor and criterion scores. A validity coefficient is an index of how well criterion scores can be predicted from the instrument score. Two techniques that can test this validity are simple correlation, for testing a summated scale with a single outcome, and canonical correlation, for testing a summated scale, or a battery of summated scales, with multiple outcomes. It can also be tested using multiple correlations, canonical correlations, and LISREL.

LISREL (linear structural relations) is a proprietary statistical software package used in structural equation modeling (SEM) for manifest and latent variables. It requires a "fairly high level of statistical sophistication".

3.4.Construct Validity

Construct validity measures whether a scale is an appropriate operational definition of an abstract variable, or a construct. Two aspects of construct validity: (1) convergent validity (2) discriminant validity (Flynn et al., 1990).

Convergent validity refers to the degree to which multiple attempts to measure the same concept (e.g. different measures or different items) are in agreement. The idea is that scores obtained by two or more measures of the same thing co-vary highly if they are good measures of the same thing.

Discriminant validity refers to the degree to which measures of different concepts (or items belonging to different measures) are distinct. The idea is that, if two or more concepts are distinct (i.e. they have content domains that do not overlap), then good measures of these concepts (or items belonging to different measures of these concepts) should not correlate too strongly.

3.4.1. Methods to Test Construct Validity

Only indirect inference about construct validity can be made by empirical investigations. Appropriate construct validity procedures depend on both the nature of the construct specified and the hypothetical linkages which it is presumed to have with other constructs. To demonstrate the linkages between the construct and other constructs, a series of assessments should be made, examining the measure against criteria established from multiple hypotheses, measures of alternative constructs and samples (Flynn et al., 1990). The hypotheses should be a logical outgrowth of the proposed linkages illustrated by the nomological network.

Covariance-based structural equation modelling (CBSEM) methods (implemented in software such as LISREL, AMOS and EQS) can be used to assess the measurement quality for reflective constructs. Specifically, Peng and Lai summarized how construct reliability, convergent validity and discriminant validity are typically assessed when using CBSEM methods: Typical measures of construct reliability include Cronbach's alpha and composite reliability. Convergent validity can be assessed by checking whether the average variance extracted (AVE) of the construct is greater than 0.50 (at the construct level) and the item loadings are greater than 0.70 and statistically significant (at the item level). Discriminant validity is usually examined by comparing the square root of AVE with the correlations between the focal construct and all other constructs (Flynn et al., 1990).

Meanwhile convergent and discriminant validity can be tested through the multitrait-multimethod (MTMM) matrix method (the traditional way) or the confirmatory factor analysis (CFA) method.

References:

Flynn, B. B., Sakakibara, S., Schroeder, R. G., Bates, K. A., & Flynn, E. J. (1990). Empirical research methods in operations management. *Journal of Operations Management*, *9*(2), 250–284. https://doi.org/10.1016/0272-6963(90)90098-X

Structural Equation Modeling

1. Definition

Structural equation modeling (SEM) is a technique to specify, estimate, and evaluate models of linear relationships (interrelated dependence relationships) among a set of observed variables in terms of a generally smaller number of unobserved variables for a single independent or dependent variable (Shah & Goldstein, 2006). It can be defined as a hypothesis of a specific pattern of relations among a set of measured variables (MVs) and latent variables (LVs) (Shah & Goldstein, 2006). The objective of using SEM is to determine whether the a priori model is valid, rather than to 'find' a suitable model (Gefen et al., 2000).

1.1. Why Use Structural Equation Modeling

First, SEM can be used for theory testing given its ability to incorporate **measurement error** and allow for constructs to be represented by several measures (indicators). Second, SEM can be used with any type of dependence relationship. In details, it simultaneously analyzes several dependence relationships and it has the ability to account for **measurement error** in the process of estimating coefficients for each independent variable.

2. Model of Structural Equation Modeling

SEM models consist of observed variables, also known as manifest or measured variables (MVs) and unobserved variables, also known as underlying or latent variable (LVs) that can be independent (exogenous) or dependent (endogenous) in nature (Shah & Goldstein, 2006). LVs are hypothetical constructs that cannot be directly measured and they are measured by multiple MVs that are used as indicators of the underlying constructs in SEM.

Path analysis and confirmatory factor analysis are two special cases of SEM. Path analysis (PA) models specify patterns of directional and non-directional relationships among MVs. The only LVs in such models are error terms (Haddock & Quinn, 2015). Therefore, PA can be used to test structural relationships among MVs when MVs are of primary interest or when multiple indicators for LVs are not available. Confirmatory factor analysis (CFA) requires that LVs and their associated MVs be specified before analyzing the data, accomplished by restricting the MVs to load on specific LVs and by designating which LVs are allowed to correlate (Shah & Goldstein, 2006). A CFA model allows for directional influences between LVs and their MVs and (only) nondirectional (correlational) relationships between LVs (Shah & Goldstein, 2006).

2.1. Estimation Methods in SEM

Many estimation methods can be used in SEM. And the selection of estimation methods should be based on the distributional properties of the MVs.

The maximum likelihood ratio (ML) assumes data are univariate and multivariate normal and requires that the input data matrix be positive definite, but it is relatively unbiased under moderate violations of normality. Generalized least square (GLS) assumes normality but does not impose the restriction of a positive definite input matrix. Asymptotically distribution free (ADF) has few distributional assumptions but requires very large sample sizes for accurate estimates. Ordinary least square (OLS), the simplest method, has no distributional assumptions and is computationally the most robust, but it is scale invariant and does not provide fit indices or standard errors for estimates (Binder, 1969).

3. Assumption in SEM

To apply SEM, several assumptions should be satisfied. The assumptions are: (1) A reasonable sample size (2) Continuously and normally distributed endogenous variables (3) Model identification (identified equations) (4) Complete data or appropriate handling of incomplete data (4) Theoretical basis for model specification and causality.

An underlying assumption of SEM analysis is that the items or indicators used to measure a LV are reflective (i.e. caused by the same underlying LV) in nature.

Another underlying assumption for SEM is that the theoretical relationships hypothesized in the models being tested represent actual relationships in the studied population.

4. Procedures to Implement SEM

Step 1: Developing a theoretically base model. SEM is a confirmatory method and not a exploratory method. The assertion of causation is applicable in SEM only when and because the data analysis corroborates theory-based causation hypotheses. And the theory may be based on empirical research from academic or derived from practical experience. In SEM, the excess of 20 variables (5-6 constructs each measured by 3-4 indicators) are difficult to manage.

Step 2: Identify exogenous and endogenous constructs and relationship among these constructs for the path diagrams. Exogenous variables are independents and endogenous ones are the dependents in a model. And the endogenous constructs may be determinants for other endogenous constructs.

There are two assumptions underline the path diagrams. The first one is completeness, which means that all causal relationships are specified in the path diagram. The second one is linearity, i.e., the relationships among the constructs are assumed to be linear.

Step 3: Convert the path diagram into a set of SEMs. Structural model defines the relationship between constructs (exogenous and endogenous) in a separate equation. And measurement model identifies the individual variables which are used to measure exogenous and endogenous constructs.

Step 4: Choose the Input matrix and estimating the proposed model. The focus of SEM is on the pattern of relationships across respondents. And SEM employs matrices as input data (covariance or correlation matrix). As for the sample size, the sample size should be large enough to include 5 observations for each estimated parameter. And as for the minimum sample size, it should be around 100. And the optimal or critical sample size is around 200.

Meanwhile, a computer program LISREL or AMOS and others (SAS, EQS) should also be selected to implement the model.

Step 5: Assess the identification of the structural model. If a model is not identified, then it is not possible to determine the model parameter. If t parameters are to be estimated, the minimum condition for identification is: $t \leq s$, Where s=1/2{(p+q) (p+q+1)}, p is the number of dependent variables (indicators) and q the number of independent variables (indicators).

Step 6: Evaluate Goodness-of-Fit Criteria:

(1) Likelihood of Chi-square to distribution function
    (the acceptable range is <=3.0)
(2) Goodness-of-fit (GFI) (from 0 to 1 and the closer to 1 the better the model fit. The marginal acceptance level is 0.9)
(3) Root-mean-square residuals (RMSR)
    (The closer to zero the better the fit. A marginal acceptance level 0.08)

Step 7: Interpret and modify the model. Whether modifications to the model have to be made should be considered. And it should also be considered whether the modifications (if needed) can be made by deleting or adding parameters. Theory should always be the basis for model specifications. And modification indices (MI) can be used to decide which parameters should be added to the model.

5. When To Use SEM

SEM can be used with any type of dependence relationship. It simultaneously analyzes several dependence relationships. It has ability to account for measurement error in the process of estimating coefficients for each independent variable.

And SEM is not recommended for exploratory research when the measurement structure is not well defined or when the theory that underlies patterns of relationships among LVs is not well established (Shah & Goldstein, 2006).

6. Advantages and Disadvantages

There are many advantages of SEM. First, it can be used with any type of dependence relationship, including correlation and casual relationships. Second, it simultaneously analyzes several dependence relationships and multiply dependent variables can be included in the model. All causal relationships should be specified in the path diagram. Third, it has ability to represent unobserved concepts in these relationships and correct the measurement errors in the estimation process, resulting reliable result.

There are also some limitations of this method. First, SEM can only be used as a confirmatory method rather than the exploratory method. Therefore, the theory base is required to apply SEM. Second, specification errors may exist in SEM. That is an important construct may not be specified in the model and/or observed measurements may not be specified enough for each construct. Bias created by either of these problems can result in an incorrect interpretation of the results, as in other types of statistical analysis. Third, SEM cannot handle the nonlinear relationship. The important assumption in SEM is that the relationship between the observed variables and their constructs and between one construct and another is linear. SEM has no established tools for handling variations from this assumption, unlike linear regression that has established methods of identifying and proven remedial data transformational methods for handling data that has nonlinear relationships. Using linear regression is recommended to be used in these cases. Fifth, the excess of 20 variables (5-6 constructs each measured by 3-4 indicators) are difficult to manage by SEM.

References:

Gefen, D., Straub, D., & Boudreau, M.-C. (2000). Structural Equation Modeling and Regression: Guidelines for Research Practice. *Communications of the Association for Information Systems*, *4*. https://doi.org/10.17705/1CAIS.00407

Haddock, R. L., & Quinn, M. S. (2015). Recreational Access Management Planning: Understanding Perceptions Regarding Public Forest Lands in SW Alberta. *Open Journal of Forestry*, *05*(04), 387–401. https://doi.org/10.4236/ojf.2015.54033

Shah, R., & Goldstein, S. M. (2006). Use of structural equation modeling in operations management research: Looking back and forward. *Journal of Operations Management*, *24*(2), 148–169. https://doi.org/10.1016/j.jom.2005.05.001

Common Method Bias

1. Definition

Common method bias, also known as common method variance (CMV), refers to the amount of spurious covariance shared among variables because of the common method (same method) used in collecting data and measuring each variable (Buckley et al., 1990; Craighead et al., 2011). These method biases are problematic because the actual phenomenon under investigation becomes hard to differentiate from measurement artifacts. Common method bias threats the the validity of conclusions about the constructs' association and creates the systematic bias in a study either by inflating or deflating the correlations.

All research contains measurement error since measurement is never perfect. Measurement error includes random error and systematic error, which cause negative influence on the conclusion of casual relationship between independent variable and dependent variable. And the latter has more significant threat to the potential conclusion. And the common method bias is a major cause of the systematic error.

Common method bias has the potential to affect the results of a single-method study. And it is normally prevalent in studies where data for both independent and dependent variables are obtained from the same respondent in the same measurement context using the same item context and similar item characteristics. Therefore, for a survey research, it is very important to control common method bias because common method bias can cause erroneous conclusions about relationships between variables by inflating or deflating results.

2. Source of Common Method Bias

Various sources can result in common method biases. The sources are (Tehseen et al., 2017):

(1) Independent and dependent variables are used with the **same item**.
(2) The presence of errors in the measurement items
(3) The context in which the measurement instruments are obtained such as social desirability, leniency bias, consistency motif, mood state of respondents, transient mood state of respondents, etc.
(4) Other sources, such as ambiguous wording and scale length, can also cause common method biases.

3. Endogeneity caused by Common Method Bias

It is caused by measurement method instead of the constructs the measures stand for, therefore, CMV is closely related with the causes of measurement errors. Ullah et al. (2018) summarized the causes of CMV, which includes common-rater effects (e.g., only collecting information from similar respondents), common measurement content (e.g., time, location used to collect data), common-item context or item characteristics (e.g., wording, length and clarity), scale types, respondents, response formats and the general content. And they argue that CMV can have sever negative influence on the relationships between measures or constructs, thus it is a source of endogeneity (Ullah et al., 2018).

4. Types of Common Method Bias

There are four categories: common rater effects, item characteristic effects, item context effects, and measurement context effects. According to the literature, these types of effects differentially influence how the rater responds to questions, thereby resulting in method biases (Malhotra et al., 2006).

5. Remedies to Common Method Bias

3.5.Single Source Study

Remedies for single source studies can be divided into two groups. The first group involves remedies that attempt to statistically detect and/or control for common method biases after the data have been collected. The second group involves remedies that are implemented prior to/during data collection. In essence, these remedies attempt to create some type of separation between the collection of the data for the independent and dependent variables.

3.6.Single Source Studies—No Common Method Bias Remedy

Meta-analyses, which aggregate results across multiple studies, are effective in removing biases (artifacts) inherent in individual studies. The meta-analysis's ability to remove systematic error, of which common method bias is a major contributor, is beneficial in reexamining prior studies where common method bias was not addressed.

However, a meta-analysis requires multiple studies in a topic area and therefore may not be appropriate for important, yet niche, or emerging areas. Under these situations, replication (i.e., duplication of a past study) can be used. If the replication study, addressing common method bias, confirms the prior research, more confidence can be placed on the findings.

3.7.Single Source Studies—Statistical Remedies

Statistical remedies are post hoc statistical tools that attempt to detect or eliminate the presence common method bias.

3.7.1. Harman's Single-Factor Test

The commonly used method is Harman's single-factor test. Generally, in this single-factor test, all of the items in a study are subject to exploratory factor analysis (EFA). Then, common method bias is assumed to exist if (1) a single factor emerges from unrotated factor solutions, or (2) a first factor explains the majority of the variance in the variables. As an alternative to EFA, CFA can be used when implementing Harmon's single-factor test. In particular, in the CFA approach, all of the manifested items are modeled as the indicators of a single factor that represents method effects. Method biases are assumed to be substantial if the hypothesized model fits the data.

The advantage of Harman's single-factor test is that this is a simple and straightforward method that does not require additional resources and can be used in almost any research design. Besides, it can be applied after data collection. While, it also has some limitations. First, it is insufficient sensitivity to detect moderate or small levels of common method bias effects, because as the number of latent factors increases, one factor is less likely to account for the majority of the variance in the manifested variables. And it has been proven that this single-factor technique was extremely unreliable.

Therefore, the Harman's single-factor test is recommending use of this test only "as a last resort" because this technique does not offer an acceptable means to estimate and control for method effects.

3.7.2. Marker-Variable Technique

Marker-variable technique can address the problems related to Harman's test in a single-method research design. The marker-variable technique use a special variable that is deliberately prepared and incorporated into a study along with the research variables. In this approach, a marker variable is implemented in the study such that the marker variable is **theoretically unrelated** to at least one variable in the study. Because the marker variable is assumed to have no relationship with one or more variables in the study, common method bias can be assessed based on the correlation between the marker variable and the theoretically unrelated variable. Such a correlation is denoted by r_M , is treated as an indicator of common method bias. To acquire a reliable estimate of common method bias, a marker variable should be carefully identified **before the start of data collection**. Meanwhile, it is possible to estimate r_M in a **post hoc fashion** without the marker variable identified a priori. Because an uncorrected correlation is influenced not only by true covariance but also by common method bias, the smallest positive value in the correlation matrix, or r_M, would be a conservative estimate of r_M. Although fairly reasonable, the post hoc approach has the potential to capitalize on chance factors.

Within the framework of marker-variable analysis, a method factor is assumed to have a constant correlation with all of the measured items. Under this assumption, a CMV-adjusted correlation between the variables under investigation, will be computed by partialling out r_M from the uncorrected correlation.

There are many advantages of marker variable techniques. It is a flexible method. First, this marker-variable method, unlike the traditional MTMM approach, yields a specific estimate of common method bias along with the statistical significance of the CMV-adjusted correlation between the variables. Second, this method does not force investigators to use multiple methods. Thus, based on theoretical and practical considerations, the marker-variable method seems to be an appealing alternative to assess common method biases.

The marker may be selected prior or post hoc and may be operationalized as a multi-item scale, a single-item scale, or an objective item. However, its validity and efficacy need to be further assessed.

3.7.3. Single Source Studies—Separation Remedies

The problem in common method bias is caused by the independent and dependent variable being captured with the same method. In studies utilizing only one source, this is certainly the situation. However, there are remedies that may alleviate this situation that focus on how the data are collected. In essence, these techniques attempt to create some type of separation when collecting the data.

Psychological separation of measurement is created when the researcher makes it appear that the independent and dependent variables are not related via a cover story or other means. While potentially valuable for addressing common method bias, this form of separation may lengthen the time to complete the survey and thus negatively affect the response rate.

Another approach is to create methodological separation of measurement, which is when participants respond to the independent and dependent variables in various formats, such as Likert scales and open-ended questions. When creating methodological separation of measurements, researchers should use various response formats rather than measures that lie on the objective and subjective continuum. Common method bias can be remedied to the extent some of the measures could be more objective as opposed to purely subjective. For example, using both objective and subjective measures of cycle time can mitigate the effect of common method bias.

Temporal separation of measurement is created when the independent and dependent variables are measured at different points in time. For example, using a longitudinal survey design and administered three waves of data collection procedures by survey. By introducing a time lag between when the independent and dependent variables are measured, the researchers can create the separation of measurement and alleviated the potential threat of common method bias. However, temporal separation does present the challenge of attrition wherein a respondent may complete the first, but not the second portion of the survey. events could occur during the time lag between surveys that could alter the dependent variable and therefore contaminate the results.

3.7.4. Single Source Studies—Statistical and Separation Remedies Combined

Combining statistical and separation methods is the ideal situation because it helps overcome limitations inherent in each approach and provides additional validation to the research conclusions. For example, to address common method bias, the combination of psychological separation and Harman's single-factor test can be used.

The remedies to common method biases in single source studies are summarized as follows (Craighead et al., 2011):

STRATEGIES TO ADDRESS CMV IN SINGLE SOURCE RESEARCH

CMV Remedy	Adaptation to the IT, OM, and SCM Context/Additional Considerations
Harman's Single-Factor Test with CFA setting	• Harman's can be performed with EFA and CFA setting. The CFA setting is recommended. • Since CMV poses a serious threat and replication research in IT, OM, and SCM research is rare, researchers should, at a minimum, use Harman's single-factor test with CFA setting.
Correlated/CFA Marker Variable Technique	• The marker variable technique has performed better in comparison to other post hoc statistical remedies. • It can be used not only to detect, but also to correct CMV. • IT, OM, and SCM research often uses respondents that have some type of "ownership" or responsibility of the focal resource or capability which may cause CMV to inflate theorized relationships, which matches well with the marker variable technique.
Methodological Separation	• Methodological separation of measurement is created when the survey is designed to capture the independent and dependent variables in various formats. • Creating separation in how the data are collected can be beneficial; however researchers should consider the effects of this technique on the statistical analysis of the results.
Psychological Separation	• Psychological separation of measurement is created when the researcher makes it appear that the independent and dependent variables are not related via a cover story or other means. • Creating separation in how the data are collected can be beneficial in single source studies, yet this technique may lengthen the time to complete the survey and thus affect the response rate.
Temporal Separation	• IT, OM, and SCM research often involves a resource or capability. In addition to its effectiveness in combating CMV, temporal separation may capture (at least partially) the lag effect of resources and capabilities on performance. • It may create a challenge of attrition in the survey response.
Separation and Statistical	• Combining the statistical and one or more separation strategies is the ideal situation.

3.8.Multiple Sources Studies

Although single source studies are more efficient in terms of time and cost, multiple sources studies are also very important. Similarly to single source remedies, multiple source remedies can be divided into remedies that attempt to statistically detect and/or control for common method biases after the data have been collected and remedies that are implemented prior to/during data collection (i.e., separation remedies).

3.8.1. Multiple Source Studies—No Common Method Bias Remedies

Using multiple responses to capture both the independent and dependent variables instills more confidence in the research findings, but the study's results still suffer from limitations inherent in single method studies, i.e., the common method biases. Therefore, researchers should also use remedies that can either detect/control or eliminate common method bias when utilizing multiple sources.

3.8.2. Multiple Source Studies—Single Source Statistical and Separation Remedies

The single source statistical and/or separation remedies to common method bias, such as Harman's single-factor test or temporal separation, can also be used in the multiple source studies. Using single source remedies in multiple source studies certainly instills more confidence in the research findings than using no common method bias remedies. However, using multiple source statistical and/or separation remedies will be better and more efficient.

3.8.3. Multiple Source Studies—Multiple Source Statistical Remedies

Multiple source statistical remedies, which are statistical techniques that use both sources to detect and isolate common method bias.

3.8.3.1.Traditional MTMM Procedure

In the MTMM procedure, researchers measure each of the research variables using multiple methods and use the data collected to create an MTMM matrix. The MTMM matrix is a table of correlations among the combinations of traits and methods. In this technique, the extent of common method bias is estimated by the difference between the monomethod-heterotrait (MH) correlations and the heteromethod-heterotrait (HH) correlations.

Accordingly, common method bias is assumed to exist if the average MH correlation is considerably greater than the average HH correlation. Likewise, common method bias is assumed to be trivial if the average MH correlation is comparable to the average HH correlation.

This method has some limitations. First, its results depend heavily on the types of methods employed in a particular study. For example, when measurement methods closely resemble each other, the HH correlations tend to be similar to the MH correlations. In this case, the similarity between HH and MH correlations does not necessarily indicate that the measures are free of common method bias. Second, because the validity of this technique relies on the methods employed, no formal means are available to assess the level of common method bias in the study under scrutiny. As a result, the traditional MTMM approach does not allow the researcher to systematically account for method biases. Finally, this procedure is restrictive in actual application because it requires measurement of each of the traits using at least two methods.

3.8.3.2.CFA-Based MTMM Technique

The CFA technique is known to address this shortcoming that the traditional MTMM technique lacks of capability to assess accurately the extent to which common method bias is present. CFA allows the researcher to model explicitly the variance in a measure as a function of three components that are <u>the "true" score variance, the variance due to method effect, and random error</u>.

Thus, unlike the traditional method, the CFA technique makes it possible to estimate how similar or dissimilar the methods adopted in the MTMM study are. Consequently, it allows estimation of the true relationships between latent factors that are free from method biases and random error.

This method also has some limitation. First limitation concerns the identification of the CFA model. The CFA technique involves complex model specification and thus produces a highly parameterized model. Therefore, the resulting model is said to be often under-identified or to result in invalid parameter estimates. Second, the limitation of this approach is essentially intrinsic to MTMM itself (i.e., the employment of multiple methods). MTMM requires measuring each of the traits using multiple methods, however, in many cases the use of multiple methods for data collection is not viable. Therefore, this sophisticated technique does not seem to be a realistic option for evaluating common method biases.

3.8.4. Multiple Source Studies—Multiple Source Separation Remedies

Single source separation remedies attempt to create some type of separation in how the data are collected, but they still rely on the same source for the dependent and independent variables—conversely, multiple source separation strategies do not. Two common remedies are: (1) escalating the unit of analysis; and (2) using an alternate method to capture either the independent or dependent variable (i.e., methodological separation).

The first multiple source separation remedy, escalating the unit of analysis, consists of two phases. First, a large sample of individuals (e.g., all employees in a company) is reduced to smaller, but still meaningful, subunits (e.g., employees in individual departments). Next, a researcher randomly selects participants from each subunit to estimate values for certain variables and uses the remaining participants in the subunit to estimate values for the other variables. Escalating the unit of analysis decreases the potential of common method bias and generally increases the overall response rate because there are fewer questions on each survey.

Complete methodological separation of measurement involves obtaining measures of the independent and dependent variables from different sources through different methods. Complete methodological separation of measurement is the most rigorous approach to combat CMV. It is recommended to design studies where the independent and dependent variables are captured through multiple sources with multiple methods when possible.

3.8.5. Multiple Source Studies—Statistical and Separation Remedies

Like the single source studies, the combination of statistical and separation remedies can be used to solve common method biases in multiple sources studies. Although using an alternate method to capture either the independent or dependent variables alleviates most concerns of common method bias, using single source statistical remedies may be used when predictor and criterion variables are not captured from the alternate source.

References:

Buckley, Cote, & Comostock. (1990). Measurement Errors in the Behavioral Sciences: The Case of Personality/Attitude Research. *Educational and Psychological Measurement*, *50(3)*.

Craighead, C. W., Ketchen, D. J., Dunn, K. S., & Hult, G. T. M. (2011). Addressing Common Method Variance: Guidelines for Survey Research on Information Technology, Operations, and Supply Chain Management. *IEEE Transactions on Engineering Management*, *58*(3), 578–588. https://doi.org/10.1109/TEM.2011.2136437

Malhotra, N. K., Kim, S. S., & Patil, A. (2006). Common Method Variance in IS Research: A Comparison of Alternative Approaches and a Reanalysis of Past Research. *Management Science*, *52*(12), 1865–1883. https://doi.org/10.1287/mnsc.1060.0597

Tehseen, S., Ramayah, T., & Sajilan, S. (2017). Testing and Controlling for Common Method Variance: A Review of Available Methods. *Journal of Management Sciences*, *4*(2), 142–168. https://doi.org/10.20547/jms.2014.1704202

Moderating Factors

1. Definition:

A moderator variable is a qualitative or quantitative variable that affects the relationship between a predictor variable (X) (i.e., independent variable) and an outcome variable (Y) (i.e., dependent variable). Moderator variables commonly affect the strength of the relationship between X and Y (Gellman & Turner, 2013). In some cases, a moderator can affect the direction of association between X and Y(Gellman & Turner, 2013).

2. Test of Moderation Effect

To make illustration, the following linear regression model with moderating factor is used.

$$Y = aX + bM + cXM + e \tag{1}$$

The Eq. (1) can be rewritten as:

$$Y = bM + (a + cM)X + e \tag{2}$$

For fixed M, this is a linear regression of Y on X. The relationship between Y and X is characterized by the regression coefficient $a + cM$, which is a linear function of M. <u>M is the moderator variable and the magnitude of the moderating effect is measured by c.</u>

<u>To infer that a variable is a moderating variable, there must be a significant statistical interaction between the predictor and the moderator (i.e. p < .05).</u> For the model Eq. (1), the analysis of the moderating effect focuses on testing and estimating c. If c is significant, then the null hypothesis ($H_o: c = 0$) will be rejected and the moderating effect of M is significant.

Moderating effect analysis for two types of variables are considered. The first type is for observable variables, that is the dependent variable, independent variable(s) and the moderator variable(s) are all observables. The second type is for variables, including, independent variable(s) and the moderator variable(s), that contain at least one latent variable (i.e., unobservable variable).

2.1. Test for Observable Variables

Variables X and M can be classified into two categories. On is the category variables, including definite and sequential variables. The other category is continuous variables, including fixed-distance and fixed-ratio variables. When fixed-order variables have more values and are evenly spaced, they can also be treated approximately as continuous variables. The methos are summarized as follow:

Moderator Variable (M)	Independent Variable (X)	
	Category Variable	Continuous Variable
Category Variable	Analysis of variance (ANOVA) with interaction effect of two factors. Interaction effect is moderated effect.	Grouped regression: The sample data is grouped based on the value of M. Regress Y on X. If the difference in regression coefficients is significant, the moderating effect is significant. If the difference in regression coefficients is significant, then

		the moderating effect is significant.
Continuous Variable	Pseudo-variables were used for the independent variables, centering the independent and moderator variables. Centering means subtracting the mean value from every value of a variable. Performing hierarchical regression analysis for $Y = aX + bM + cXM + e$ (3) Regress Y on X and M to obtain R_1^2 (4) Regress Y on X, M and XM to obtain R_2^2. If R_2^2 is significantly higher than R_1^2, then the moderating effect is significant. Or perform the regression coefficient test of XM. if it is significant, then the moderating effect is significant.	Centering the independent variable (X) and the moderator variable (M). Performing hierarchical regression analysis for $Y = aX + bM + cXM + e$ (2) Regress Y on X and M to obtain R_1^2 Regress Y on X, M and XM to obtain R_2^2. If R_2^2 is significantly higher than R_1^2, then the moderating effect is significant. Or perform the regression coefficient test of XM. if it is significant, then the moderating effect is significant. Besides considering the interaction effect of XM, the higher-order interaction effect terms can be considered (e.g., XM^2 for the nonlinear moderating effect; M X^2 for moderating effect of the curvilinear regression).

When using **regression analyses** to detect a moderator effect, multicollinearity problem can be solved by "centering" the predictor (independent variable) and moderator variables. Centering simply means that these variables are transformed by subtracting each variable by its sample mean (e.g., X-mean of X).

Three-Stage Least Squares (3SLS): 3SLS is an instrumental variables estimator for structural equations in which at least one equation contains endogenous independent variables. It is similar to the 2SLS estimation, with the difference being that moderator variables are used as instrumental variables to obtain residuals for the endogenous independent variable. The detailed steps are:

Step 1: Regress each predictor on all moderators, confirm significant relationship between moderatos and the predictor, and obtain residuals for the predictor.
Step 2: Replace each predictor with the corresponding residual and regress dependent variable on obtained residuals.
Step 3: Add the interaction terms to the model.

2.2.Test for Latent Variables

Moderating effects analysis for latent variables needs to use structural equation models (SEM). The measurement of latent variables introduces measurement error, so latent variables are considered to be continuous. Moderated effects models with latent variables usually only consider two scenarios: either the moderator is a categorical variable, and the independent variable is a latent variable, or both the moderator and the independent variable are latent variables.

When **the moderator variable is a categorical variable**, group analysis in SEM is used. The steps are: First, the regression coefficients obtained from the two-group analysis using SEM is restricted to be equal, yielding a chi-square value and corresponding degrees of freedom. Then, this restriction is removed and re-estimating the regression coefficients based on the same model to obtain another chi-square value and corresponding degrees of freedom. The previous chi-square is subtracted from the subsequent chi-square to obtain a new chi-square whose degrees of freedom are the difference between the degrees of freedom of the two models. If the chi-square test is statistically significant, the moderating effect is significant.

When **both the moderator and independent variables are latent variables**, there are many different methods of analysis. For example, the constrained approach of Algina and Moulder (2001) allows to ignore the intercepts and latent means by centering the product of indicators (Steinmetz et al., 2011). This method is palpable for the normal distribution. And the generalized product indicator (GAPI) method of Wall and Amemiya (Wall & Amemiya, 2001), also for the non-normal distribution. Both these two methods are constrained approach and require non-linear constraints, which are cumbersome and error prone.

And there is unconstrained approach proposed by Marsh et al., (2007), which does not require any constraint, greatly simplifying the procedure and easily to be applied in research. This is the making it one of the most convenient methods.

References:

Gellman, M. D., & Turner, J. R. (Eds.). (2013). *Encyclopedia of Behavioral Medicine*. Springer New York. https://doi.org/10.1007/978-1-4419-1005-9

Marsh, H. W., Wen, Z., Hau, K.-T., Little, T. D., Bovaird, J. A., & Widaman, K. F. (2007). Unconstrained Structural Equation Models of Latent Interactions: Contrasting Residual- and Mean-Centered Approaches. *Structural Equation Modeling: A Multidisciplinary Journal*, *14*(4), 570–580. https://doi.org/10.1080/10705510701303921

Preacher and Hayes—2004—SPSS and SAS procedures for estimating indirect ef.pdf. (n.d.).

Steinmetz, H., Davidov, E., & Schmidt, P. (2011). Three Approaches to Estimate Latent Interaction Effects: Intention and Perceived Behavioral Control in the Theory of Planned Behavior. *Methodological Innovations Online*, *6*(1), 95–110. https://doi.org/10.4256/mio.2010.0030

Mediating Factors

1. Definition

A mediator variable is the variable that causes mediation in the dependent and the independent variables, explaining the relationship between the dependent variable(s) and the independent variable(s). Two types of the mediation process are complete mediation and the partial mediation: (1) The process of complete mediation is defined as the complete intervention caused by the mediator variable. This results in the initial variable no longer affecting the outcome variable. (2) The process of partial mediation is defined as the partial intervention.

The mediation caused by the mediator variable is developed as a mediation model, where the mediator variable is assumed to cause the effect in the dependent variable and not vice versa. A set of causal relationships with mediation effect can be defined as follow:

$$Y = \alpha_0 + \alpha_1 D + \varepsilon_{Y_1} \tag{1}$$

$$Y = \beta_0 + \beta_1 D + \beta_2 M + \varepsilon_{Y_2} \tag{2}$$

$$M = \gamma_0 + \gamma_1 D + \varepsilon_M \tag{3}$$

Where Y is the dependent variable, D is the treatment variable (independent variable) and M is the mediator variable. Eq. (1) represents a direct casual effect of D on Y. Eq. (3) describes the casual effect of D on M. And the Eq. (2) establishes the path D→M→Y that implies a casual effect of M on Y and D can also influence Y independently without the help of M. This set of casual relationships can be reflected by Figure 1.

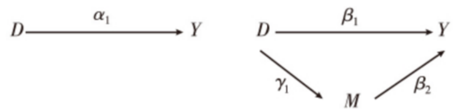

Figure 1. The Mediation Effect

α_1 is the total effect of D on Y, and β_1 is the direct effect of D on Y. $\beta_2\gamma_1$ is the indirect effect that D influence Y via the mediator M. Therefore,

$$\alpha_1 = \beta_1 + \beta_2\gamma_1 \tag{4}$$

2. Test of Mediation Effect

Mediation effects are indirect effects that can be analyzed using structural equation modelling no matter the latent variables exist or not. If all variables are observable, the three regression processes can be performed accordingly based on relationship shown by Figure 1 to test the mediation effect.

2.1.Casual Stop Approach

Baron and Kenny (1986) proposed a set of testing procedures for mediation effects, known as the casual stop approach. To verify the existence of the mediation effect, four conditions should be satisfied: (1) $\alpha_1 \neq 0$ (2) $\gamma_1 \neq 0$ (3) $\beta_2 \neq 0$ (4) $\beta_1 = 0$ or at least $|\beta_1| < \alpha_1$.

The four steps of the testing procedure are:

Step 1: Perform the regression of Eq. (1). That is regressing the dependent variable Y on the treatment variable D and all other control variables to obtain the estimated value $\widehat{\alpha_1}$, representing the total effect of D on Y (not including the mediator M). If the value of $\widehat{\alpha_1}$ is statistically significant, then it implies a potential mediation effect.

Step 2: Perform the regression of Eq. (3) to obtain $\widehat{\gamma_1}$. If $\widehat{\gamma_1}$ is statistically significant, then it implies that the treatment variable D has influence on the mediator M.

Step 3: Perform the regression of Eq. (2) to obtain $\widehat{\beta_1}$. If $\widehat{\beta_1}$ is statistically significant, then it implies that the mediator M affects the outcome (dependent variable) Y.

Step 4: If $\widehat{\beta_1}$ is statistically insignificant, it implies that the complete mediation process exists in the relationship between M and D, otherwise, M represents a partial mediation process.

2.2.Statistical Test

2.2.1. Sobel Test

Use the Sobel test calculator.

2.2.2. Bootstrap Test

Use the p-value of direct and indirect effects.

References:

Baron, R. M., & Kenny, D. A. (1986). *The Moderator-Mediator Variable Distinction in Social Psychological Research: Conceptual, Strategic, and Statistical Considerations*. https://doi.org/10.1037/0022-3514.51.6.1173